SECTION 1 GENERAL

INTERPRETATION OF DOCUMENTS

Definitions 1.01

(1) The *conditions of contract* referred to in this contract are the NEC4 Engineering and Construction Contract Option A: Priced contract with activity schedule (June 2017, with amendments October 2020) as identified in the Contract Data Part One.

(2) The General Specification (GS) referred to in this contract is the General Specifications for Civil Engineering Works, 2020 Edition (the Government of the Hong Kong Special Administrative Region) and the Corrigendum together with any latest amendment issued up to and including the day before the closing date of tender.

(3) The Specifications referred to in this contract shall comprise the General Specification as specified in sub-clause (2) and this Particular Specification.

(4) Not used;

(5) Not used;

(6) Not used;

(7) Not used;

(8) The Construction Site Safety Manual (CSSM) referred to in this Contract is the CSSM, Chapter 3 – Contractual Provisions on Construction Safety, and Chapter 12 – Pay for Safety Scheme (PFSS), version November 2000 (Works Bureau) together with all amendments and additions issued up to and including the date before the close of tender and as set out in this contract.

(9) The latest version of Domestic Arbitration Rules 2014 adopted to take effect by the Hong Kong International Arbitration Centre shall apply to any arbitration.

(10) The Particular Specification (PS) shall be read in conjunction with the *conditions of contract* and *additional conditions of contract*, the Contract Data Part One, Works Information, Site Information and the Construction Site Safety Manual.

Abbreviations 1.02

(3) In addition to the abbreviations defined in GS Clauses 1.02(1) and 1.02(2) which shall also be applicable to PS, abbreviations used in this PS shall have the following meaning:

AMD : Amendments

ASD : Architectural Services Department

CATV : Hong Kong Cable Television Limited

CDR : Compact Disc Recordable

CEDD : Civil Engineering and Development Department

CLP : CLP Power Hong Kong Limited

DSD : Drainage Services Department

EMSD : Electrical and Mechanical Services Department

EN : European Standard

ETWB: Environment, Transport and Works Bureau

ETWBTC: Environment, Transport and Works Bureau Technical Circular

EVA : Emergency Vehicle Access

FSD : Fire Services Department

GMB : Green Mini Bus

GRP : Glass-fibre Reinforced Plastic

HGC : Hutchison Global Crossings

HKG : Hong Kong and China Gas Co. Ltd.

HKPF : Hong Kong Police Force

HKT : PCCW Hong Kong Telecom

HyD : Highways Department

HyD/L : Highways Department (Lighting Division)

LCSD : Leisure and Cultural Services Department

NBWSC : Noise Barrier Works subcontractor

NWTC : New World Telecom (HK) Ltd

PLB : Public Light Bus

SCT : Special Conditions of Tender

TCSS : Traffic Control and Surveillance System

TD : Transport Department

TIA : Traffic Impact Assessment

TMLG : Traffic Management Liaison Group

TPDM : Transport Planning and Design Manual

TTA : Temporary Traffic Arrangement

TTM : Temporary Traffic Management

ULG : Utilities Liaison Group

WBTC : Works Bureau Technical Circular

WSD : Water Supplies Department

Glossary of terms 1.03

(3) Survey Datum Levels are referred to either:

Principal Datum (PD) is defined as the Hong Kong Principal Datum which is the height datum generally used in Hong Kong and referred by the benchmarks of lands Department; or

Chart Datum (CD) is 0.146m below Principal Datum.

(4) Co-ordinates shall refer to the Hong Kong Metric Grid 1980.

(5) "Site" as defined elsewhere in this contract shall comprise "Works Area" and "Works Site", which are referred to in sub-clauses (6) and (7) below.

(6) "Works Area" shall refer to the portions of the Site designated as Works Area on the Drawing. The Works Areas shall be used for the Project Manager's Site Accommodations, Contractor's Site Accommodations and ancillary support facilities such as storage etc.

(7) "Works Site" shall refer to whole of the Site but excluding the Works Area. Limits of Works Site are as shown on the Drawings.

Trials and approval 1.04

(1)***S*** Reference in the Contract to the approval of the *Project Manager* shall mean approval given by the *Project Manager* in writing. Materials, methods of construction and any other matters, which have been approved by the *Project Manager*, shall not be changed without the approval of the *Project Manager* to the proposed changes.

British Standards, European Standards adopted as 1.05

(1)***S*** Unless otherwise stated in the Contract, reference in the Contract to British Standards, British Standard Codes of Practice and similar standards shall be to that edition of the document stated in GS Appendix 1.1, including all amendments issued by the relevant authority up to and

including the closing date of the tender, unless otherwise specified.

British Standards, Codes of Practice and other standards

(3) Except as specified elsewhere in this contract, for any British Standards, European Standards as adopted as British Standards or Codes of Practice not listed in Appendix 1.1 to the GS, the editions of the British Standards, European Standards as adopted as British Standards and Codes of Practice that apply to this contract shall be those editions including amendments which are listed in the British Standards Institution Catalogue.

(4) The *Contractor* shall provide for the use of the *Supervisor's* Representative such British Standards, European Standards as adopted as British Standards, Code of Practice and other national standards in English as he may call for as being required for the execution of this contract. Upon completion of the *works* or at such earlier date as the *Supervisor* may notify in writing, such British Standards, European Standard as adopted as British Standards, Code of Practice and other national standards shall be reverted to the *Contractor*.

PROGRAMME

Programme requirements and progress reporting 1.08***S***

(1) The *Contractor* shall prepare and submit the following works programmes in accordance with the requirements as stipulated in PS Appendix C1.

(i) Accepted Programme (the programme referred to in Clause 31.1 of the *conditions of contract*);

(ii) Executive Summary Programme; and

(iii) Three Month Rolling Programme.

(2) In compiling his Accepted Programme and in subsequent updating and reporting, the *Contractor* shall make provision for the time required for determining and complying with the requirements of Government Departments and entities whose consent, permission, authority or licence is required prior to the execution of any work, for order and delivery periods for any manufactured parts or materials, for the obtaining and preparation of any lands for use in fabricating or preparing any materials or parts offsite and for the transport of such materials and parts to the Site and for any other purposes in connection with the *works*.

(3) The *Contractor* shall submit to the *Project Manager* Monthly Progress Report (MPR) as detailed in PS Appendix C2, describing the current status of the *works*. One copy of the report shall also be submitted in CD-ROM format. The MPR shall address the matters set out

in PS Appendix C2.

(4) The MPR shall be submitted by the end of each calendar month, or at a date as agreed by the *Project Manager* or the *Supervisor*. Unless otherwise agreed, the MPR shall account for all works actually performed up to the date of the report and work forecast to be accomplished.

(5) Not used

CONTRACTOR'S SUPERINTENDENCE

Surveyor 1.09

(1A) The *Contractor* shall submit for the approval of the *Project Manager* within 7 days of the *starting date* the following particulars of his proposed Surveyor:

(i) Name in English and Chinese;

(ii) Copy of identification documents;

(iii) Qualifications with dates of courses and diplomas, degrees etc (whichever applicable), together with certified true copies of certificate; and

(iv) Past experience with contract titles, positions held and dates, together with copies of any documentary proof.

(2)***S*** The *Contractor*'s Surveyor shall possess a Degree in Land Surveying and Geo-informatics from the Hong Kong Polytechnic University or equivalent (local or overseas), plus a minimum of 2 years of experience appropriate to the nature of the survey work included in the Contract, OR Higher Certificate/Higher Diploma in Land Surveying or Geomatics from the Hong Kong Polytechnic University or equivalent (local or overseas), plus a minimum of 4 years of experience appropriate to the nature of the survey work included in the Contract.

(3) The *Project Manager* shall not carry out any checking of the setting out work performed by the *Contractor* until the *Contractor*'s Surveyor or his subsequent replacement is approved.

(4) The *Project Manager* shall have the authority to withdraw the approval of the *Contractor*'s Surveyor and to require the *Contractor* to replace him. Upon notification of replacement of the *Contractor*'s Surveyor, the *Project Manager* shall not carry out any further checking of the setting out survey work performed by the *Contractor* until the replacement is approved.

(5) The *Contractor*'s Surveyor shall be authorised to receive from the *Project Manager* all survey data relevant to the Contract. The *Contractor*'s Surveyor

shall be available on Site for this purpose and/or attending the site meetings when required by the *Project Manager*.

(6) The *Contractor* shall give at least 24 hours' notice to the *Project Manager* when making a request for checking of the survey work. In case the checking of the survey work is required to be carried out on General Holiday or on the day immediately following the General Holiday, the *Contractor* must give notice at least 48 hours before the commencement of the General Holiday.

(7) The *Contractor*'s Surveyor shall be full time on Site during the Contract.

Foreman for concrete, roads and drainage works 1.10A

Identify GS Clause 1.10 as sub-clause (1) and add the following:

(2) The Contractor shall employ on Site in connection with the execution of the works a Foreman who has a minimum of 3 years' experience in roads and drainage works. The Foreman shall be on the Site at all times when roads and drainage works are in progress.

The Contractor's key staff and specially required staff 1.12A

(1) <u>Staffing Proposal – General</u>

(a) Within 7 days of *starting date*, the *Contractor* shall submit his staffing proposal to the *Project Manager* for approval. Staffing shall be of suitably qualified and experienced to manage and supervise the Contract throughout the execution of the *works*. The proposal shall include:

(i) The *Contractor*'s proposed staff organisation showing in chart form the names of his proposed staff for each position;

(ii) Copy of the Hong Kong Identify Card;

(iii) CVs of the *Contractor*'s proposed staff, with adequate details on the qualification and experience of each staff to substantiate that he is competent for undertaking the proposed position;

(iv) The scope of responsibilities of each staff and the reporting lines between individual staff;

(v) The documents that each staff is authorised to sign on behalf of the *Contractor*.

(vi) The documentary proof on the employment status of the staff, such as employment contracts, tax returns, payment of salaries and the like upon the request by the *Project Manager* or a formal declaration to the effect such that a staff is indeed under the direct employment of the *Contractor*.

The declaration shall be signed by a person authorised to sign tenders on behalf of the *Contractor*.

(vii) The estimates man hours and proposed remunerations of staff working for the project site office

(b) The staff organisation shall cover the *Contractor*'s key staff and specially required staff as stipulated in this Clause, as well as other working-level staff with authorities and responsibilities in providing the *works* on Site, or in deciding technical details of the *Contractor*'s proposals.

(c) The key staff in the staffing proposal are prohibited to given a sub-contract to any part of the *works* or to have a vested interest in any of the sub-contractors irrespective of ties including Special Sub-contractors and Nominated Sub-contractors under this Contract.

(d) The *Contractor* shall inform the *Project Manager* forthwith of any changes made to any of the staff in the proposal.

(2) Key Staff

(a) Without prejudice to the generality of GS Clause 1.12, the *Contractor*'s staffing proposal shall include at least the following key people. One person shall only fill up one post.

(i) a Construction Manager responsible for overseeing and managing the *works* under this Contract;

(ii) a Site Agent who shall assist the Project Manager and responsible for day-to-day management to provide the *works*;

(iii) a Quantity Surveyor is responsible for quantity surveying and all relevant payments and assessment of costs for contract.

Unless otherwise specified, minimum requirements on qualifications of the Project Manager, Site Agent and Quantity Surveyor shall be as below:

Position	**Minimum Professional / Academic Qualification Requirement**	**Minimum Post-qualification Experience Requirement**
Construction Manager or equivalent	Corporate Member of the Hong Kong Institution of Engineers (Civil	(i) 10 years relevant post-qualification experience, among which at least 2 years of post-professional qualification experience was gained as Construction

	Discipline) elected after 5 December 1975 or the Institution of Civil Engineers or equivalent	Manager; and (ii) at least 1 year of proven experience as Construction Manager / Project Manager / Site Agent in the use of NEC ECC contract.
Site Agent or equivalent	(i) Corporate Member of the Hong Kong Institution of Engineers (Civil Discipline) elected after 5 December 1975 or the Institution of Civil Engineers or equivalent; or	(i) 5 years relevant post-qualification experience, among which at least 2 years of post-professional qualification experience was gained as Site Agent; and (ii) at least 1 year of proven experience as Construction Manager / Project Manager / Site Agent in the use of NEC ECC contract.
	(ii) University degree or equivalent in Civil Discipline	(i) 10 years relevant post-qualification experience, among which at least 2 years of post-professional qualification experience was gained as Site Agent; and (ii) at least 1 year of proven experience as Construction Manager / Project Manager / Site Agent in the use of NEC ECC contract.
Quantity Surveyor or equivalent	(i) Corporate Member of an appropriate professional institution (e,g, HKIS or RICS) or equivalent; or	No relevant post-qualification experience is required.
	(ii) University degree or equivalent in an appropriate discipline; or	5 years relevant post-qualification experience
	(iii) Diploma/Higher certificate or equivalent in an appropriate discipline	7 years relevant post-qualification experience

(b) All the foregoing staff shall be a holder of a recognized university degree in civil engineering or equivalent in appropriate discipline. In addition, they shall either be a Corporate Member of the Hong Kong Institution of Engineers in respective disciplines or have a similar chartered engineer qualification as agreed by the *Project Manager*.

(c) The Construction Manager shall be authorised to represent the *Contractor* on all aspects under this contract and shall bear overall responsibilities on these aspects.

(d) The Site Agent shall be authorised to represent the *Contractor* on all aspects under this contract and shall be bestowed with adequate authority to receive and carry out the directions and instructions from the *Project Manager*.

(e) The above key staff shall be assisted by engineers and foremen of adequate number and experience for ensuring that the *works* will meet the requirements of the Contract from quality, progress and safety points of view. Without prejudice to the generality of the above, the *Contractor*'s Staffing Proposal shall also include Specially Required Staff as stipulated below.

(f) All the foregoing key people shall be allocated to this Contract on a full-time basis on Site.

(g) Site Superintendent must have an acceptable diploma or an equivalent qualification in Civil Engineering and shall have at least 3 years of relevant experience in roads and drainage works and construction of highway structures projects relevant to their duties.

(h) Unless otherwise specified, minimum requirements on qualifications of the Site Engineers shall be as below:

Position	**Minimum Professional / Academic Qualification Requirement**	**Minimum Post-qualification Experience Requirement**
Site Engineer or equivalent	(i) Corporate Member of the Hong Kong Institution of Engineers (Civil Discipline) elected after 5 December 1975 or the Institution of Civil Engineers or equivalent; or	No relevant post-qualification experience is required.
	(ii) University degree or equivalent in an appropriate discipline (Civil Discipline)	5 years relevant post-qualification experience
	(iii) Diploma / Higher certificate or equivalent in Civil Discipline	7 years relevant post-qualification experience

(3) <u>Specially Required Staff</u>

(a) Surveyor – The *Contractor*'s staffing proposal shall include at least one Surveyor, whose responsibility and minimum requirements/ experience shall be as specified in PS Clause 1.09.

(b) Environmental Officer – The *Contractor*'s staffing proposal shall include at least one Environmental Officer, whose responsibility and minimum requirements/experience shall be as specified in PS Clause 1.104.

(c) Safety Officer – The *Contractor*'s staffing proposal shall include at least one Safety Officer, whose responsibility and minimum requirements/ experience shall be as specified in PS Appendix E1 of the Specification.

(d) *Contractor*'s Labour Officer – The *Contractor*'s staffing proposal shall include at least one *Contractor*'s Labour Officer, whose responsibility and minimum requirements/experience shall be as specified in PS Clause 1.100.

(e) Traffic Liaison Officer – The *Contractor*'s staffing proposal shall include a Traffic Liaison Officer during the Contract Period. The officer shall be competent and possess a recognised university degree or an equivalent qualification in civil/structural engineering with at least five years relevant experience in coordinating works with interfacing projects, utility undertakings, traffic liaison works with Government departments etc. He shall be responsible for all aspects of traffic matters including implementation and maintenance of temporary traffic arrangements and control. He shall also be responsible for liaison with the public including but not limited to the relevant residents, members of District Council and complainants etc in connection with the execution of the whole construction works. The *Contractor* shall ensure that his traffic liaison officer shall be supported in all aspects for satisfactory discharge of his duties.

(f) Temporary Traffic Arrangement and Control Scheme Patrol Team – The *Contractor* shall provide and maintain a Temporary Traffic Arrangement and Control Scheme Patrol Team to carry out patrol and instant rectification works for the TTA with the RSS twice a day. The Patrol Team shall include at least the Site Agent/Sub-agent, Traffic Liaison Officer and General Foreman/ foreman supported by at least 2 labourers who shall all be delegated full-time to the Patrol Team. The main duties of the Patrol Team shall include the up keeping of proper lighting, signing, guarding provisions, carry out instant rectification works for the TTA, report flooding incidents and site cleanliness as well as other related duties as required from time to time by the *Project Manager* as specified in PS Clause 1.15.

Employment of qualified tradesmen /qualified skilled workers and intermediate tradesmen/ 1.12B

(1) The minimum number of ***Qualified Tradesmen /Qualified Skilled Workers and Intermediate Tradesmen/Qualified Semi-skilled Workers*** to be employed by the *Contractor* in accordance with Clause E2 of *additional conditions of contract* shall be calculated in accordance with the following stated percentages of the total local workforce on the Site (i.e. excluding those labour who are imported with the

qualified semi-skilled workers

approval of the Director of Immigration) at any time in each of the following specified trades and rounded up to the nearest whole number :

(a) Building and Civil Engineering Trades

Specified Trade	Minimum Percentage of Qualified Tradesmen/Qualified Skilled Workers	Minimum Percentage of Qualified Tradesmen/Qualified Skilled Workers plus Intermediate Tradesmen/Qualified Semi-skilled Workers
Asphalter (Other Construction) (C301)	20%	20%
Asphalter (Roadworks) (C302)	5%	5%
Bamboo Scaffolder (C303)	60%	80%
Bar Bender and Fixer (C304)	35%	55%
Bricklayer (C305)	60%	75%
Carpenter (Fender) (C306)	5%	5%
Carpenter (Formwork) (C307)	40%	60%
Concretor (C309)	30%	60%
Construction Plant Mechanic (C310)	20%	20%
Drainlayer (C314)	20%	20%
General Welder (C318)	60%	70%
Ground Investigation Operator / Driller / Borer (C320)	80%	80%
Joiner (C322)	40%	60%
Leveller (C323)	60%	70%
Marble Worker (C324)	60%	70%
Marine Construction Plant Operator (C325)	5%	5%
Metal Scaffolder (C327)	30%	30%
Metal Worker (C328)	30%	50%
Painter and Decorator (C329)	55%	75%
Piling Operative (C330)	40%	40%
Pipelayer (C331)	20%	20%
Plant and Equipment Operator (Piling) (C335)	40%	40%
Plasterer (C337)	45%	70%
Plumber (C338)	55%	70%
Shotcretor (C342)	10%	10%
Tiler (C347)	55%	70%
Window Frame Installer (C350)	20%	40%

(b) Electrical and Mechanical Trades

Specified Trade	Minimum Percentage of Qualified Tradesmen/Qualified Skilled Workers	Minimum Percentage of Qualified Tradesmen/Qualified Skilled Workers plus Intermediate Tradesmen/Qualified Semi-skilled Workers
Electrician/ Electrical Fitter (E305)	30%	40%
Building Services Mechanic (E302)	0%	20%
Fire Services Mechanic (E306)	0%	20%

Lift Electrician (E308)	10%	15%
Lift Mechanic (E309)	10%	15%
Mechanical Fitter (E310)	10%	20%
Refrigeration/ Air-Conditioning/ Ventilation Mechanic (E314)	10%	20%
Sheet Metal Worker (E315)	0%	20%

(2) Job descriptions of the specified trades shall be as shown in the Final Report for the Review of Trade Classification in the Construction Industry published by the Works Bureau.

(3) Trade test certificates, certificates of completion of apprenticeship and intermediate trade test certificates are relevant if they are in trades or types of work specified in the following table:

(a) Qualified Tradesmen/Qualified Skilled Workers

Specified Trade	**Relevant Trade Test Certificate (issued either jointly or separately by Vocational Training Council (VTC) & Construction Industry Council (CICTA))**	**Relevant Certificate of Completion of Apprenticeship**
Asphalter (Other Construction)	Asphalter (Waterproofing)	Not available
Asphalter (Roadworks)	Asphalter (Road Construction)	Not available
Bamboo Scaffolder	Bamboo Scaffolder	Bamboo Scaffolder
Bar Bender and Fixer	Bar Bender and Fixer	Not available
Bricklayer	Bricklayer	Bricklayer/Plasterer/Tiler
Carpenter (Fender)	Carpenter (Fender)	Not available
Carpenter (Formwork)	Carpenter, Carpenter (Formwork), Carpenter (Formwork-Building Construction) or Carpenter (Formwork-Civil Construction)	Carpenter/Joiner or Carpenter (Formwork)
Concretor	Concretor	Not available
Construction Plant Mechanic	Construction Plant Mechanic	Construction Plant Mechanic
Drainlayer	Drainlayer	Not available
General Welder	General Welder	Not available
Ground Investigation Operator / Driller / Borer	Ground Investigation Operator	Not available
Joiner	Joiner	Carpenter/Joiner
Leveller	Leveller	Leveller
Marble Worker	Marble Worker	Marble Worker
Marine Construction Plant Operator	Marine Construction Plant Operator	Not available
Metal Scaffolder	Metal Scaffolder	Not available
Metal Worker	Metal Worker	Metal Worker
Painter and Decorator	Painter and Decorator	Painter/Decorator/Sign Writer
Piling Operative (Piling)	Piling Operative (H-Pile) or Piling Operative (Bored Pile)	Not available

Specified Trade	Relevant Trade Test Certificate (issued either jointly or separately by Vocational Training Council (VTC) & Construction Industry Council (CICTA))	Relevant Certificate of Completion of Apprenticeship
Pipelayer	Pipelayer	Not available
Plant and Equipment Operator	Plant and Equipment Operator (H-Pile) or Plant and Equipment Operator (Bored Pile)	Not available
Plasterer	Plasterer	Bricklayer/Plasterer/Tiler
Plumber	Plumber	Plumber
Shotcretor	Shotcretor	Not available
Tiler	Tiler	Bricklayer/Plasterer/Tiler
Window Frame Installer	Window Frame Installer	Not available
Electrician/Electrical Fitter	Electrician (issued by VTC)	Electrical, Electrician or Electrical Fitter
Building Services Mechanic	Not Applicable	Building Services Mechanic
Fire Services Mechanic	Not Applicable	Fire Services Mechanic
Lift Electrician	Not Applicable	Lift Electrician
Lift Mechanic	Not Applicable	Lift Mechanic
Mechanical Fitter	Mechanical Fitter (issued by VTC)	Mechanical, Mechanical Fitter
Refrigeration/Air-Conditioning/Ventilation Mechanic	Refrigeration/Air-Conditioning /Ventilation Mechanic (issued by VTC)	Air-Conditioning, Refrigeration/ Air-Conditioning/ Ventilation Mechanic
Sheet Metal Worker	Not Applicable	Steel Metal Worker

(b) Intermediate Tradesmen/Qualified Semi-skilled Workers

Specified Trade	Relevant Intermediate Trade Certificate	
	Trade or Type of Work	Issued by
Bamboo Scaffolder	Bamboo Scaffolder	CICTA
Bar Bender and Fixer	Bar Bender and Fixer	CICTA
Bricklayer	Bricklayer	CICTA
Carpenter (Formwork)	Carpenter, Carpenter (Formwork), Carpenter (Formwork-Building Construction) or Carpenter (Formwork-Civil Construction)	CICTA
Concretor	Concretor	CICTA
Construction Plant Mechanic	Construction Plant Mechanic	CICTA
General Welder	General Welder	CICTA
Joiner	Joiner	CICTA
Leveller	Leveller	CICTA
Marble Worker	Marble Worker	CICTA
Metal Worker	Metal Worker	CICTA
Painter and Decorator	Painter and Decorator or Painter (Texture-Spray)	CICTA
Plasterer	Plasterer	CICTA
Plumber	Plumber	CICTA
Tiler	Tiler	CICTA

Specified Trade	Relevant Intermediate Trade Certificate	
	Trade or Type of Work	**Issued by**
Window Frame Installer	Window Frame Installer	CICTA
Electrician/Electrical Fitter	Electrical Work	VTC
Building Services Mechanic	Any one of the following : Electrical Work, Sheet Metal Work, Mechanical Fitting, Pipe Works (E & M) or Welding (E&M)	VTC
Fire Services Mechanic	Any one of the following : Electrical Work, Mechanical Fitting or Pipe Works (E & M)	VTC
Lift Electrician	Electrical Work	VTC
Lift Mechanic	Mechanical Fitting	VTC
Mechanical Fitter	Mechanical Fitting	VTC
Refrigeration/Air-Conditioning/ Ventilation Mechanic	Any one of the following : Electrical Work, Sheet Metal Work, Mechanical Fitting, Pipe Works (E & M) or Welding (E&M)	VTC
Sheet Metal Worker	Sheet Metal Work	VTC

(4) For the purpose of complying with the requirements set out in paragraph (1)(b) above, Qualified Tradesmen/Qualified Skilled Workers of the other related E&M trades can also be counted as Intermediate Tradesmen/Qualified Semi-skilled Workers for the specific trade as set out below:

Specified Trade for Intermediate Tradesmen/Qualified Semi-skilled Workers	**Qualified Tradesmen/Qualified Skilled Workers of related trades**
Electrician/ Electrical Fitter	Lift Electrician
Lift Electrician	Electrician/ Electrical Fitter
Building Services Mechanic	Any one of the following except the specified trade for Intermediate Tradesmen/Qualified Semi-skilled Workers: Electrician/Electrical Fitter, Building Services Mechanic, Fire Services Mechanic, Lift Electrician, Lift Mechanic, Mechanical Fitter, Refrigeration/ Air-Conditioning/ Ventilation Mechanic or Sheet Metal Worker
Fire Services Mechanic	
Lift Mechanic	
Mechanical Fitter	
Refrigeration/ Air-Conditioning/ Ventilation Mechanic	
Sheet Metal Worker	

(5) The *Contractor* shall keep daily records of the names and trades of the Qualified Tradesmen/Qualified Skilled Workers and Intermediate Tradesmen/Qualified Semi-skilled Workers working at the Site and shall make such records available for inspection by the *Project Manager* as required.

(6) The Day of Checking is defined as the 15th of each month or the subsequent working day if the 15th is a General Holiday. The *Contractor* shall submit to the *Project Manager*, within 3 days (General Holiday not to be counted) from and including the Day of Checking, a name list of the Qualified Tradesmen/Qualified Skilled Workers and Intermediate Tradesmen/Qualified Semi-skilled Workers working at

the Site in the morning of the Day of Checking. The List shall be submitted in the forms shown in the table in PS Appendix B1. The *Contractor* shall also direct the workers to sign against their names on the forms in PS Appendix B1. In addition, the *Contractor* shall send pages PSB1/2 and PSB1/4 (in Excel format) of the completed forms direct to the CICTA by e-mail (E-mail address: VERIFY@CICTA.edu.hk) on the same day of the submission to the *Project Manager*. Likewise completed forms should be sent to VTC by e-mail (E-mail address: aputtr@vtc.edu.hk) and to EMSD (E-mail address: contract@emsd.gov.hk), if applicable. The submission of this information in electronic format is to facilitate CICTA / VTC / EMSD in processing the submitted forms. Checking for compliance with the contractual requirements is based on the hard copies of the submitted forms. Submission of the forms in PS Appendix B1 will not be required after the issue of the Certificate of Completion.

(7) Random site checks will be carried out by the *Project Manager*'s site supervisory staff on the workers working on Site. The *Contractor* shall ensure that all Qualified Tradesmen/Qualified Skilled Workers and Intermediate Tradesmen/Qualified Semi-skilled Workers will be able to produce their trade test certificates, intermediate trade test certificates, trade test certification cards, intermediate trade test certification cards, certificates of completion of apprenticeship or registration certificates under the Electricity Ordinance (Cap. 406) upon request.

CONSTRUCTION SAFETY AND SAFETY OF THE PUBLIC

Construction safety and safety of the public

1.13 (1)***S*** The *Contractor* shall comply with the current Construction Site Safety Manual (CSSM) issued by the Environment, Transport and Works Bureau (ETWB) of the Government of the Hong Kong Special Administrative Region (HKSAR) as well as the safety requirements specified in PS Appendix E1, in additional to all relevant safety legislations and regulations in force, and codes of practices promulgated by the Labour Department or other statutory safety and health institutions. The *Contractor* shall keep one set of the current CSSM, the relevant safety legislations, regulations and/or codes of practices in the *Contractor*'s Office and the *Project Manager*' Accommodations.

(8) The *Contractor* shall prepare and submit to the *Supervisor* a Safety Plan in accordance with Clause D6 of *additional conditions of contract* in respect of all aspects of the *works*. The *Contractor* shall ensure that the Safety Plan is regularly reviewed and updated to suit the progress of the *works*.

(9) In addition to the safety requirements stated in the Contract, the Safety Officer shall prepare and submit to the *Supervisor* weekly, from the commencement of the *works* up to the date of substantial completion of the *works* or at a date proposed by the *Contractor* and agreed by the *Supervisor*, a safety inspection report endorsed by the Safety Officer. These reports shall be in the form of checklist as shown in PS Appendix B2 or in any form and detail as required by the *Supervisor*.

(10) If the two-month moving average of the accident rates is above the contemporary average accident rate of Highways Department or the

cumulative number of accidents continue to rise, the *Contractor* will be required to submit monthly reports in writing to detail his latest remedial measures and efforts to reduce the high accident rate. This will continue until the two-month moving average is lower than the contemporary average accident rate of Highways Department.

Site safety cycle 1.13A (1) The *Contractor* shall practise "Site Safety Cycle" (SSC) to improve and promote the safety and health of the Site. Site Safety Cycle shall begin when there are workers working on the Site, and shall cease by the date of substantial completion of the *works*, or at a date proposed by the *Contractor* and approved by the *Supervisor*. The activities for Site Safety Cycle for one day, one week and one month are referred to as the "Daily Cycle", "Weekly Cycle" and "Monthly Cycle" respectively. Details of the activities and the provisions for holding activities are described in PS Appendix E2. For simplicity, the Pre-work Exercise and Safety (PES) meeting, Hazard Identification Activity (HIA) meeting and Pre-work Safety Checks of the Daily Cycle are collectively referred to as the "Pre-work Activities".

(2) The *Contractor* shall arrange Pre-work Activities to be held for attendance by persons employed on the *works* (excluding clerical and administrative staff in site office), irrespective of whether they are in the employment of the *Contractor* or his subcontractors. For the avoidance of doubt, persons employed on the *works* are those persons whose number of man-hours worked on the Site are to be included in the number of man-hours worked for the Contract. The *Contractor* is encouraged to arrange Pre-work Activities to be held daily, but in any case the frequency of Pre-work Activities for attendance by each person employed on the *works* shall be not less than once in a week (commencing on Monday). Where necessary, the *Contractor* can arrange more than one session of Pre-work Activities to be held in a day in order to suit the large workforce or the different times of arrival of workers at the Site. The Pre-work Activities shall be carried out prior to any work carried out by the persons attending the Pre-work Activities on that day. Furthermore, the number of persons attending the Pre-work Activities in a session shall be governed by the hard-paved area which shall be sized based on the rate for a person specified in PS Appendix E2. Details of the Pre-work Activities are described in PS Appendix E2.

Report on Safety Performance and Payment for Performance-tied Payment Items 1.13B (1) The payments on safety cover both the task-tied payment items for the implementation of stipulated safety measures such as provision and updating of safety plan, attending safety meetings and safety walks, provision of safety officers and safety training etc. and the performance-tied payment items which are measured according to the *Contractor*'s achievements in the performance indicators such as no reportable accident in a month, accident frequency rate, no notice of safety and environmental prosecution, etc. The task-tied payment items are payment items under the Pay for Safety Scheme (PFSS) and are listed in Schedule No. 5 of the Activity Schedule. The performance-tied payment items are payment items under the Pay for Safety Performance Merit Scheme (PFSPMS) and are listed in the Schedule of Performance-tied Payment Items of the PFSPMS in Appendix A to Contract Data Part One. Measurement and payment for the performance-tied payment items shall be in accordance with the

additional conditions of contract Clause D28.

(2) In the performance-tied payment items of the PFSPMS, "reportable accident" means accidents arising from activities of the *works* resulting in any person suffers death or injury with incapacity for more than three days but excluding the following types of accidents:-

(a) accident occurred in a site office but was not related to any activity of the *works*;

(b) accident concerning the injury of a government staff;

(c) accident concerning the injury of resident site staff of the HKSAR Government or its agents including those employed by consultants; and

(d) accident concerning the injury of visitors or the public to the Site and was not related to any activity of the *works*.

(3) The accident frequency rate is calculated as the number of reportable accidents per 100,000 man-hours worked in which:-

(a) wherever appropriate, the rules for counting reportable accidents for accidents statistics in Section 9.3 in Chapter 9 of Construction Site Safety Manual published by the Development Bureau shall be followed; and

(b) "man-hours worked" is the man-hours worked by all persons employed by the *Contractor* and his sub-contractors who are exposed to risk, including the *Contractor*'s site Supervisory staff, site agent and engineer(s), workers and watchmen etc., and the man-hours of *Supervisor's* site staff are to be excluded.

(4) The *Contractor* shall submit a "Monthly Report on Safety Performance" (the Monthly Report) to the *Supervisor* using the prescribed proforma in Appendix E3 of this Particular Specification on the measurement of the performance-tied payment items of the PFSPMS for each month falling within the period which commences on the *starting date* and ends on the date six months after the completion date or on an earlier or later date notified by the *Supervisor* provided that such date shall not be earlier than the last date of the time for completion of the *works*. The first month shall commence on the *starting date* and end on the last day of the calendar month within which the *starting date* falls. The second month shall commence on the first day of the second calendar month and end on the last day of the second calendar month and the subsequent months shall run in a similar way. The last month shall end on the date on which the said period ends. The Monthly Report shall be prepared and signed by the Safety Officer and endorsed by the Site Agent of the *Contractor*. For the purposes of this Clause, the terms "calendar month", "earliest date of possession of the Site", "time for completion" and "last Section" shall have the meaning assigned to those terms in the Measurement Rules for the Performance-tied Payment Items of the PFSPMS in Appendix A to Contract Data Part One.

(5) The *Contractor* shall submit documentary proof for the performance achievements stated in the "Monthly Report on the Safety Performance" to the *Supervisor*.

(6) No measurement and payment on the performance-tied payment items shall be assessed by the *Supervisor* unless the performance achievements on the corresponding indicators reported by the *Contractor* in the duly completed and signed Monthly Report has been agreed by the *Project Manager's* Representative.

(7) If it is subsequently aware by the *Contractor* or the *Supervisor* that adjustment(s) on any of the Monthly Reports on the Safety Performance, including those submitted and agreed by the *Supervisor* previously, is/are required, it is the responsibility of the *Contractor* to revise and re-submit the concerned Monthly Report with adjustment(s) to the *Supervisor* for correction and agreement. Any previous measurement(s) and payment(s) made to the *Contractor* shall be deducted or adjusted accordingly.

WORK ON ROADS

Approval for temporary traffic arrangements and control 1.14

(3) The relevant authorities for the purpose of this clause shall be those given in PS Clause 1.16***S*** (8) or those that the *Supervisor* may advise from time to time.

(4) The *Contractor* shall follow the procedures and requirements as the *Supervisor* may require for obtaining the prior endorsement from the relevant authorities and the approval from the *Supervisor* with respect to any temporary traffic arrangements and controls. Proposals submitted for approval shall include details of all lighting, signing and guarding to be implemented.

Temporary traffic arrangements and control 1.15

(1)***S*** The *Contractor* shall construct and maintain diversion ways wherever the *works* will interfere with existing roads, footways or other ways over which there is a public or private right of way. Temporary traffic arrangements (TTAs) and control shall not be implemented without the prior endorsement of the relevant authorities and shall be agreed by the *Supervisor*. The *Contractor* shall provide such evidence as the *Supervisor* may require that the endorsements of the relevant authorities have been obtained. No works requiring temporary traffic arrangements and control shall commence until such arrangements and controls have been implemented to the satisfaction of the relevant authorities and the *Supervisor*. The *Contractor* shall provide, construct and maintain, unless otherwise specifically provided for in the Contract, all necessary lighting, signing, guarding and things all to the satisfaction of the relevant authorities and agreed by the *Supervisor*.

(2)***S*** Temporary traffic arrangements and control for work in roads and footways or other ways shall comply with the requirements, which are deemed to be the minimum requirements, contained in the fifth issue 2017 edition of the Code of Practice for the Lighting, Signing and Guarding of Road Works issued by the Highways Department of HKSAR at the time of implementing the TTA.. The *Contractor* shall maintain two copies of the Code of Practice on site at all times, one copy of which shall be for the exclusive use of the *Supervisor*.

(7)***S*** The *Contractor* shall inspect and regularly maintain the temporary traffic arrangements and control, both day and night. He shall keep the traffic lights, lights and signs clean and easy to read, and shall immediately repair or replace the equipment that is damaged, dirty, incorrectly positioned or not in working order. Any equipment which, in the opinion of the *Supervisor*, is in an unsatisfactory condition or is damaged shall be replaced immediately. All unsatisfactory equipment shall be removed off the Site by the *Contractor*.

(8) The temporary traffic arrangement proposals should be designed in accordance with the Transport Planning and Design Manual and any other highway standards and guidance notes issued by the Highways Department.

(9) In addition to the minimum requirement stipulated in the 'Code of Practice for Lighting, Signing and Guarding of Road Works', the *Contractor* shall comply with the following requirements for all works executed in this Contract:

(a) At the commencement of the Contract, all lighting, signing and guarding equipment shall be new (or renovated to almost new conditions to the satisfaction of the *Supervisor*), clean and in good working condition and marked with manufacturing dates (or renovated date), in month/year or stamped by a punch without damaging them or other method approved by the *Supervisor*. The letter size shall not be smaller than 12mm (h) and 8mm (w).

(b) All directional sign faces shall be illuminated unless otherwise agreed by the *Supervisor*. Other sign faces shall also be illuminated where considered necessary by the *Supervisor*.

(c) For every week or at any time when ordered by the *Supervisor*, when lighting, signing and guarding equipment, plastic banner(s) and rubbish bin(s) are in use on Site, the *Contractor* shall clean and maintain such equipment on Site in good conditions to the satisfaction of the *Supervisor*. The *Contractor* shall submit for the approval of the *Supervisor* the cleaning method. All cleaning water shall not be drained to the public carriageway as surface runoff. All damaged lighting, signing and guarding equipment, plastic banner(s) and rubbish bin(s) shall be removed off the Site and replaced immediately.

(d) Every three months, all lighting, signing and guarding equipment, plastic banner(s) and rubbish bin(s) shall be cleaned thoroughly with approved detergents, re-painted, renovated and refurbished (i.e. free of dent, bent, dirt, rust and paint damages) with the date of reconditioning marked, or amended as the case may be, adjacent to the manufacturing dates) and maintained in good working conditions to the satisfaction of the *Supervisor*.

(e) The *Contractor* shall carry out daily inspection of the lighting, signing, and guarding equipment for each temporary traffic arrangement and control implemented to ensure that all equipment are in working order, clean and aesthetically pleasing to the satisfaction of the *Supervisor*.

(f) The defects so identified in (e) shall be rectified within the following specified time limit:

(i) immediately upon instruction by the *Supervisor* if the defects are urgently required to be rectified for safety reason.

(ii) within two hours if the defect will void the endorsement of the traffic authority for the temporary traffic arrangements or the traffic lane/ road closure.

(iii) within 24 hours in normal situation.

(g) The *Contractor*'s name shall be distinctively shown in both English and Chinese on every temporary barrier, traffic cone, on the back of every temporary traffic sign and on the rubbish bin.

(h) Each temporary barrier shall contain a HyD logo plate and shall be of the type shown on Drg. No. H6112 and H6144 to H6147 or of any other type and size approved by the *Supervisor*. Pedestrian barriers placed on carriageways adjacent to running traffic lanes should be of plastic type the details of which shall be submitted to the *Supervisor* for agreement before use.

(i) Notice boards as detailed on Drg. No. H6112 and H6144 to H6147 erected on approved supports shall be supplied and maintained for display on site for every Works Order. Notice boards shall either be new or properly renovated and refurbished to remove all dirts and rusty marks. Damaged painting work must be repainted.

(j) Steel plates supplied and maintained over trench opening as vehicle crossing shall be treated with anti-skid dressing as detailed on Drg. No. H6136.

(k) The *Contractor* shall remove a closure on a road or footpath immediately subsequent to completion of works, or when the *works* in one stage are completed and the *works* in the following stage cannot be continued immediately.

(l) After each day or night's work and during non-working periods every day, the *Contractor* shall relocate the lighting and guarding equipment for each closure on a road or footpath such that only the minimum area necessary for enclosing the *works* (including areas under curing) is closed to vehicular or pedestrian traffic. No allowance in a closed area during non-working hours in the closure shall be left for working space.

(m) Unless otherwise approved by the *Supervisor* in writing, each lane closure on a high-speed road shall be manned by a traffic controller and an assistance traffic controller at all times, including meal time, (i.e. full-time watching required) during the period of closure. The traffic controller shall be responsible for coordinating the entry/exit of all vehicles/plant in connection with the lane closure. The *Contractor* shall provide a

telecommunication system e.g. mobile telephones, trunked-walkie-talkies for the concerned traffic controller, assistant traffic controller, plant operators, vehicle drivers and site supervisors. A summary showing the details of the telecommunication system provided for each closure shall be reported daily to the *Supervisor* and shall be submitted together with the notification of progress of works. Unless otherwise agreed by the *Supervisor* in writing, a lane closure shall be removed immediately on completion of each day or night's work.

(n) Unless otherwise approved by the *Supervisor* in writing, any intermittent full road closure of a local road and the temporary traffic arrangement therefore shall be manned by a traffic controller, an assistance traffic controller and a Site Agent at all times including meal time. The traffic controller and the assistant traffic controller shall be responsible for coordinating the entry/exit of all vehicles/plant in connection with the full road closure. The Site Agent shall be responsible for supervising temporary traffic arrangement and managing Stop/Go sign at the time of implementing full road closure. The *Contractor* shall also provide a telecommunication system e.g. mobile telephones, trunked-walkie-talkies for the concerned traffic controller, assistant traffic controller, Site Agent, plant operators, vehicle drivers and site supervisors. A summary showing the details of the telecommunication system provided for each full road closure shall be reported daily to the *Supervisor* and shall be submitted together with the notification of progress of works.

(10) Electric lamps complying with BS 3143 shall be used when a connection to a mains electricity supply is available and feasible. Supply voltage for such lamps shall be stepped down to 50 volts or less. When a mains electricity supply is not available, battery powered lamps shall normally be used. Kerosene lamps may only be used in emergencies and the *Supervisor* shall be informed immediately of their use.

(11) Temporary road markings shall be hot-applied thermoplastic material and comply with the requirements of GS Clause 12.23.

(12) Temporary traffic signs shall be retro reflective and comply with the requirements of GS Clause 12.09.

(13) The *Contractor* shall also provide and maintain licensed flashing arrow signs on vehicles or trailers on a 24 hour basis for the temporary traffic diversions and approved measures as appropriate in connection with PS Clauses 1.16***S*** and 1.16A below.

(14) If the condition of any lighting, signing and guarding equipment used on site is considered not acceptable by the *Supervisor* under the above sub-clause (9) and which is considered not due to the *Contractor*'s default, then, when directed by the *Supervisor*, such lighting, signing and guarding equipment shall be replaced by new or satisfactorily refurbished equipment to the satisfaction of the *Supervisor* and the unsatisfactory equipment shall be transported off Site for renovation and refurbishment.

(15) During transportation of the renovated and refurbished lighting, signing and guarding equipment including the plastic banner(s) and rubbish bin(s), the *Contractor* shall take good care of the equipment. The equipment shall not be subject to rough handling, shock loading or dropping from height. Instead the equipment shall be placed in an orderly manner which will not result in damage or deformation.

(16) Different types and sizes of lighting, signing and guarding equipment shall be placed separately during transportation. Painting to temporary barriers, notice boards and publicity broads for motorist shall be well protected from damage or being scratched or peeled off during transportation. Reflective jackets of traffic cone shall be protected from damage.

(17) If the *Contractor* fails to replace any unsatisfactory equipment for traffic control, lighting and guarding when instructed to do so by the *Supervisor* or fails to provide approved equipment at locations as instructed by the *Supervisor*, then without prejudice to any other provisions of the Contract, the *Contractor* shall not be entitled to payment under items for traffic control, lighting and guarding for the period during which the *Contractor* fails to comply with the requirements of the Contract, and the *Client* shall be entitled to have such work carried out by a person other than the *Contractor*. All additional costs properly incurred by the *Client* in so doing shall be recoverable from the *Contractor* in accordance with the Contract.

Works executed on High Speed Roads 1.15A

(1) The *Contractor* shall comply with the following additional requirements for all works executed on High Speed Roads. Such requirements will include but not be limited to the following:

(a) Requirements put forth by the authorities on restricted working hours, lane working and temporary traffic arrangements and control.

(2) Cyclic lane closure programmes are implemented by HyD Regions for the regular maintenance of the High Speed Roads. The *Contractor* is encouraged to carry out any necessary expressway works in conjunction with the cyclic lane closure programmes. Cyclic lane closure programmes shall be circulated in advance to the *Contractor* for their planning purpose.

Planning requirements for working on red and pink route 1.15B

The *Contractor* shall comply with the planning requirements as given in Highways Department Technical Circular (HyDTC) No. 8/2003 or its updates when undertaking road works on red and pink routes. The current requirements in HyDTC No. 8/2003 are described below. The location of red and pink routes is shown on Highways Department Technical Circular No. 5/2001 and may be updated from time to time by the *Client*.

(a) Short-term road works on red and pink routes:

(i) Short-term road works refers to those works requiring closure of road lanes for a period of less than twenty-four hours.

(ii) All short-term road works on red and pink routes shall be prepared for evacuation at any time for unexpected incidents.

Partial reopening of one or more lanes within 30 minutes and complete evacuation within one hour is considered acceptable. The roadwork site shall be confirmed to be safe before being reopened for public use.

(iii) Prior to commencing any short-term road works on red and pink routes, the *Contractor* shall submit and agree with the *Supervisor* **a detailed programme, a contingency plan and a checklist** to ensure that the route can be reopened on time and that traffic will not be blocked in case of the road works being delayed.

(iv) The **detailed programme of works** shall be prepared at half-hour intervals with key milestones for individual operation/phase of works. Milestones or critical timings ale used to determine whether the planned activities shall be revised or cancelled due to delays occurring.

(v) The **contingency plan** shall cover different scenarios such as car accidents happening near the Site, breakdown of plant, inadequate supply of material or adverse weather condition. The contingency plan shall include the following issues:

- number and type of critical standby plant and their stationed area;
- standby materials and the stockpile area;
- any alternative traffic diversion routes; and
- contingency or temporary measures to reopen the road and the time required for partial and complete evacuation.

All standby plant and material are required to be ready on site within 30 minutes upon request by the *Supervisor*.

(vi) The **checklist** shall include at least the following items:

- checking of plant;
- estimate of and provision of materials used for road works;
- standby material and equipment;
- the detailed programme;
- the contingency plan;
- a contact list showing the contact points of the *Contractor*'s Site Agent, foreman and ganger responsible for the road works.

The *Contractor* shall agree the checklist with the *Supervisor*. Prior to the commencement of the *works*, the *Contractor* shall submit the checklist for verification by the *Supervisor*. The *Contractor* shall only be allowed to commence works when the items on the checklist are confirmed by the *Supervisor*.

(vii) For road works which can be reopened to traffic within one hour or emergency road works, the requirement of a detailed programme, a contingency plan and a checklist can be waived. However, the *Contractor* shall stand ready to evacuate the site quickly when required.

(b) Road works other than short-term road works or emergency works on red and pink routes:

(i) The *Contractor* shall be required to submit a contingency plan as described in sub-clause (a)(v) above for roadworks other than short-term road works or emergency works on red and pink routes. In addition, the *Contractor* shall ensure that the contingency plan shall cover situations where serious traffic congestion due to the *works* is envisaged.

(ii) Prior to the commencement of the road works, the *Contractor* shall provide the *Supervisor* a contact list showing the contact points of his Site Agent, foreman and ganger responsible for the road works.

Unsatisfactory traffic control, watching and lighting

1.15C (1) If the *Contractor* fails to provide approved traffic control, lighting, signing and guarding equipment on locations as directed by the *Supervisor* or fails to rectify unsatisfactory equipment or traffic control measures when ordered to do so by the *Supervisor*, the *Supervisor* may take remedial action as he deems appropriate, including stopping the *works* affected until the *Contractor* complies in all respects with the requirements. Under such circumstances the *Contractor* shall not be entitled to any extension of time for completion as stated in the Contract nor compensation of expense in connection thereto.

(2) Notwithstanding any other provisions of the Contract, the *Supervisor* shall also be entitled to have such work carried out by the *Client*'s own workmen or by other persons, all at the *Contractor*'s expense.

(3) All costs and charges properly incurred by the *Client* in so doing shall on demand be paid to the *Client* by the *Contractor*, or may be deducted by the *Client* from any moneys due or becoming due to the *Contractor* under this or any other contract with the *Client*.

Particulars of temporary traffic arrangements and control

1.16***S*** (1) The *Contractor* shall engage a traffic consultant not an associated body of the *Contractor* for ensuring compliance with the traffic-related requirements stipulated in the contract and advising on all traffic-related matters, including but not limited to: -

Preparing and updating Traffic Impact Assessment (TIA) and Traffic Management Contingency Plan (TMCP), as well as proposing Temporary Traffic Arrangements and Control (TTA) for acceptance in accordance with the contract.

Forecasting the traffic conditions (such as VC ratio and queue length) after implementation of the proposed TTA modelled by sophisticated traffic software.

Attending TMLG meeting and other meetings with the relevant

authorities, parties, and public, presenting the TTA proposals and other traffic-related proposals/reports, and handling all comments, complaints and enquiries.

Formulation of trial runs proposal before acceptance of the TTA and monitoring plan after the implementation of the TTA, evaluating and assessing the traffic impacts based on the observations and data collected.

Provide technical inputs on all traffic related matters.

(2) The proposed traffic consultant for acceptance by the *Supervisor* shall have proven experience and history as a traffic consultant responsible for the TTA for at least three roadworks projects in urban areas in the last 5 years, the contract sum of each project exceeds HK$100M.

(3) The traffic consultant shall propose a project traffic engineer responsible for this project. Minimum requirements on the project traffic engineer's qualifications/experience are: -

Recognised university degree or an equivalent qualification in civil engineering and a Corporate Member of the Hong Kong Institution of Engineers or have a similar chartered engineer qualification agreed by the *Supervisor*;

At least 10 years of experience in carrying out traffic impact assessment study and proposing temporary traffic management and control measures for roadworks projects.

(4) Besides the project traffic engineer, the *Contractor* shall also provide professionals and technicians of adequate number and experience to assist the project traffic engineer in advising all traffic-related issues.

(5) The following particulars of proposed temporary traffic arrangements and control shall be submitted to the *Supervisor* for approval at least 7 days before the traffic arrangements and control are implemented:

(a) Details of traffic diversions and pedestrian routes,

(b) Details of lighting, signage, guarding and traffic control arrangements and equipment, and

(c) Any conditions or restrictions imposed by the Commissioner for Transport, the Commissioner of Police, the Director of Highways or any other relevant authority, including copies of applications, correspondence and approvals.

(6) All temporary traffic arrangements and control shall be designed by an independent, experienced and qualified Traffic Consultant appointed by the *Contractor*.

(7) The Traffic Consultant shall design temporary traffic arrangements and control to fulfil the traffic diversion requirements under the Contract, with supporting traffic counts, traffic results, signal phasing diagrams, capacity, queue length calculations, swept path analysis and turning radii. The *Contractor*'s Traffic Consultant shall also be responsible for

checking that the temporary traffic arrangements and control measures are installed and maintained in accordance with the approved design and plans to the satisfaction of the *Supervisor*. Any changes to the approved design will require resubmission to Traffic Management Liaison Group (TMLG) for vetting.

(8) Notwithstanding sub-clauses (2) and (3) above, the *Contractor*'s Traffic Consultant shall monitor constantly the traffic conditions and amend his temporary traffic arrangements and control design during the implementation in order to meet the prevailing traffic demand or the requirements raised by Hong Kong Police Force, Transport Department, Highways Department, the relevant authorities and the TMLG as stipulated in sub-clause (8) below.

(9) The *Supervisor* may not consider the *Contractor*'s proposals for temporary traffic arrangements and control before he has given approval of the *Contractor*'s proposed Traffic Consultant.

(10) The *Supervisor* shall have the authority to withdraw the approval of the approved Traffic Consultant and to require the *Contractor* to replace him within 7 days of the *Supervisor*'s notification.

(11) Immediately after the date for commencement of the *works*, the *Contractor* shall prepare detailed construction sequences with associated temporary traffic arrangements and control and obtain endorsements from the relevant authorities and the *Supervisor*'s Representative within one month. All temporary traffic arrangements and control proposals shall be designed to the requirements as stated in Clause 1.16A, and any conditions or restrictions described on the Drawings or imposed by Hong Kong Police Force, Transport Department, Highways Department, the relevant authorities and the TMLG as stipulated in sub-clause (8) below.

(12) All temporary traffic arrangements and control shall be commented on and endorsed by a Traffic Management Liaison Group (TMLG) established by the *Contractor*. The *Contractor* shall not carry out any roadworks or demolition works prior to the implementation of the endorsed temporary traffic arrangements and control. The TMLG shall comprise representatives from various government departments and authorities which may include, but are not limited to, the following :-

(a) Transport Department

(b) Hong Kong Police Force – Road Management Office (RMO)

(c) Highways Department including representatives from NT Region, and Bridges & Structures Division (TMCA)

(d) District Offices

(e) Drainage Services Department

(f) Electrical and Mechanical Services Department

(g) Environmental Protection Department

(h) Fire Services Department

(i) Food and Environmental Hygiene Department

(j) Leisure and Cultural Services Department

(k) Water Supplies Department

(l) The *Supervisor*'s Representative

(m) The *Contractor*

(n) Traffic Consultant for the *Contractor*

In addition, the *Supervisor* shall invite other affected parties to attend the TMLG meetings on an as need basis. These affected parties may include local community organisations or any other affected bodies who will attend as observers to voice any concerns or provide comment on the proposed temporary traffic arrangements and control under review at the meeting.

(13) The TMLG shall meet monthly or at intervals agreed by the TMLG members. The meeting shall be convened by the *Contractor* and chaired by the *Supervisor*'s Representative. The function of the TMLG shall be to scrutinize detailed temporary traffic arrangements and control proposed by the *Contractor*'s Traffic Consultant as well as to oversee and ensure that adequate unimpeded safe routes through the Site are available for both vehicular, emergency and pedestrian traffic at all times. In order to ensure the proper functioning of the TMLG, the *Contractor* shall circulate proposed traffic management measures to group members at least 7 days prior to the intended date of the TMLG meeting at which the proposals are to be scrutinized. If changes in traffic signal operations are involved, at least 21 days shall be allowed for the part of traffic signal modification works to be carried out by the Traffic Control Division of Transport Department and Electrical and Mechanical Services Department. The endorsement of the temporary traffic arrangements and control shall be determined by all the members of TMLG, excluding the *Contractor* and his Traffic Consultant. The *Contractor* shall obtain an approval of the submission and report on the progress to the *Supervisor*. Decisions of the TMLG shall be submitted to the *Supervisor*. The *Contractor* shall also produce minutes of meeting within 7 working days after each meeting and circulate the minutes to the members of the TMLG for comments and agreement.

(14) For all adjacent interfacing contracts with their respective TMLGs come into operation, for the case when the TMLGs are consolidated together, the decision of the relevant authorities and the consolidated TMLG will be binding on the *Contractor* who shall be required to modify his overall construction sequences to comply. The *Contractor* shall allow for such flexibilities in his programme and construction sequences.

(15) The temporary traffic arrangements and control proposals shall be presented to the TMLG meeting at least 6 weeks prior to the

implementation of the scheme and construction works. Details of temporary traffic diversions to be circulated to members of the TMLG shall include all details as described in PS Clause 1.16A(3) and shall be prepared on 1:500 scale drawings or such larger scale as may be necessary to properly describe the proposals as directed by the *Supervisor*. Where major traffic diversions or lane closures or amendments to clearway restrictions, speed limit or expressway restrictions need to be authorized through publication in the Government Gazette and/or to be publicized through a press release, the time between submission of drawings/document and implementation shall be increased as advised by the Commissioner for Transport and the *Contractor* shall make due allowance for this in his submission.

(16) A monthly report shall be prepared by the *Contractor* giving the status of all temporary traffic arrangements and control in operation, the findings of the *Contractor*'s Traffic Consultant's checks described in sub-clause (20), effects on traffic and the traffic arrangements to be implemented in the following three months. This report shall be circulated to all TMLG members 2 weeks prior to the TMLG meeting and presented to the meeting by the *Contractor* or his Traffic Consultant.

(17) Any delay to the *works* resulting from the *Contractor*'s delay in submission, or from his failure to submit sufficient and correct design details, shall be the responsibility of the *Contractor*.

(18) The number of works areas and extent of each works area on roads shall be proposed by the *Contractor* for the endorsement from TMLG. No works shall be allowed to be carried out on roads unless endorsement of TMLG has been given to the *Contractor*'s submission or any amendment thereof. The *Contractor* shall, when required, provide such evidence to the *Supervisor* that the endorsements have been obtained from the relevant authorities.

(19) The *Contractor* shall not vary the traffic management, construction sequence or any works area in any form, without prior consultation with the TMLG and the approval of the *Supervisor*.

(20) The *Supervisor*, Transport Department and Hong Kong Police Force shall have the right to modify and lift any endorsed temporary traffic arrangements and control during implementation or whilst in operation, in order to suit the local traffic, road conditions and community events. The *Contractor* shall not be entitled to additional time and cost due to such modifications and lifting.

(21) The *Contractor* shall be responsible for the implementation of the temporary traffic arrangements and control whether the scheme is within the site boundary or otherwise. The *Contractor* shall also be responsible for coordinating the *works* required to be carried out by Hong Kong Police Force, relevant Government Departments and utilities undertaker to facilitate the implementation of the temporary traffic arrangements and control. The *Contractor*'s programme shall take into account the time required by these departments and others in implementing the temporary traffic arrangements and control, which may be in excess of the lead time for calling the TMLG meetings. No

works requiring temporary traffic arrangements and control shall commence until such temporary traffic arrangements and control have been implemented to the satisfaction of the relevant authorities, government departments and the *Supervisor*. The *works* required by other government departments and authorities include, but are not limited to, the following :-

(a) Transport Department - Transport Operations Division

(i) Publish gazettals and press notices

(ii) Rerouting of Bus, PLB and GMB Services

(iii) Relocation of Bus Stops, roadside carparks, PLB Stands, GMB Stands, Taxi Stands and subsequent reinstatement

(b) Transport Department - Traffic Engineering Division/ Traffic Control Division

(i) Modification to traffic signal timing, linking and method of control

(c) Electrical and Mechanical Services Department

(i) Modification to the cables for the traffic signals, public lighting and bollards.

(ii) Modification to signal aspect configuration

(iii) Modification to the controller wiring

(iv) Relocation of signal

(d) the Power Companies and other parties involved

(i) Provision and disconnection of power supply to new, existing or relocated signal controllers, public lighting and traffic bollards

(e) Hong Kong Police Force

(i) Provide assistance in setting up the temporary traffic arrangements and control measures

(f) Highways Department

(i) Relocation of street lights etc.

(22) The *Contractor* shall arrange and conduct road trials to be carried out for the proposed temporary traffic arrangements and control and shall make such modifications as may be necessary as a result of these trials to the satisfaction of the relevant authorities, prior to commencing the *works*.

(23) The *Contractor* shall be responsible for any temporary traffic arrangements and control required for the *works* involving the diversion

of utilities required to carry out the *works*.

(24) Prior to and at regular intervals after the commencement of any temporary traffic arrangements and control, the *Contractor*'s Traffic Consultant shall check and ensure that all temporary traffic arrangements and control are in accordance with the plans endorsed by the TMLG.

(25) Unless otherwise advised by the *Supervisor*, the *Contractor* shall be responsible for the erection, maintenance, relocation and subsequent removal of all necessary traffic aids and signals for temporary traffic arrangements and control including but not limited to signs, traffic signals, road markings, temporary concrete profile barriers, lighting and guarding as well as carrying out any excavation or reinstatement works outside the contract limits that may be required as part of the endorsed temporary traffic arrangements and control. The *Contractor* shall take note that the sole authority for authorizing the erection, relocation and removal of traffic signs and road markings on public highways, is the Commissioner for Transport.

(26) All temporary traffic arrangements and control should be inspected and certified by the Independent Checking Engineer (ICE) or his site representative before implementation. Detailed inspection list should cover signing, lighting, guarding and railing, superelevation, road width, sight lines, conditions of temporary barrier and all other special temporary traffic arrangements and control requirement as stipulated Clause 1.16A. The ICE site representative should have minimum of 10 years in traffic/ highway engineering design and 3 years site experience.

(27) The *Contractor* shall assist the *Supervisor* and the *Client* with the preparation of plans and diagrammatic representations to accompany publicity material relating to endorsed temporary traffic arrangements and control. The *Contractor* and his Traffic Consultant shall also provide assistance to the *Supervisor* to attend meetings and reply to queries on temporary traffic arrangements and control and related works progress/programme raised by members of the public, the media, District Councils and other interested parties from time to time as required by the *Supervisor*.

(28) Execution of the *works* shall under no circumstances cause obstruction to the nearby fire hydrants and their ground valves in-situ. Should any relocation or shut-off of fire hydrants be necessary, prior consent from the Fire Services Department shall be sought.

(29) The *Contractor* shall be responsible for monitoring and on-going adjustment of his temporary traffic arrangements and control, as necessary during construction to ensure the safety of the scheme, to ensure the maximum traffic capacities are maintained and to minimize any delay/queuing/inconvenience to road users.

(30) If the *Contractor* fails to replace any unsatisfactory equipment for traffic control, lighting and guarding when instructed to do so by the *Supervisor* or fails to provide approved equipment at locations as instructed by the *Supervisor*, then without prejudice to any other provisions of the Contract, the *Contractor* shall not be entitled to

payment under items for traffic control, lighting and guarding for the period during which the *Contractor* fails to comply with the requirements of the Contract, and the *Client* shall be entitled to have such work carried out by a person other than the *Contractor*. All additional costs properly incurred by the *Client* in so doing shall be recoverable from the *Contractor* in accordance with the Contract.

(31) Any *Contractor*'s proposed temporary traffic arrangements and control shall be subject to the following restrictions or other requirements of the TMLG and relevant departments, whichever is more onerous :

(a) Existing traffic flows shall be maintained along the existing roads.

(b) Any use of existing slopes to form temporary roads or footways shall require adequate strengthening and protection works which shall be designed by the *Contractor* and commented by GEO and approved by the *Supervisor*. The *Contractor* shall reinstate the slopes to the satisfaction of the *Supervisor*.

(c) Detailed temporary traffic arrangements and control proposals on lane/road closure for works shall be submitted to Transport Department (TD), Hong Kong Police Force and other relevant authorities for comment/endorsement. No lane/road closure shall be implemented without the written endorsement of all relevant authorities. The *Contractor* shall allow adequate lead-time for all lane/road closures. Submission to TD shall not be less than 8 weeks prior to the scheduled commencement date of the *works*. After commented by all concerned departments and the approved by the *Supervisor*, the *Contractor* shall notify TD at least 14 working days before commencement of works.

(d) The *Contractor* shall note that any closure of lane/road should follow "Highways Department Guidance Notes No. RD/GN/021 - Guidelines on Traffic Impact Assessment & Daytime Ban Requirements for Road Works on Traffic Sensitive Routes". The *Contractor* shall be deemed to have allowed sufficient allowance in his Prices for complying with this requirement. No claim for extension of time or for additional payment will be entertained as a result of complying with the restrictions stipulated.

Requirements for temporary traffic arrangement design

1.16A (1) During the course of the *works*, the *Contractor* shall plan and arrange his work to reduce disturbance to traffic to a minimum. The *Contractor* shall comply with all the requirements as referred to in this PS clause and shall abide by any further requirements as may be imposed by the relevant authorities from time to time. The *Contractor* shall also closely liaise with the relevant authorities and coordinate all works being carried out by the utility undertakings and other contractors employed by the *Client* on or adjacent to the Site to minimize traffic disruption. The *Contractor* shall attend ad hoc public and interested party's meetings, such as District Councils, to explain proposed traffic arrangements.

(2) All temporary traffic arrangements and control shall be designed in accordance with the Transport Planning and Design Manual, the Code

of Practice for Lighting, Signing, Guarding of Road Works and other Highway Standards and shall be submitted to TMLG for endorsement. The design of temporary traffic arrangements and control shall also take into account the surge of cross-boundary traffic during the golden weeks, festive holidays, special events/incidents, etc.

(3) For any temporary traffic diversions, the following recommended particulars of the proposed temporary traffic arrangements and control shall be submitted to the *Supervisor* for approval at least two months before the traffic arrangements and control are to be implemented and the *Contractor* shall allow sufficient time for the construction and implementation of his traffic diversion proposals :-

(a) details of traffic diversions, pedestrian routes and precautionary measures;

(b) details of lighting, signing, guarding, fencing, road markings, street furniture, all traffic control arrangements and equipment, etc. The *Contractor* shall carry out and arrange for the installation and maintenance of such required temporary traffic aids and subsequent reinstatement in areas beyond the contract limits;

(c) details of road geometry, drainage and lighting, pavement thickness, composition and associated traffic aids and protection works for proposals involving the construction of temporary roads or footpaths or the alteration of existing roads or footpaths; detailed calculations shall also be submitted if required by the *Supervisor*;

(d) any conditions or restrictions imposed by the TMLG, including copies of applications, permits or license, correspondence and endorsements; and

(e) where conditions on site are different to those assumed at the time of presentation of the proposed temporary traffic arrangements and control to the TMLG, and may have some impact on the effectiveness of the proposed arrangements, the *Contractor* shall resubmit his proposed arrangements to the TMLG for endorsement prior to submitting the said arrangements to the *Supervisor*.

(4) To fulfil all the traffic requirements, the *Contractor* shall carry out by himself or make arrangements for, as appropriate, all works necessary for this purpose, such as the construction of temporary carriageway pavement, provision of temporary pedestrian access, resiting or provision of street lights, traffic signals/bollards, fire hydrants, sign gantries and other street furniture, temporary protection to or alteration of underground utilities, transplantation of plants, temporary noise barriers and other environmental mitigation measures as required etc. He shall maintain temporary drainage and shall undertake to reinstate the footways, amenity areas, road dividers etc. where affected to their original state immediately after any phase of temporary traffic arrangements and control has no longer been required as a result of the completion of a corresponding phase of works. The temporary occupation of footways, amenity areas, road dividers, etc. shall be

permitted solely for the purpose of temporary traffic diversions and shall not be used for works operation, storage of plant and materials or similar activities. All the costs so incurred shall be deemed to be covered in the Prices in the Activity Schedule.

(5) Temporary traffic and road construction standards

(a) Standard Road Widths

Unless endorsed otherwise by the TMLG, minimum lane widths shall be :

One lane	4.0 metres
Two lanes	6.75 metres
Three lanes	10.3 metres

Note: where curves of radius of 400m or less are used, the road widths should be increased in accordance with Transport Planning and Design Manual Volume 2 Chapter 3 Table 3.4.4.1.

Footpaths shall be 1.5 metres minimum but shall be increased to 2.5 metres at pedestrian crossings and bus-stop locations if required by the TMLG or the *Supervisor*.

(b) Headroom beneath temporary falsework

This shall not be less than 4.7 metres. Temporary height gauges should be provided to the satisfaction of TD if so required.

(c) Maximum length and width of occupation of carriageway

The maximum length and width of occupation of carriageway shall be subject to the endorsement of the TMLG.

(d) Temporary road construction and other associated work

(i) The *Contractor* shall submit a design check certificate for any temporary road construction and thickness and associated works to the *Supervisor* for approval at least 4 weeks before his proposed date of commencement. The design check certificate shall be signed by the temporary works Independent Checking Engineer in accordance with ACC Clause F5.

(ii) The running surface shall be within the tolerance same as that applied to new carriageway and as specified in Section 9 and Section 10 of the Specification. All manhole covers shall be flush with the running surface.

(iii) The minimum thickness for rigid and flexible construction shall be not less than the following unless otherwise approved by the *Supervisor*:

Temporary flexible pavement

The temporary road surface for public traffic shall be formed by flexible pavement composed of 30mm friction course + 45mm wearing course + 65mm bituminous base course + 115mm road base + 220mm sub-base.

Temporary rigid pavement

190mm Grade D 40/20 unreinforced concrete with suitable expansion and contraction joints + 130mm sub-base.

Temporary footpaths

50mm Grade S 30/20 concrete + 75mm sub-base.

(iv) No ramping is allowed at carriageway which is open to public traffic.

(v) Notwithstanding the above requirements, if the construction is subsequently found to be inadequate for traffic using it, then the *Contractor* shall be responsible for strengthening and repairing the roads/footpaths and the associated works. The proposals for temporary pavement shall be submitted by the *Contractor* to HyD for comment and the *Supervisor* for approval.

(vi) No additional surfacing will be allowed to be placed to the existing structure without a comprehensive check of the capacity of the structure to sustain the loading of the additional surfacing. This check shall be carried out by the *Contractor* and certified by the Independent Checking Engineer in accordance with ACC Clause F5. The proposal for additional surfacing shall be approved by the *Supervisor* prior to the placing of any surfacing.

(e) Temporary pedestrian fencing and screens

(i) Temporary pedestrian fencing shall be of a type approved by the *Supervisor*. Temporary pedestrian fencing shall be continuous with no gap except at designated road crossing points. For pedestrian ways diverted onto carriageway the arrangement stipulated in Highways Department Standard Drawing No. H1133 shall be followed and the temporary barriers as given in Highways Department Standard Drawing No. H6112 shall be used unless otherwise agreed by the *Supervisor*.

(ii) Unless agreed otherwise by the *Supervisor*, a continuous temporary site hoarding with transparent panels as per Highways Department standard drawing no. H6155 shall segregate pedestrian traffic with works site along all footpath and cycle tracks.

(iii) The *Contractor* shall provide barriers, screens, netting and

the like to prevent objects from the *works* falling on roads and footways

(f) Access points from construction site and prevention of illegal occupation

Access points for construction traffic onto the highways, including access to the Site, shall be kept to the necessary minimum, be made conspicuous to motorists and shall be subject to the endorsement of the TMLG.

To prevent illegal access and parking in the Site, the *Contractor* shall erect, maintain and remove temporary fencing to the satisfaction of the *Supervisor*.

(g) Temporary lighting

The *Contractor* shall design, provide, install and commission temporary lighting in accordance with the relevant standards and codes of practices.

(h) Speed limit on carriageway

The *Contractor* shall not reduce the speed limits imposed on the roads unless supported by detailed justifications and agreed by TMLG. Under normal circumstances, reduction of speed limits on carriageway is unacceptable. However, if and when required by the TMLG, the *Contractor* shall make necessary arrangements for the lowering of speed limits.

(6) Special Temporary Vehicular Barrier

(a) In any case where the existing central barriers will need to be taken down to facilitate shifting of the traffic lanes or other works, the *Contractor* shall replace the existing barriers dividing opposite traffic with Special Temporary Vehicle Barriers which shall meet the requirements as stipulated in this clause.

(b) The Special Temporary Vehicle Barriers shall be of a profile type at least 800mm high, and it shall comply with the requirements of test level TL-3 of NCHRP 350. The deflection of the Special Temporary Vehicle Barriers, when tested under test level TL-3 of NCHRP 350, shall not be more than 0.5 metre. It should preferably be a detachable type capable of being erected and dismantled quickly. It should consist of special sections for emergency opening. The special section for emergency opening shall also comply with the requirements of test level TL-3 of NCHRP 350. For the passage of emergency vehicle, the *Contractor* should demonstrate to the *Supervisor*'s satisfaction that an 8m wide clear passage within the emergency opening can be formed manually by four people within one minute and fifteen seconds.

(c) The *Contractor* shall allow adequate time in his programme for obtaining the approval of the *Supervisor* for the proposed Special Temporary Vehicle Barriers. Such period shall in no case be less

than 8 weeks prior to its intended use.

(d) The Special Temporary Vehicle Barriers may be purposely designed barriers or proprietary products. For purposely designed barriers, the barriers shall be subject to physical tests at an approved testing laboratory to demonstrate that it meets the requirements of sub-clause (b) above. The testing procedure shall be subject to the approval of the *Supervisor*. Also, the design and testing shall be certified by an approved independent checking engineer with proven relevant experience. For proprietary products, the same approval procedure as that for purposely designed barriers shall apply except that the *Supervisor* may accept testing certificates as adequate proof of meeting the acceptance criteria if the *Supervisor* is satisfied that the test was carried out under the required conditions and that the certificate had been accepted before by a traffic authority as a valid proof of meeting the NCHRP 350 requirements.

(7) The temporary traffic arrangements and control proposals shall be submitted with supporting traffic analysis including junction capacity calculations, the traffic counts for the road links and the traffic movements at the junctions in number of vehicles and in Passenger Car Units (PCUs). The traffic counts shall be conducted within three months of the intended commencement date of the schemes. On request by the *Supervisor* or relevant authorities, the *Contractor* shall submit the queue length and delay calculations, and the signal junctions linking with time distance diagrams.

Particulars of Traffic Impact Assessment 1.16B (1) Within 3 weeks after the Contract Date, the *Contractor* shall submit a draft TIA for comment and endorsement by the *Project Manager* and the relevant authorities, and for consultation with the public in District Offices and District Boards, and in meetings with individual District Councillors, representatives from Estate Owners' Committees, Area Committees, local organisations, or alike. The draft TIA shall comply with all requirements as specified in PS Appendix F1. After submission of the TIA. The *Contractor* shall take into account comments and incorporate the comments into the TIA accordingly. Within two months after the Contract Date, the *Contractor* shall finalise the TIA for comment and endorsement by the *Project Manager* and the relevant authorities.

(2) If the traffic impacts after implementation of the TTA are unacceptable, the *Contractor* shall also revise the TIA and resubmit it for comment and endorsement. Unless otherwise agreed by the *Project Manager*, the *Contractor* shall reiterate the consultation process. The *Contractor* shall not implement the revised TTA before endorsement to the revised TIA is received.

(3) The *Contractor*'s attention is drawn to the fact that endorsement of the TIA does not imply the endorsement of the TTA and the *Contractor* is required to submit for endorsement in accordance with Clauses 1.14 to 1.16C of the Specification.

TTA for Yuen Long Highway 1.16C (1) The TTA schemes shall take into account any permanent and temporary traffic management schemes implemented by Transport Department, Highways Department, utility companies or other

Government departments and authorities in the vicinity.

(2) The *Contractor* shall obtain Expressway Works Permit (HYD 90) from Region Office of Highways Department and the *Contractor* shall comply with HyD Technical Circular No. 12/2009 – Expressway Legislation.

(3) In the event of any vehicle breakdown or accident or any other event occurring to disrupt the flow of traffic between the Shap Pat Heung Interchange and Tong Yan San Tsuen Interchange of the Yuen Long Highway where construction is in progress, the *Contractor* shall provide assistance to the HKPF to expedite resumption of normal traffic flow.

(4) The *Contractor* may be required by the TMLG to present the TTA scheme at District Council and to obtain District Council's endorsement on the TTA scheme and the contingency plan. The *Contractor* is deemed to have allowed sufficient rate and time in compliance with this requirement.

(5) It is the responsibility of the *Contractor* to allow sufficient time in his programme to plan, prepare, design and submit temporary traffic arrangements and control proposals, seeking approvals, implementing the schemes, monitoring and modification of the schemes or construction sequences when appropriate.

Use of roads and footways 1.17 (1)***S*** The *Contractor* shall not make use of public or private rights of way, roads and footways for depositing or storing plant or materials, other than such plant, materials, tools and implements as shall from time to time be required for immediate use on the several sections of the *works*. Plants, materials and Temporary Works shall be placed in such a way as to cause minimum interference with the use of any right of way by the public and the *Contractor* shall maintain those parts of the right of way, roads and footways not temporarily occupied by the *works* in a clean, passable and safe state at all times.

(2)***S*** Measures shall be taken to prevent excavated material, silt or debris from entering drainage systems in roads, footways or other paved areas; entry of water to the gullies and catchpits shall not be obstructed.

(5) The use of public roads by dumping lorries on General Holidays will not be permitted.

(6) The *Contractor* shall assist HKPF in the maintenance of normal traffic at roads by all means.

(7) Unless otherwise agreed by the *Supervisor* in writing, the use of all public surfaced roads will be limited to pneumatic tyre and fully licensed vehicles only. Tracked vehicles will not be allowed to run on the roads. The *Contractor* shall be held responsible for any damage to roads caused by the above or illegal movement of his vehicles and shall be required to make good immediately any such damage at his own expense.

Work on roads 1.18 (1)***S*** Work on roads on the Site shall be carried out in sections such that the length of road occupied at any time does not exceed that allowed by the

and footways

relevant authorities and the width of road occupied at any time does not exceed the width of one traffic lane unless permitted by the *Supervisor*. Works on each section shall be completed and the road shall be reinstated and opened to traffic before work commences on the next section. Works on any section, including loading and unloading, shall be carried out in such a manner that traffic and utilities on the adjacent road and pedestrian access in the adjacent footway shall be adequately maintained.

(7) Prior to the commencement of work on any road crossings, the *Contractor* shall make available on Site steel trench covers designed in accordance with the requirements of GS Clause 1.18(4) to the full width of the carriageway not available to traffic. These covers shall be used for emergencies or where directed by the *Supervisor* to meet the requirements of the Contract with regard to maintenance of land traffic flow. The *Contractor* shall submit the design, working drawings and method of working for the fabrication and installation of temporary steel covers, for each location where such cover may be required, for the approval of the *Supervisor* two weeks prior to the commencement of excavation. When installed the steel trench covers shall not result in any noise nuisance caused by them rocking under the section of traffic.

(8) The *Contractor* shall agree with the *Supervisor* the existing road furniture to be retained/ demolished/ reprovided.

(9) The *Contractor* shall be responsible for obtaining all necessary licences and permits from all relevant authorities, government departments and affected parties including but not limited to, Hong Kong Police Force, Transport Department, Highways Department and Environmental Protection Department.

(10) The *Contractor* shall submit application for Road Works Advice, together with relevant document, to Hong Kong Police Force at least 7 days prior to commencing any roadworks.

(11) The *Contractor* shall inform the Transport Operations Division and the Traffic Engineering Division of Transport Department at least 3 weeks before implementation of any works which involve arrangements of bus stop relocation, publication of Traffic Advice/Gazette Notice.

(12) In compliance with the Code of Practice for Lighting, Signing and Guarding of Roadworks, the *Contractor*'s name and contact telephone number shall be distinctively shown on the name plate of each barrier by a means approved by the *Supervisor*. In addition, the Contract details and enquiry telephone numbers shall be shown on a publicity board approved by the *Supervisor*. The *Contractor*'s name shall also be clearly shown on the traffic cones by either marking with indelible ink or by fixing securely a durable sleeve to each traffic cone placed on the carriageway.

(13) Further to sub-clause (12) of this clause, the *Contractor* shall provide a publicity board for motorist showing the contract information, the date of anticipated completion of the trench works, enquiry telephone number and any other information required by the *Supervisor* at the front end of each section of trench work. Construction details of the publicity board shall be as agreed by the *Supervisor* but shall largely

follow the details shown in the Highways Department Guidance Notes "RD/GN/019A" and Standard Drawings H6143, H6114 to H6149. The date of anticipated completion shown in the publicity boards for motorist and any amendment thereof shall be as agreed by the *Supervisor*.

(14) In case where the *Contractor* has been served with a summons under the Road Traffic (Traffic Control) Regulations or Road Traffic (Expressway) Regulations, the *Contractor* shall submit copy of the summons to the *Supervisor*'s Representative for record and information.

Reinstatement of roads and footways 1.19***S***

(1) Temporary diversions, pedestrian access and lighting, signing, guarding and traffic control equipment shall be removed immediately when they are no longer required. Roads, footways and other items affected by temporary traffic arrangements and control shall be reinstated to the same condition as was existing before the work started or to such other condition as may be agreed or instructed by the *Supervisor* or, where such work is covered under an Excavation Permit, to the condition as stipulated under the conditions of Excavation Permit.

(2) Any damage done by the *Contractor* to adjoining roads, highway structures, roadside slopes, footways, street furniture, etc. arising from the execution of the *works* due to the default of the *Contractor* shall be made good to the satisfaction of the Director of Highways at the expenses of the *Contractor*.

(3) The *Contractor* shall strictly follow the guidelines "HQ/GN/05A - Guidelines on Good Practice for Laying Unit Pavers" as promulgated by the Highways Department at no additional cost to the Contract.

CARE OF THE *WORKS*

Protection from weather 1.21

(3) The *Contractor* shall make necessary arrangements and take necessary precautions to protect the *works* and Specialist Works and all plant, temporary buildings, goods or materials against the effects of inclement weather.

DAMAGE AND INTERFERENCE

Damage and interference 1.23

(5) No damage to or interference with the existing fire hydrants and valves shall be caused unless with the prior approval of the Director of Fire Services.

Structures, roads and other property 1.26***S***

(1) The *Contractor* shall immediately inform the *Supervisor* if any structures, roads or other properties are damaged, whether or not the damage is due to the execution of the *works* by the *Contractor* or other parties.

(2) The *Contractor* shall carry out topographic surveys of all roads, buildings, structures, services, etc. within and in the vicinity of the Site immediately after taking possession of the Site. The *Contractor* shall take all necessary precautions to protect roads, buildings and any other structures, or services under or above ground adjacent to or within the Site from the effects of vibration, undermining, or other

earth movements caused by the carrying out of the *works*.

(3) Should the effects of ground movements be monitored in any adjacent roads, buildings or any other structures, the *Contractor* shall modify his method of constructions as instructed by the *Project Manager*. Under such circumstances, the *Project Manager's* instruction shall not constitute a compensation event.

(4) The *Contractor* shall be responsible for the following measures regarding the existing roads and footways within the Site. Unless otherwise specified, the costs of such repair and maintenance works shall be deemed to be covered in the Prices in the Activity Schedule.

(a) The *Contractor* shall carry out repairs to all damages to the existing road and footway surfaces, traffic signs, directional signs, fencing, railings, street furniture, etc. caused by traffic accidents.

(b) The *Contractor* is responsible for the maintenance of all the existing roads and footways and to ensure that the existing roads and footways are cleared of obstructions. The *Contractor* shall carry out repairs to all damages and defects to the existing roads and footways as and when directed by the *Project Manager* except for the damages and defects caused by traffic accidents as mentioned in sub-clause (a) above. The damages and defects where the *Contractor* shall carry out repair works shall include, but not limited to, potholes, uneven road surface, damaged carriageway, damaged pavement surface, damaged channel, unpaved trench works, defective lane surface, earth/debris deposit, defective manhole/cover, missing manhole cover, defective/missing grating, damaged railing, damaged TCSS civil provisions, damaged TCSS field equipment, damaged/missing street name plate, defective traffic sign, road markings faded/blackened, choked drain, choked catch-pit, flooding, fallen scaffolding, etc.

(c) The *Contractor* shall maintain all temporary traffic signs and TCSS civil provisions, guarding and lighting and alterations to existing furniture necessitated by the project to the satisfaction of the *Supervisor*.

(d) The *Contractor* shall be responsible for the maintenance of all roadside slopes within the Working Areas in accordance with Geoguide 5 and the Highways Slope Manual (2000) produced by GEO to the satisfaction of the *Project Manager* and GEO. The *Contractor* shall submit method statements and schedules for the maintenance regime of each individual slope. The *Contractor* shall minimize the disturbance to the existing slope in related to their temporary works. In case of disturbance of existing slopes by temporary works within the slope boundary line, the *Contractor* shall reinstate all damages to the existing slopes to the satisfaction by the *Project Manager* and GEO and the costs of such reinstatement works will be borne by the *Contractor*.

(e) The *Contractor* shall maintain all soft landscape within the Working Areas including but not limited to grass cutting, tree trimming, watering etc. The contractor shall coordinate with LCSD and/or relevant parties on the maintenance responsibilities of the soft landscaping.

(f) The *Contractor* shall be responsible for the maintenance of all drainage services and structures within the Working Areas in accordance with DSD's standard practice.

Condition survey and monitoring of structures, roads and other property 1.26A

(1) The *Contractor* shall carry out a survey to record the existing conditions of all roads, buildings, structures, services, etc. within and in the vicinity of the Site. For roads, the survey shall be in form of "Detailed Inspection" in accordance with "Road Inspection Manual" published by Highways Department, unless otherwise approved by the *Project Manager*. For structures and other property, the survey shall be in form of "General Inspection" in accordance with "Inspection Manual for Highways Structures" published by Highways Department, unless otherwise approved by the *Supervisor*. After completion of the survey, the *Contractor* shall submit a report documenting the findings of the survey for record by the relevant authorities and the *Supervisor*.

(2) During the course of the *works*, The *Contractor* shall continue to monitor the conditions of all roads, buildings, structures, services, etc. within and in the vicinity of the Site. For roads, the monitoring shall be in form of "Safety Inspection" daily and "Detailed Inspection" monthly in accordance with "Road Inspection Manual" published by Highways Department, unless otherwise approved by the *Supervisor*. For structures and property, the monitoring shall be in form of "General Inspection" monthly in accordance with "Inspection Manual for Highways Structures" published by Highways Department, unless otherwise approved by the *Supervisor*. After completion of the *works*, the *Contractor* shall submit another report updating the findings of the monitoring for record by the relevant authorities and the *Supervisor*.

(3) The scope and the extent of the survey and monitoring shall be determined on Site and accepted by the *Supervisor* but shall in no case, less than 50m from the works limit which may be affected, and include but not limited to the following.

(a) Existing structures along Yuen Long Highway;

(b) Existing retaining walls and slopes; and

(c) Existing traffic signs and sign gantries and its associated structures.

The scope and extent shall be increased, subject to the nature of the *works* and the construction methodology. This shall not constitute a compensation event.

(4) All survey and monitoring results, including qualitative and quantitative descriptions of the findings, photographs taken for the observations / defects, shall be in the formats as required by the

"Inspection Manual by Highways Structures" or "Road Inspection Manual", and shall be submitted to the *Supervisor* within 24 hours after the survey.

(5) Any defects which are only not identified in the initial condition survey but are discovered afterwards are due to the *works* of the *Contractor* unless the *Contractor* can otherwise justify. The *Contractor* shall be fully responsible for the defects and propose the remedial works for comment by the relevant authorities and comment and approval by the *Project Manager*. The remedial proposal shall be certified by the Independent Checking Engineer as if it is a *Contractor*'s Design under Clause F2 of *additional conditions of contract*. The *Contractor* shall complete the remedial works to the satisfaction of the relevant authorities and the *Project Manager*.

Access 1.27***S*** (1) Alternative access shall be provided if interference with existing public or private vehicular or pedestrian access is necessary to enable the execution of the *works*. The arrangements for the alternative access shall be as agreed by the *Supervisor*. The permanent access shall be reinstated as soon as practicable after the *works* are complete and the alternative access shall be removed as soon as practicable after it is no longer required.

(2) The *Contractor* shall be responsible for the provision and maintenance of all temporary access roads to and within the Site. Details of all proposed temporary access roads and connections to existing public roads which may be required by the *Contractor* from time to time during the *works* shall be submitted to the *Project Manager* for approval. In addition, the *Contractor* shall submit proposed points of ingress/egress of the Site to TD and HyD (where appropriate) and the Police for comment/approval before operation of the temporary access roads. Unless otherwise instructed by the *Project Manager*, design, construction and reinstatement of the temporary access roads shall be certified by the independent checking engineer in accordance with Clause F7 of *additional conditions of contract*.

(3) The *Contractor* shall construct temporary concrete run-in at all ingress/egress to the Site.

(4) All temporary access roads shall be provided with drainage with desilting facilities along entire length and before connecting into existing drainage system. The surfaces of all temporary cut and fill slopes shall be protected against erosion.

(5) The *Contractor* shall be responsible for the maintenance of access roads, both existing and those formed by the *Contractor* during the period of the Contract and shall at the completion of the *works* reinstate all access roads and footpaths to the condition that prevailed at the start of the Contract, remove temporary access roads and concrete run-in, make good any damage done to the adjoining public roads, footpaths and street furniture and reinstate the ground to the original condition and contours. Any trees and planting which are removed for the formation of temporary access roads shall be replaced and compensated to a scheme approved by the *Project Manager* and

relevant authorities after the reinstatement of the access roads.

(6) The *Contractor* shall maintain reasonable pedestrian access to all public footpaths through the Site, providing temporary diversions where necessary, to the satisfaction of the *Supervisor*.

(7) The *Contractor* shall provide reasonable access to other contractors, utility undertakings and other duly constituted authorities. The *Contractor* shall notify the *Supervisor* and other concerned contractors at least 14 days in advance should he wish to alter these access arrangements during the course of the *works*.

(8) Without prejudice to the generality of the sub-clauses above, the *Contractor*'s attention is drawn to the following specific requirements of site access for this Contract.

(a) Unless otherwise approved by the *Project Manager*, the *Contractor* shall only be allowed to access into or exit from the or the Working Areas using designated Site access as shown on the Drawings or specified in the Contract.

(b) The *Contractor* shall be responsible to obtain approval by the *Project Manager* and the relevant authorities for any other Site access necessary for the *works* but not specified in the Contract.

Trees 1.27A Trees which are to be retained or which are not required to be removed in order to carry out the *works*, shall be protected from damage at all times by methods agreed by the *Supervisor*. Materials, including excavated materials, shall not be banked around such trees and they shall not be trimmed or cut without the approval of the *Project Manager*. The *Contractor* shall be responsible to provide temporary supports for the trees which have potential danger to the public. The requirements and specification for preservation and protection of trees shall refer to Section 26 of the Specification.

LIAISON WITH OTHERS

Liaison with others 1.31 (3) The *Contractor*'s attention is drawn to the fact that there will be other contractors/parties working within or in the vicinity of Site. They include but not limited to those as listed in Clause 1.31A of the Specification.

(4) The *Contractor* shall be responsible for liaising and convene regular meetings in compliance with the requirements specified in PS Clause 1.31A with other contractors/personnel as to the time and manner of executing their works and shall copy all correspondence to the *Project Manager*. Any agreed programme of works with other contractor/personnel shall be submitted to the *Project Manager* for information. The *Project Manager* shall also be informed immediately of any delays in works by other contractors which may affect progress of the *works*. The *Contractor* shall refer to PS Clause 1.31B for the requirements for preparing a Preliminary Interface Management Plan and a Detailed Interface Management Plan with each individual interface contractors.

Interfacing 1.31A (1) The *Contractor* shall liaise and coordinate with all interfacing parties whose works will involve interfacing with the *works* under this

parties

Contract, and shall agree with these parties the details of interfacing construction arrangement including, but not limited to, timings of interfacing site activities, temporary traffic arrangements, access arrangement and working spaces. The *Contractor* shall document his coordination efforts in this regard and shall submit such documents to the *Project Manager* for information. These interfacing parties include, but not limited to: -

(a) All works departments, FSD, Police, their contractor(s); and

(b) Other contractor(s) and party/parties working in or adjacent to the Site.

(2) The *Contractor* shall prepare Interface Management Plan for each of these interfacing parties addressing the interfacing issues and requirements. The interfacing aspects to be addressed in the *Contractor*'s liaison and coordination with these parties are described further as follows:

Works by the contractor(s) other than Utility Undertakings as mentioned in PS Clause 1.31A in or adjacent to the Site

(3) The *Contractor* shall convene regular meetings with the contractor(s) listed in (1) above, so that interfacing aspects are agreed adequately in advance of construction. Examples of topics to be addressed include:

(a) Channels of communication;

(b) Programme of each party's activities at/near the interface, and requirements as regards access and working space; and

(c) The amount of advance notice to be given if one party's activities may affect the other party's.

Works by other contractor(s) in or adjacent to the Site

(5) The *Contractor* shall convene regular meetings with other contractors including those employed by the *Client*, so that interfacing aspects are agreed in advance of construction. Examples of topics to be addressed are as those cited in (2) above.

Interfacing requirements 1.31B

(1) Pursuant to Clause 1.31 of the Specification, the *Contractor* shall liaise with all interfacing and adjacent contractors, relevant authorities and parties working inside or adjacent to the Site, and they are collectively and individually referred to as interface party under this clause. The *Contractor*'s works having interface with the *works* of the interface party is referred to as interface works.

(2) The *Contractor* shall form an Interface Management Liaison Group (IMLG) and convey regular meetings with all interface parties to discuss the interface matters.

(3) At least 60 days before commencement of the interface works, or within 30 days after an interface party is identified, whichever is the earlier, the *Contractor* shall coordinate with the interface party and submit a Preliminary Interface Management Plan which is mutually

agreed by the interface party for record by the *Project Manager*. The Interface Management Plan shall include but not limited to the following: -

(a) the channels of communication, and the amount of advance notice to be given if one party's activities may affect the other party's;

(b) the programme and sequence of each party's activities at/near the interface, and requirements with regard to the access and working space;

(c) the information to be exchanged between the *Contractor* and the interface party for the purpose of preparing the Detailed Interface Management Plan; and

(d) any other interface arrangements and provisions necessary for the completion of the interface works.

The *Contractor* shall seek comment by the *Project Manager* before agreement of the Preliminary Interface Management Plan with the interface parties.

(4) At least 30 days before commencement of the interface works, or within 60 days after an interface party is identified, whichever is the earlier, the *Contractor* shall submit a Detailed Interface Management Plan which is mutually agreed by the interface party for approval by the *Project Manager*. The Detailed Interface Management Plan shall include but not limited to the following: -

(a) contain all details necessary for the accomplishment of the interface arrangement and provisions agreed in the Preliminary Interface Management Plan;

(b) contain a detailed works programme;

(c) the information to be exchanged between the *Contractor* and the interface party during the course of execution of the interface works.

The *Contractor* shall seek comment by the *Project Manager* before finalization of Detailed Interface Management Plan. The *Contractor* shall update the Detailed Interface Management Plan during the course of execution of the Contract.

(5) Once the Detailed Interface Management Plan has been agreed by the *Project Manager*, the *Contractor* shall execute the *works* in accordance with the Detailed Interface Management Plan. The *Contractor* shall advise the *Project Manager* immediately of any difficulty in developing a mutually acceptable Interface Management Plan.

(6) The requirements of the Drawings and the Specification associated with the interfaces shall form the basis of the Detailed Interface Management Plan, but does not relieve the *Contractor*'s obligation to identify and address any new interface issues to meet the Contract

requirements.

(7) The *Contractor* shall give the *Project Manager* a minimum of 7 days' notice of all meetings to be held with the interface parties and allow the *Project Manager* to attend such meetings. The *Contractor* shall prepare the minutes of meetings of all Interface Management Liaison Meetings and copied to the interface parties and the *Project Manager* for records before 14 days of each meeting. The *Contractor* shall also copy any correspondences with the interface contractor related to the interface for record by the *Project Manager*. Any revision to the Detailed Interface Management Plan shall be submitted for review by the *Project Manager*.

SITE CLEANLINESS

Site cleanliness 1.32***S***

(1) The Site shall be maintained in a clean and tidy condition. Materials, including materials required for Temporary Works, shall be stored in an orderly manner. Rubbish and debris shall be disposed of at least once a week.

(2) The *Contractor* shall ensure that no earth, debris or rock is deposited on public or private rights of way as a result of the *works*, including any deposits arising from the movement of plant and vehicles.

(3) The *Contractor*'s attention is drawn to the requirements as stipulated in PS Clause 1.32A, ETWB TC(W) 8/2010 for tidying up of the Site. In general, the Site shall at all times be maintained in a clean, safe, tidy and passable state. Materials, including materials required for Temporary Works shall be stored in an orderly manner. All waste or superfluous material on the Site including rubbish, cement bags, disused formwork, debris, illegal dumpings and the like irrespective of their nature and source shall be cleared away and disposed of by the *Contractor* at least once a day and the *works* are cleaned by flushing with water necessary so that the Site is kept constantly clean and tidy.

(4) The *Contractor* shall clean all vehicles and plant before they leave the Site to ensure that no earth, mud, debris and the like is deposited by them on the roads. If, despite the above, any vehicle found to be depositing any form of dirt on the public road then at the direction of the *Supervisor*, the *Contractor* shall remove that vehicle or plant from the Site forthwith and shall not thereafter allow that vehicle or plant to re-enter the Site. The *Contractor*'s attention is brought to the requirements of the Environmental Management Plan (EMP) as given in PS Clause 1.104.

Enhanced site cleanliness and tidiness – daily cleaning and weekly tidying 1.32A

(1) The *Contractor* shall perform Daily Cleaning and Weekly Tidying of the Site including Public Cleaning Areas. The extent of the Public Cleaning Areas required for cleaning shall be within 10 metres on the periphery outside the barriers or hoardings, or determined by the *Supervisor* on site for each work location based on the requirements in the Specification, and taking into account the actual site condition before work commences.

(2) The *Contractor* shall draw up, in the EMP where appropriate, a system on waste management, and maintenance of cleanliness and

tidiness of the Site including, but not limited to, the methods and provisions including the proposed areas for on-site sorting, separation, storage and disposal of waste materials, and the proposal of how to maintain the Site in clean and tidy condition. The *Contractor* shall provide the necessary facilities, receptacles and transport for the temporary storage, disposal and removal of different types of wastes. The EMP shall be in accordance with PS Clause 1.104.

(3) Daily Cleaning

(i) "Daily Cleaning" shall include cleaning and tidying up after work of tools, equipment, unused materials, storage areas and common areas such as passageways, daily removal of waste materials from works areas, removal of any rubbish and debris dumped into the Site by the public and, without derogating from the generality of the foregoing, cleansing and/or re-conditioning of hoardings, barriers, guarding, lighting, signage and /or traffic cones, shall include, but not limited to, all the items subject to checking as listed in sub-clause (3) (ii).

(ii) The *Contractor* shall develop inspection checklist for Daily Cleaning for the approval of the *Supervisor*. The inspection checklist shall be reviewed and updated whenever there is a change in work nature or work location and re-submitted for approval by the *Supervisor*. The inspection checklist shall include an assessment on the cleanliness and tidiness of all work locations, plus the Public Cleaning Areas. Items to be checked against for each work location shall include, but not limited to, the following:

(1) Maintenance of passageways, common accesses and public areas free of obstruction;

(2) Proper storage and stacking of materials;

(3) Proper placement and storage of tools and equipment after work;

(4) Proper sorting, storage and/or disposal of waste materials in accordance with the EMP;

(5) Proper securing of hoarding, barriers, guarding, lighting, and signing of works;

(6) Prevention and removal of water ponds and flooding;

(7) Clearing of stockpiling and wastes arising from the *works*;

(8) Conditions of cleanliness and tidiness of the Site including Public Cleaning Areas in the perspective of the general public; and

(9) Other cleaning requirements as instructed by the *Supervisor*.

(iii) The *Contractor* shall assign person(s) to inspect the Site after

each Cleaning Day's work. The assigned person shall check and ensure the cleanliness and tidiness of the Site, complete the inspection checklist, record the areas requiring improvements, and take photographs of areas where cleaning and tidying up works have been done and where improvement actions are required.

(iv) The *Contractor* shall notify the *Supervisor* the time schedule for Daily Cleaning on every Cleaning Day and the name of assigned person as referred to in sub-clause (3)(iii) above responsible for inspection and checking after each Cleaning Day.

(v) The *Contractor* shall submit, at or before 10:00am of the day (which is not a General Holiday) following a Cleaning Day, the inspection checklist, records and photographs prepared pursuant to sub-clause (3)(iii) above to the *Supervisor* for checking and recording. Submission should be in soft copy with three sets of hard copies.

(vi) The *Project Manager* or the *Supervisor* may carry out inspections and surprise checks to verify the *Contractor*'s performance on cleanliness and tidiness of the Site before the noon of the day (which is not a General Holiday) following a Cleaning Day. The *Contractor* shall not be entitled to any payment of the item for "Daily Cleaning" for a Cleaning Day if the performance of the *Contractor* on cleanliness and tidiness for any part of the Site on that Cleaning Day is not carried out to the satisfaction of the *Supervisor*. The *Supervisor* shall notify the *Contractor* and record in the Site Diary for such non-compliance and the areas of dissatisfaction for improvement by the *Contractor*.

(vii) The *Contractor* shall in the morning of every Cleaning Day before work commences inspect and remove any rubbish and debris that may be littered by the public over the night within any area which cleanliness and tidiness the *Contractor* is required to maintain under this Contract before the inspection by the *Supervisor*.

(viii) If a tropical cyclone warning signal (No. 8 or above), a Black Rainstorm warning or other special circumstances renders Daily Cleaning not possible or practical on a Cleaning Day (or if more than one Cleaning Day, Cleaning Days), the *Contractor* shall submit the detailed particulars of the circumstances for the consideration of the *Project Manager*'s Representative. The *Project Manager*'s Representative may, at his discretion, waive the *Contractor*'s duties and obligations in respect of Daily Cleaning on that particular Cleaning Day (or if more than one Cleaning Day, those particular Cleaning Days) and in such event, the *Contractor* shall not be entitled to any payment in respect of Daily Cleaning on that particular Cleaning Day (or if more than one Cleaning Day, those particular Cleaning Days).

(4) <u>Weekly Tidying</u>

(i) "Weekly tidying" shall include the cleansing and tidying up of the common areas and accesses, cleansing and/or re-conditioning of hoardings, barriers, guarding, lighting, signage and /or traffic cones, cleansing of external covers for plant and equipment, removal of waste and debris etc. so as to ensure that the plant and equipment, hoardings, as well as the Site as a whole, to be clean and tidy in the perspective of the general public and, without derogating from the generality of the foregoing, shall include, but not limited to, all the items subject to checking as listed in sub-clause (4)(ii).

(ii) The *Contractor* shall develop inspection checklist for Weekly Tidying for the approval of the *Supervisor*. The inspection checklist shall be reviewed and updated whenever there is a change in work nature or work location and re-submitted for approval by the *Supervisor*. The inspection checklist shall include an assessment on the cleanliness and tidiness of the Site conditions at various work locations, plus including the Public Cleaning Areas. Items to be checked against for each work location shall include, but not limited to, the following:

(1) Thorough cleansing of passageways, common accesses and public areas;

(2) Re-organizing of storage materials for better utilization of storage spaces and safe stacking if appropriate;

(3) Maintenance and re-conditioning of tools and equipment;

(4) Cleansing of external covers for plant and equipment;

(5) Collection and removal of disposed waste materials off site in accordance with the EMP;

(6) Cleansing, re-conditioning and/or replacement of hoarding, barriers, guarding, lighting, and signage of works to good working condition;

(7) Clearing of drains and channels to prevent flooding; and

(8) Other cleansing requirements as instructed by the *Supervisor* from the perspective of the general public.

(iii) The *Contractor* shall assign person(s) to inspect the Site after each Cleaning Week Day's work. The assigned person shall check and ensure the overall cleanliness and tidiness of the Site, complete the inspection checklist, record the areas requiring improvements, and take photographs of areas where overall site cleaning and tidying up actions have been done and where improvement actions are required.

(iv) The *Contractor* shall notify the *Supervisor* the time schedule for Weekly Tidying on every Cleaning Week Day, and the name of the assigned person as referred to in sub-clause (4)(iii) above responsible for the inspection and checking after each Cleaning

Week Day.

(v) The *Contractor* shall submit, at or before 10:00am of the day (which is not a General Holiday) following the Cleaning Week Day, the inspection checklist, records and photographs prepared pursuant to sub-clause (4)(iii) above to the *Supervisor* for checking and recording.

(vi) The *Supervisor* shall, together with the Site Agent, check and inspect the overall cleanliness and tidiness of the Site on the day (which is not a General Holiday) following the Cleaning Week Day. The *Supervisor* shall advise the *Contractor* whether his performance is to his/her satisfaction, or if not, where improvement actions are required. Submission should be in soft copy with three sets of hard copies.

(vii) The *Contractor* shall promptly rectify the defects identified by the assigned person as referred to in sub-clause (4)(iii) above, and/or the *Supervisor*. The *Contractor* shall not be entitled to any payment of the item for "Daily Cleaning" for a Cleaning Day if the *Contractor* fails to rectify the identified defects pursuant to sub-clauses (4)(iii) and (4)(vi) above to the satisfaction of the *Supervisor* before the end of the day (which is not a General Holiday) following the Cleaning Week Day.

(viii) The *Contractor* shall in the morning of every Cleaning Week Day before work commences inspect and remove any rubbish and debris that may be littered by the public over the night within any area which cleanliness and tidiness the *Contractor* is required to maintain under this Contract before the inspection by the *Supervisor*.

(ix) If a tropical cyclone warning signal (No. 8 or above), a Black Rainstorm warning or other special circumstances renders Daily Cleaning not possible or practical on a Cleaning Day (or if more than one Cleaning Day, Cleaning Days), the *Contractor* shall submit the detailed particulars of the circumstances for the consideration of the *Project Manager*'s Representative. The *Project Manager*'s Representative may, at his discretion, waive the *Contractor*'s duties and obligations in respect of Daily Cleaning on that particular Cleaning Day (or if more than one Cleaning Day, those particular Cleaning Days) and in such event, the *Contractor* shall not be entitled to any payment in respect of Daily Cleaning on that particular Cleaning Day (or if more than one Cleaning Day, those particular Cleaning Days).

Control of rodents on Site 1.32B The *Contractor* is reminded that even a minor degree of rodent infestation is very dangerous as this will always develop into a serious rodent problem quickly and pose a severe threat to the health of workers and the public. To control the rodent problem, the Site must be maintained in a tidy state at all times to discourage rodent harbour aging and detect rodents at an early stage. Refuse and food must be kept in containers with well-fitted covers/rodent proof store rooms to deprive rodents of food. The *Contractor* shall regularly engage specialist rodent control service providers to control the rodent problem on Site.

Prevention of mosquito breeding 1.33 (3) The posters in (2), which may be obtained from Leisure and Cultural Services Department shall be removed by the *Contractor* on completion of the *works*.

Tidying up the Site 1.33A (1) The *Contractor* shall arrange for tidying up the following areas at the end of each day's work or at other times as may be directed or agreed by the *Supervisor*:

(a) the respective areas where works are in progress (including idling areas). These areas are to include but not limited to all fenced off areas and/or all areas with stockpile of materials, equipment or tools;

(b) all ground level soft landscaping areas, raised planters and soft landscaping areas within 1.5 m of the toeline of slopes within the Contract area;

(c) publicly accessible areas within the Site and lying within 2.5 m of the areas in (a) and (b) above. Areas with running vehicular traffic shall not be regarded as accessible to the public but the *Contractor* will still be required to clean the edge of the roads bordering fenced off areas.

(2) Tasks to be carried out shall include the cleaning and tidying up tools, equipment, un-used materials, collection of rubbish and removal of waste materials including rubbish from the Site on a daily basis unless otherwise agreed by the *Supervisor*. The *Contractor* shall plan in advance places and methods for the storage of materials, tools, machinery, equipment and the disposal of waste materials and the division of responsibilities for common areas among the subcontractors. The *Contractor* shall provide suitable facilities, receptacles and transport for the temporary storage, disposal and removal of different types of waste.

(3) When construction works are near completion, the *Supervisor* may notify the *Contractor* in writing that daily tidying up of the Site pursuant to this Clause is no longer required. Notwithstanding any notice given under this Clause, the *Contractor* remains at all times responsible for cleanliness of the Site pursuant to Clause D10 of *additional conditions of contract*.

(4) Rubbish and other waste materials collected by the *Contractor* pursuant to this Clause shall be disposed of in accordance with the EMP of the *Contractor* where such Plan is required under the Contract.

TESTING

Testing 1.40 (5) The *Contractor* shall refer to PS Appendix B4 to observe the requirements in carrying out the tests with the Public Works Laboratory.

SITE ESTABLISHMENT

Works Areas 1.45A (1) The *Contractor* shall be responsible for all obligations and liabilities

of the use of the Works Areas as explicitly or implicitly required by or referred in the Engineering Conditions from time to time issued by the District Lands Office and this specification. The Works Areas will be available for the *Contractor* up to the end of the last Maintenance Period or the dates as specified in the Engineering Conditions, whichever is the earlier. The *Contractor* shall assume full responsibility under the Engineering Conditions as if the *Contractor* is the "allocatee" in the Engineering Conditions. The *Contractor* is required to indemnify the *Client* and the *Project Manager* against any charge or responsibility for non-compliance with the Engineering Conditions. This shall not constitute a compensation event and the Contractor shall not be reimbursed for any cost or time due to the *Contractor*'s non-compliance with the Engineering Conditions.

(2) The *Contractor* shall only be allowed to use Works Areas exclusively for this contract.

(3) The *Project Manager* reserves the right to terminate the *Contractor*'s right to use the proposed works area if the *Contactor* fails to comply with the Engineering Conditions or if the works area are found to be used for purpose not related to this contract. In such case, the termination is not a compensation event.

Submission of particulars 1.46***S***

(1) The following particulars shall be submitted to the *Supervisor* for acceptance not more than 14 days of the commencement of the *works*:

(a) Drawings showing the layout within the Site of the project signboards, access roads and major facilities required early in the Contract,

(b) Drawings showing the layout and the construction details of the joint site accommodation for the *Project Manager*, the *Supervisor* and the *Contractor*, and

(c) Drawings showing the details to be included on project signboards.

Survey of the Site 1.47***S***

(1) The *Contractor* shall carry out all surveys including the provisions and installation of survey markers for monitoring/ positioning/ record arising from and/or in connection with the *works*. However; the *Contractor* shall, when notified by the *Supervisor* in writing, be required to verify surveys carried out by the *Supervisor* or to carry-out a joint survey with the *Supervisor*'s surveyor prior to or during the progress of the *works*.

(2) The *Contractor* shall afford the *Project Manager* all facilities· and equipment to attend and check each survey, and shall give a minimum of 48 hours notice when a check survey is required. However, it is the sole responsibilities of the *Contractor* to ensure all setting out survey done by him is of good quality and is free from mistake. The *Supervisor* is not obliged to check any and all of the setting out survey work done by the *Contractor*.

(3) The *Contractor* shall carry out an Initial Survey jointly with the *Supervisor* and agree the result as soon as practicable after

completion of site clearance of the Site, prior to commencing other works in the area surveyed. The original ground levels and details of existing features on the Site should be established by a survey carried out jointly between the *Contractor* and the *Supervisor*. The results of the survey should be agreed between the *Contractor* and the *Supervisor* and recorded in the form of drawings with scale of 1 to 500. The *Contractor* shall also agree with the *Supervisor* the existing road features to be retained/demolished/re-provided.

(4) The *Contractor* shall comply with the requirements for survey of the Site and setting out as specified in PS Appendix I1.

(5) The Site Survey Stations shall be set out and agreed at site meetings attended by the Agent and the *Supervisor*. The *Contractor* shall be responsible for setting out the Survey Stations using surveying instruments, and as such it shall ensure that the Survey Stations are positioned according to the plans, or according to the locations as set out on site, unless alternative positions are approved by the *Supervisor*. The *Contractor* shall also conduct an independent survey to establish the co-ordinates (Hong Kong 1980 Grid System) and elevations (Hong Kong Principal Datum) of completed Investigation Stations for inclusion in the Final Fieldwork Report. All surveys shall be conducted by the approved surveyor.

(6) Unless otherwise requested by the *Supervisor*, the *Contractor* shall submit within 7 days from the Contract Date a drawing showing the location of any points, levels and lines of reference to which the setting out of the *works* will be based.

(7) Digital photographs plus three sets of 4R printed photographs recording the existing buildings, road furniture, roadworks, bridges and other features in the vicinity of the Site shall be submitted to the *Supervisor* within 7 days after the Contract Date. A key plan showing the locations and direction of photo taking shall also be submitted with the photos. The *Contractor* shall supply approved albums of a 300mm x 200mm size to accommodate the photographs. The digital photograph record shall include, besides those described, Reference number, Name of photographer, Date and time taken, Location and Identification of subject matter, Weather conditions and any other points of interest. The digital photograph record shall be part of the as-constructed survey records.

As-built Survey 1.47A (1) As-built survey shall be executed by the *Contractor* on all completed phases of the *works*. The Contractor shall be responsible for submission a set of as-built survey plans in both hardcopy and soft copy to the *Project Manager* submitted upon phase completion of the *works*. The *Contractor* shall be responsible for submitting survey information which was presented in an acceptable professional standard and in an appropriate scale specified by the *Project Manager*. A copy of field notes, field data and resultant plans arising from these surveys shall be handed over to the *Project Manager* in transparency and/or microfilms and digital format upon completion.

(2) The *Contractor* shall be responsible for submitting drawing computer files which shall be delivered on compact disc in MicroStation (.dgn) file format unless otherwise specified by the Director's

Representative. The survey drawings shall be in compliance with the CAD Standard for Works Project (CSWP) version 1.03.00 (or later versions as agreed between the *Project Manager* and the *Supervisor*/ *Contractor* from time to time) as posted on the Environment, Transport and Works··Bureau's web-site and the Drafting Specification for Engineering Survey Rev 2.0 (or later versions as agreed between the *Project Manager* and the *Supervisor*/ *Contractor* from time to time) as posted on Civil Engineering and Development Department's web-site.

Fences and signs on the Site 1.48 (2A) The *Contractor* shall provide and erect two numbers of size C project signboards as shown on the latest HyD Standard Drawings H6106 and H6107 at locations to be directed by the *Supervisor*. The *Contractor* shall also be responsible for the maintenance, updating the details and subsequent dismantling and removal of the project signboards and project notice boards at the *defects date* or such earlier date as directed by the *Supervisor*. The wording of the project signboards will be prescribed by the *Supervisor* but the pictorial description and the calligraphy and characters shall be designed by the *Contractor*.

(7) The *Contractor* shall provide sufficient numbers of clean and legible publicity boards for motorists and notice boards for pedestrians as detailed on the latest HyD Standard Drawings at all times that carriageway and footway space is occupied. The *Contractor* shall also be responsible for the maintenance, updating the details and subsequent dismantling of the publicity boards and the notice boards. Locations of these boards shall be approved by the *Supervisor*. The guidelines for the display of publicity board for motorists given in HyD Guidance Notes No. RD/GN/019 shall be followed.

(8) The *Contractor* shall provide and erect display boards in accordance with HyD Guidance Notes No. RD/GN/024B for apparently unattended road works as detailed on the latest HyD Standard Drawings H6133, H6134, and H6138 where a site has to be left unattended apparently for a whole working day or longer because of a good reason for the proper execution of the *works*. The wording and the location of the board shall be approved by the *Supervisor*. The display boards shall be covered up or removed when not in use to avoid confusion. The *Contractor* shall also be responsible for relocating, maintenance and subsequent dismantling of the display boards.

(9) The *Contractor* shall, provide and maintain a telephone line operated by a competent person during office hours and maintain an answering machine for after hours to receive enquires and complaints from the public. The telephone number shall be prominently displayed on the Project signboards on the Site. Every telephone call received shall be individually logged and recorded in a form acceptable to the *Supervisor*. The *Contractor* shall notify the *Supervisor* within 1 hour of each of the call and keep the *Supervisor* fully informed of action being taken or to be taken to address the enquiry or complaint. The *Contractor* shall ensure that resources are mobilised expeditiously to address any such complaint, which is in the opinion of the *Supervisor* a default due to the *Contractor* under the Contract, in order that any

inconvenience to the public is kept to minimum.

A public complaints and enquiries cum information centre will be set up at the *Project Manager* and the *Supervisor*'s principal office or secondary office as directed by the *Project Manager*. In case of enquiries or complaints received, the *Contractor* shall arrange meetings with the complainants, in the presence of the *Project Manager* and his staff, in order to alleviate the concerns/ complaints, if upon requested.

Sufficient indicators, directional signs and notice boards shall be required to escort the complainants as directed by the *Project Manager*.

The Temporary Accommodation for the Project Manager

1.49 G.S. Clause 1.49(1) is deleted and replaced by:

(1) The *Project Manager*'s Site Accommodation shall be a rented office to be provided for the *Project Manager* and *Supervisor* and his staff and shall be available for occupation within the period from the time as instructed by the *Project Manager* after the Contract Date as notified by the *Project Manager* until the *defect date* or such earlier date as directed by the *Project Manager*. The *Contractor* shall ensure that the rented site office is allowed for the purposes/uses of Commercial (excluding any residential, service apartment, hotel and eating place) or Office(s) or others that permitted to set up any Office(s) or the like in the Deed of Mutual Covenant, if any, of the private unit and registration under the Land Registry. The rented *Project Manager*'s Site Accommodation and all items of furniture, equipment, accessories, and supporting personnel etc for the rented *Project Manager*'s Site Accommodation as listed in PS Appendix J1 shall be provided. All equipment, furniture and accessories shall be returned to the *Contractor* after removal of the rented *Project Manager*'s Site Accommodation.

The following is added after G.S. Clause 1.49(5):

(6)(a) The *Project Manager*'s Site Accommodation shall be for the exclusive use of the *Project Manager* and *Supervisor* and their staff or any parties authorized by the *Client* without interruption.

(6)(b) The *Contractor* shall obtain the *Supervisor*'s acceptance before relocating or decommissioning of accommodation and equipment.

(7) The *Contractor* shall make the *Project Manager's Site Accommodation* habitable and operational at all times throughout the course of the *works* up to the completion of the *defects date* or any extension as instructed by the *Project Manager*.

(8) The *Contractor* shall provide and maintain in good working order and repair if necessary, all equipment, furniture, accessories, and supporting personnel etc. as specified in PS Appendix J1. All equipment, furniture and accessories shall be returned to the *Contractor* after the decommissioning of the accommodation.

(9) The *Contractor* shall provide and maintain in good working order and repair if necessary, computing and networking facilities as specified in

PS Appendix J2. Apart from the hard-disks, all computing and networking facilities shall be returned to the *Contractor* after the decommissioning of the *Project Manager*'s Site Accommodation.

(10) The *Contractor* shall provide and maintain in good working order and repair if necessary, all testing equipment as specified in PS Appendix J4. All testing equipment shall be returned to the *Contractor* after the decommissioning of the *Project Manager*'s Site Accommodation.

(11) Within 2 weeks from the *starting date*, the *Contractor* shall provide and erect a temporary office for the *Supervisor* in accordance with PS Appendix J1. The *Contractor* shall maintain and service the temporary office until the *Project Manager*'s Site Accommodation is completed. The *Contractor* shall be responsible for transportation of documents, furniture, and equipment from the temporary office for the *Supervisor* to the *Project Manager*'s Site Accommodation and subsequent decommissioning of the temporary office and reinstatement of the properties occupied by the temporary office for the *Supervisor*.

(12) The *Contractor* shall, within 2 weeks from the Contract Date, submit to the *Project Manager* for acceptance the proposed layout and details of the *Project Manager*'s Site Accommodation in a suitably rented building building in Yuen Long or other locations as agreed by the *Project Manager*, prior to enter any rental arrangement with landlord or his agents. The rented *Project Manager*'s Site Accommodation shall have a usable floor area not less than 130 square metres.

(13) Not used.

(14) The *Contractor* shall, when advised in writing by the *Supervisor*, decommission the *Project Manager*'s Site Accommodation; reinstate the properties occupied by the *Project Manager*'s Site Accommodation to the conditions prevailing at the time of taking over the properties.

(15) The *Contractor*'s attention is drawn to the fact that the site staff for other contracts of the *Client* shall be allowed use of the *Project Manager*'s Site Accommodation and he shall allow for the extension of servicing to these site staff.

(16) All contract preliminary items shall be provided to the office of the *Supervisor*'s Representative for central acceptance and distribution. The *Supervisor*'s Representative should inform the *Contractor* of the name of the officers responsible for accepting these items. The *Contractor* shall not provide the items directly to an individual member of the site supervisory staff.

(17) Any equipment or facilities to be provided for the use of the *Supervisor* are only required to meet the minimum requirement stipulated in the *contract.* Where this is impracticable (e.g. When the model just satisfying the minimum requirements are outdated or out of stock), the *Contractor* may provide at his own cost equipment or facilities slightly exceeding the minimum requirement. Extravagant or out-of-the-norm over-provision is unnecessary and should be avoided. In the event that a much higher quality than that stipulated shall be provided for legitimate reasons, the *Contractor* shall give prior

notification to the *Client* of such an over-provision.

(18) Not used.

(19) The *Contractor* shall provide adequate lockable file cabinets placed within the main site office of the *Project Manager* and the *Supervisor*, or at other locations as directed by the *Project Manager* for storing document 1 year after the *defects date*.

(20) The *Contractor* shall maintain the *Project Manager's* Site Accommodation together with all the sustainable provisions in serviceable conditions.

(21) The *Contractor* shall ensure that the automatic devices for energy saving are not overridden manually or by other means.

(22) The *Contractor* shall prepare an inspection checklist for checking and monitoring the conditions and functions of the facilities in the *Project Manager's* Site Accommodation.

(23) The *Contractor* shall assign a staff member at supervisory level or above to conduct the checking and arrange the corrective actions as and when necessary.

(24) The *Contractor* shall post notices at conspicuous locations to remind staff to switch off the computers and other electrical appliances when not in use and before leaving.

(25) The *Contractor* shall prepare and submit monthly summary reports which should include details of the checking, the findings and the corrective actions taken, if any, in the reporting month to the *Supervisor* for record and monitoring.

(26) The *Contractor* shall ensure that facilities for natural lighting and ventilation are not adversely affected or obstructed.

(27) The *Contractor* shall provide 24-hour security guard services at the Works Areas including the *Project Manager*'s Site Accommodation and the temporary office.

Facilities to assist Project Manager's and Supervisor's off-site inspection

1.49A In event that the staff of the *Supervisor* has to carry out the inspection or other duties in connection with the works outside Hong Kong due to the *Contractor*'s proposed off-site works, the *Contractor* shall facilitate the *Project Manager* and the *Supervisor* and his staff by providing the following:

(a) Transportation or reimbursement of the expenses of transportation between the Site and the place to carry out inspection; accommodation and subsistence or reimbursement of these expenses if day-return is not practical;

(b) Reimbursement of other expenses incurred by the *Project Manager* and the *Supervisor* and his staff such as insurance and addition taxation in Mainland;

(c) Roaming services for the mobile phones;

(d) Any other assistance and facilities as requested by the *Project Manager* and the *Supervisor* and his staff to carry out their duties.

Contract Computer Facilities for the Project Manager and the Supervisor

1.49B (1) The *Contractor* shall provide and maintain, for the use of the *Supervisor*, "Contract Computer Facilities" in accordance with the following sub-clauses (2) to (10) of this Clause.

(2) The *Contractor* shall provide and maintain in good working order and repair if necessary, computing and networking facilities as specified in PS Appendix J2.

(3) The *Contractor* shall, within 14 days from the date of the acceptance of the tender, submit for *Supervisor*'s acceptance details of Contract Computer Facilities to be provided for the purposes of this contract. The proposed equipment and software shall be well proven and must not infringe any copyrights.

(4) Within 14 days of receiving the *Supervisor*'s acceptance or other such date agreed by the *Supervisor*, the *Contractor* shall deliver the Contract Computer Facilities to the *Supervisor*'s temporary accommodation. All software shall be delivered to the locations specified by the *Supervisor*, in intact shipping cartons with all seals unbroken. At the time of delivery, the originals of all warranty agreement, invoices and other supporting documents related to the price of the equipment shall be produced for inspection by the *Supervisor*.

(5) Software included in the Contract Computer Facilities shall be licensed in the name of the Government of the Hong Kong Special Administrative Region.

(6) Within five working days of delivery, the *Contractor* shall arrange for the equipment to be installed and tested to complete working order to the satisfaction of the *Supervisor*.

(7) The *Contractor* shall license and be responsible for the cost of servicing and maintenance of the Contract Computer Facilities including application software and replacement parts throughout this contract and until the *defects date*, or where there is more than one such period, the expiry of the latest Period or other agree date if so directed by the *Supervisor*. The *Contractor* shall allow for regular servicing of all items including application software and for replacement of worn out or defective parts.

(8) In the event of a breakdown in the Contract Computer Facilities, the *Contractor* shall provide a replacement unit for the period of breakdown.

(9) Not Used.

(10) In the event of a burglary incident in the Contract Computer Facilities, the *Contractor* shall provide replacement unit immediately after the incident at no extra cost to the contract.

Servicing of Contract Computer Facilities for the Project Manager and the Supervisor 1.49C

(1) The *Contractor* shall license, service and maintain the Contract Computer Facilities throughout the period of servicing.

(2) The *Contractor* shall keep the Contract Computer Facilities in first class condition and repair any defects. Replacement of parts or whole of the equipment shall be necessary if they are damaged beyond repair.

(3) The *Contractor* shall supply all consumables for the Contract Computer Facilities.

(4) On the *defects date* or at such earlier date as instructed by the *Supervisor*, the Contract Computer Facilities shall be reverted to the *Contractor*.

The Contractor's Site accommodation 1.50***S***

(1) The *Contractor* shall setup a main *Contractor*'s site accommodation next to the *Project Manager*'s Site Accommodation. The *Contractor* shall centralise all his office-based site staff including but not limited to all *Contractor*'s key staff and specially required staff as stipulated in PS Clause 1.12A to work in the *Contractor*'s site accommodation. The *Contractor* shall not be entitled to additional time and cost for the main *Contractor*'s site accommodation outside the Site. Notwithstanding this, upon instructed by the *Project Manager*, the *Contractor* shall centralise all his office-based site staff to work in the site accommodation for the *Project Manager* and the *Supervisor* within 2 months of the instruction and the *Contractor* may propose the shared use of some equipment, facilities, furniture, accessories and supporting personnel as stipulated in PS Clause 1.49 and Appendix J1 and J2 with the extended and enhanced specification and requirement to the *Project Manager* for acceptance.

(2) The *Contractor* shall provide and maintain all necessary sheds and stores and remove the same from the Site on completion of the *works*. No structures shall be erected by the *Contractor* within the Site without the *Project Manager*'s written consent; such consent will not relieve the *Contractor* of the responsibility of siting temporary structures clear of the *works*. A copy of the plan showing the extent and position of all offices, stores, sheds, etc. shall be prepared by the *Contractor* and disposal of sewage and waste water from the Works Area shall be by means approved by the *Project Manager* and relevant authorities.

(3) No temporary accommodation for workmen shall be erected by the *Contractor* within the Site without the *Project Manager*'s written consent. It shall be to the satisfaction of the *Project Manager* and kept in a clean and sanitary condition.

(4) For the purpose of this Clause, stores, sheds, structures, etc. shall be taken to include fixed Construction Plant.

(5) Immediately after the award of the Contract, the *Contractor* shall arrange to provide welfare facilities specified below for workers employed on the *works*, irrespective of whether they are in the employment of the *Contractor* or his subcontractors. The *Contractor* shall maintain the welfare facilities provided on the Site throughout

the Contract, and shall remove the facilities and reinstate the areas after removal of the facility or upon completion of the *works* where necessary. The costs for the provisions shall be priced in the Activity Schedule under the Item "Temporary Accommodation of the *Contractor*". The sufficiency of provision for the welfare facilities shall be monitored and reviewed in the Site Safety Management and Environmental Management Committee (SSEMC) meetings. The *Contractor* shall replenish the insufficiency of provision to meet the specified requirements and shall not be entitled to compensation events of such replenishment after commencement of the *works*.

(a) Storage Compartments

The *Contractor* shall provide storage compartments for use by workers. The storage compartment shall be placed at a location close to the Site entrance to facilitate workers to obtain/place their personal protective equipment such as safety helmet, reflective vest, eye protectors, safety harness etc. when they enter/leave the Site.

The *Contractor* shall ensure adequate provision of storage compartments taking into account the fluctuation in labour workforce. The *Contractor* shall determine the dimensions of the storage compartments. To facilitate monitoring and control, each storage compartment shall be designed for use by not more than five persons unless otherwise approved by the *Supervisor*. The *Contractor* shall propose the number, location and layout arrangement for the placement of storage compartments on the Site for the approval of the *Supervisor* within 14 days from the date of commencement of the *works* on the Site. The *Supervisor* shall review regularly the adequate provision of storage compartments on the Site for use by workers.

(b) Drinking Water Facilities

The *Contractor* shall provide free drinking water facilities for workers working on the Site throughout the Contract to minimize waste plastic bottles. The drinking water facility can be in the form of a water pot with a cover at the top for water refilling and a tap at the bottom for drawing water, a distilled water drinking fountain or any other form that the *Contractor* considers appropriate. The drinking water facilities shall be provided indoors and each of which shall have a storage capacity of not less than 20 litres.

(c) Toilet Facilities

The *Contractor* shall provide toilet facilities for workers working on the Site throughout the Contract, unless otherwise approved by the *Supervisor* that the provision is not necessary. The toilet facilities shall be suitable for use bisexually and placed at convenient locations close to workers' workplace. The toilet facilities shall be either one of the following types:

(i) wet type with flushing water supply and the toilet waste

properly collected and discharged into a sewerage system, septic tank, soakaway system, or in-situ sewage treatment facilities proposed by the *Contractor* and approved by the *Supervisor*;

(ii) chemical type complete with containers for regularly cleaning and removal by specialist contractor.

For connection to a public sewerage system, the *Contractor* shall obtain approval from the Drainage Services Department prior to making the connection. If a soakaway system is adopted, the *Contractor* shall carry out in-situ percolation test to prove that the soil shall have sufficient absorption capacity to treat the toilet waste flow. The test results shall be submitted to the *Supervisor* for recording before discharging. If in-situ sewage treatment facility is adopted, the plant shall be equipped with disinfection unit to sterilize the treated effluent before discharging.

The number of toilet facilities provided on the Site shall be at a ratio of not less than one for every 30 workers. The *Contractor* shall propose the toilet type, number and their locations of placement for the approval of the *Supervisor* within 14 days from the date of commencement of the *works* on the Site. The *Contractor* shall maintain the toilet facilities in clean and hygienic condition. The *Supervisor* shall review regularly the adequacy of toilet facilities provided by the *Contractor* on the Site and the cleanliness and hygienic conditions of these toilets throughout the Contract.

(d) Hand-wash Facilities

The *Contractor* shall provide hand-wash facilities in the form of water points and sinks for workers working on the Site throughout the Contract, unless otherwise approved by the *Supervisor* that the provision is not necessary The number of hand-wash facilities provided on the Site shall be at a ratio of not less than one for every 20 workers. The discharge from hand-wash facilities shall be collected to prevent spillage on the floor, and discharged to a water reception tank or flushing water supply tank for recycling/reusing as appropriate. The *Contractor* shall propose the number of hand-wash facilities, their locations and the system of how to collect the discharge from the wash-water basin for reuse/recycle on the Site for the approval of the *Supervisor* within 14 days from the date of commencement of the *works* on the Site. The *Supervisor* shall review regularly the adequacy of hand-wash facilities provided by the *Contractor* on the Site and the effectiveness of the discharge collection system throughout the Contract.

(e) Showering Facilities

When the number of workers working on the Site will likely be exceeding 100 in a day for a continuous period of twelve months or more during the Contract, the *Contractor* shall, in addition to the provision of hand-wash facilities pursuant to the

Clause above, provide showering facilities on the Site unless otherwise approved by the *Supervisor* that the provision is not appropriate. The showering facilities shall be provided indoors or inside containers with appropriate drainage connections. The number of showering points provided on the Site shall be at a ratio of not less than two for the first 100 workers and an additional one for every additional 50 workers.

The *Contractor* shall provide showering facilities at different locations/rooms for use by male or female workers separately. Each showering point shall be furnished with hot and cold water supply. The *Contractor* shall be responsible for the water and electricity charges for providing the facilities. Details of installation of the facilities shall be submitted for the approval of the *Supervisor* within 30 days after the commencement of the *works* on the Site.

(f) Rubbish Bins

The *Contractor* shall provide sufficient rubbish bins with covers at strategic locations on the Site for collection and disposal of general wastes generated by workers throughout the Contract. The rubbish bins shall be provided in pairs, one for aluminium cans and plastic bottles and the other for general refuses. The locations for the rubbish bins in pairs shall be placed at convenient locations close to the workers' workplace to facilitate use. The number of paired rubbish bins provided on the Site shall be at a ratio of not less than one pair for every 20 workers. The size of the rubbish containers shall be of minimum 1 m high and have an opening of at least 0.28 m^2 at the top for collecting wastes.

The *Contractor* shall maintain the cleanliness of rubbish bins, and arrange collection and disposal of general waste inside the rubbish bins regularly but in any case shall be not less than once in every three days. The *Contractor* shall also make arrangement for collecting papers and packaging on the Site to reduce disposal of wastes to landfills. The *Contractor* shall propose the number and the location of placement of rubbish bins provided on the Site together with the arrangement for on-site sorting of aluminium cans, plastic bottles and papers for the approval of the *Supervisor* within 14 days from the date of commencement of the *works* on the Site. The *Supervisor* shall review regularly the adequacy of rubbish bins provided on the Site by the *Contractor* and the effectiveness of on-site sorting of general wastes throughout the Contract. The *Contractor* shall include this requirement to form part of his EMP if appropriate.

Transport for the Project Manager and the Supervisor

1.52 GS Clause 1.52(2) is deleted and replaced by:

(2) (a) The transport shall be for the exclusive official use of the *Project Manager* and *Supervisor* and their staff (except that it shall be made available for use by Highways Department and/or other Government officials if and when the *Project Manager* so

informs) in connection with:-

(i) this contract, the works and related duties;

(ii) for journeys in connection with the supervision of works of other contracts in the same geographical area;

(iii) for other official journeys or by Highways Department and/or other Government officials on official business if and when the *Project Manager* so informs.

(iv) transportation of *Project Manager*'s and *Supervisor*'s site staff between MTRC Station and the *Project Manager* and *Supervisor*'s Site Accommodation.

(b) The transport shall be kept available for use at all times during the following periods:

(i) normal working hours;

(ii) such other times outside normal working hours, including 1900 hours to 0700 hours on the next day, Sunday and General Holidays when the *Contractor* is working; and

(iii) when requested by the *Project Manager* or *Supervisor* for emergency situations including but not limited to the periods of flooding warning, red or black rainstorm warnings, landslide warning, periods with the hoisting of any tropical cyclone warning signal, and periods with "extreme condition after super typhoons" announced by the Government, or for the discharge of his duties outside the period stated in sub-para (b)(i) and (ii) above. The *Contractor* shall arrive at the designated location as instructed by the *Project Manager* or *Supervisor* within one hour. The *Contractor* shall extend the insurance coverage for the transport, passengers and the driver to that effect.

(c) The *Contractor* shall not use the transport for his own purposes and he shall ensure that the transport is not used by any person outside the specified periods.

GS Clause 1.52(4) is deleted and replaced by:

(4) (a) The *Contractor* shall provide the services of competent English/Cantonese speaking drivers approved by the *Project Manager* but he shall nevertheless permit the transport to be driven in an emergency by any other qualified drivers nominated by the *Project Manager* or *Supervisor*. The *Contractor* shall provide sufficient number of drivers to cover all working hours including meal break. In particular, if the *Contractor* usually need to carry out works at night, another driver should be provided for the night shift.

(b) The *Contractor* shall make due allowance for the services of competent driver during the period specified in Clause

1.52(2)(b).

(c) The *Contractor* shall provide Construction Industry Safety Course (Green Card) training for the relevant drivers.

GS Clause 1.52(8) is deleted and replaced by:

(8)S The transport shall be provided until the *defects date* or such earlier date instructed by the *Project Manager*. The transport shall revert to the *Contractor* on the *defects date* or at such earlier date as the *Project Manager* may advise the *Contractor* in writing. The *Contractor* shall make available similar alternative arrangement when the transport provided are unavailable for any reason.

The following is added after GS Clause 1.52(8):

(9) The *Contractor* shall provide the transport for use by the *Project Manager* and *Supervisor* within 2 weeks form the date of the *Project Manager*'s instruction. The following vehicle shall be environmental-friendly vehicles that have been type-approved by Environmental Protection Department:

(a) One (1) new (or not more than 2 years old) hybrid vehicle that shall be in good roadworthy conditions having an engine capacity of not less than 1,198 c.c., with seating capacity of not less than 7 passengers including the driver, and be one of the type-approved hybrid private car models as shown in Environmental Protection Department's website at:

https://www.epd.gov.hk/epd/sites/default/files/epd/english/environmentinhk/air/prob_solutions/files/Hybrid_PRC_GV_list_e_Table_a.pdf

The vehicle is to be under a hard cover and provided with air-conditioning, seat belts for all seats. The driver of the vehicle shall be equipped with hand free type mobile telephone. The vehicle shall also be provided with 5 sets of safety helmets, fluorescent vests, umbrellas and auto-toll vehicle devices for all tunnels and bridges within the HKSAR.

The vehicle shall be unless otherwise instructed by the *Project Manager*, the vehicle for the *Project Manager* shall be painted in a standard colour and with a standard logo and legend as that detailed in the latest version of HyD Standard Drawing. The *Contractor* shall ensure that the identification sign on the transport is removed immediately after its reversion to the *Contractor*.

The vehicle must equip a Vehicle Roof Mounted Highway Maintenance Sign as shown on Highways Department Standard Drawing No. H6116 and painted in bright yellow complying with the following:-

- No. 008 E51 in British Standard 5252F: 1976; or
- No. BSC 355 in British Standard 381C: 1996; or

- No. BSC 356 in British Standard 381C: 1996

(10) The vehicle in sub-clause (9) shall be to the acceptance of the *Project Manager* or *Supervisor*. It shall display the information of this contract such as contract number/title and the *Contractor*'s name.

(11) The *Contractor* shall maintain the transport in a clean and serviceable condition. The transport shall be serviced regularly according to the respective sole agent's recommendation. The *Contractor* shall provide all necessary fuel, oil, electricity and other consumables for the transport and pay all taxes, licenses, insurances, toll charges, parking fees and mooring fees. The vehicle shall be equipped with auto-toll devices for all tunnels, bridges and expressways.

(12) The *Contractor* shall ensure that the drivers institute and maintain records of the use of the vehicle. Such a record shall include, inter alia, details, times and purpose of journeys together with appropriate odometer readings and distances travelled. The person using a vehicle or authorizing the journey shall be required to sign his name and title against the entries.

(13) The *Contractor* shall obtain log books from the *Supervisor*. The drivers shall present current log books for inspection when so required by the *Project Manager* or *Supervisor*. All completed log books shall be handed over to the *Supervisor*.

(14) The vehicle shall be covered by a fully comprehensive insurance which includes passenger liability under all weather conditions and allows it to be driven by any driver.

(15) The *Contractor* shall supply, license and maintain a mobile telephone, subscribed with service plans of unlimited local airtime and able to provide clear transmission in all areas of the Hong Kong Special Administrative Region, in the vehicle as required in this PS Clause for the exclusive use of the *Project Manager* and *Supervisor* and their staff, and the staff of Highways Department if and when the *Project Manager* so informs. The mobile telephones supplied under this sub-clause are additional to those supplied under PS Clause 1.49 for the *Project Manager*'s site accommodation, and shall be reverted to the *Contractor* on the date when the vehicle is reverted to the *Contractor*.

(16) (a) The vehicle listed in sub-clause (9) are to be parked in the *Project Manager*'s site accommodation when not in use, unless the *Contractor* is authorized by the *Project Manager* to park the vehicles elsewhere.

(b) The *Contractor* should not leave the vehicle unattended without reasonable causes when the vehicle is away from its designated base.

(c) No spare vehicle keys are to be produced without prior acceptance from the *Project Manager*.

(17) The transport for the *Project Manager* and *Supervisor* should be properly painted or affixed by adhesive plastic labels with the

contract number, the name of the *Contractor*, Department name, Department logo, Department complaint hotline (or other suitable identifications) and the phrase "For Official Use Only" "只供公務用途" in both English and Chinese characters and in good size letters for easy identification.

PHOTOGRAPHS

Photographs 1.55***S***

(1) The *Contractor* shall take digital colour progress photographs at monthly intervals in accordance with the direction of the *Supervisor*. The digital photos shall be of a resolution not less than 3M pixels. The photos shall be taken at time and locations as directed by the *Supervisor* to show the progress of the *works* and the quality of materials and workmanship.

(2) The progress photographs taken at monthly intervals by the *Contractor* shall be as follows: -

(a) A set of photographs shall consist of (i) 30 digital pictures; (ii) a 4R size print for each digital picture; and (iii) three 4R size prints for each of the not more 20 digital pictures selected by the *Supervisor*. The *Contractor* shall supply approved albums for keeping the 4R size prints of the digital pictures selected by the *Supervisor*. The *Supervisor* may order additional 8R size framed prints from the digital pictures where necessary.

(3) The photographs shall be taken by the professional photographer, who is also a professional cameraman approved by the *Supervisor*. Processing shall be carried out by a competent processing firm approved by the *Supervisor*.

(4) All prints of the photographs shall be authenticated by both the *Supervisor* and *Contractor* by signing the back of the prints. Self adhesive labels containing the information as shown on PS Appendix B5 are to be provided by the *Contractor*. The *Contractor* shall be responsible for informing the *Supervisor* when photographs are taken and for completing the information on the self adhesive label to be placed on the back of each photograph.

(5) (a) Not used.

(b) Exposed film shall be processed at a professional laboratory to ensure even results throughout the *works*. Photographs of poor quality resulting from uneven colour balance, poor focus, missing information, or other defects, will not be accepted, and the *Contractor* shall re-photograph the subject and shall not entitle to any compensation event.

MISCELLANEOUS

Assistance to the Project Manager and the Supervisor 1.56

(1) The *Contractor* shall provide safe access to all parts of the *works* and where necessary provide and maintain suitable scaffolding, ladders, and gangways etc. to facilitate inspection by the *Project Manager* and the *Supervisor*.

(2) The *Contractor* shall provide and maintain labour, temporary staging, markers, equipment and information required by the *Project Manager* and the *Supervisor* for checking the setting out, inspection, testing, measurement and supervision of the *works*.

Request for inspection 1.57 (1) The *Contractor* shall arrange and provide for the operation of a Request for Inspection and/or Survey Check RISC system using the standard RISC form in Appendix B6 to this Particular Specification for inspection and/or survey check of the *works* as required under this contract or as notified by the *Supervisor*. The *Contractor* may propose any changes to the standard RISC form to the *Project Manager* and the *Supervisor* for acceptance to suit the nature and scope of the *works*.

(2) The *Contractor* shall adopt digitization of the RISC system, and submit the proposal to the *Supervisor* for acceptance according to Clause 1.49B(11) of this Particular Specification. Upon completion of this contract, the *Contractor* shall handover the digital RISC system including the data/record contained therein to the *Client* according to Clause 1.49B(11) of this Particular Specification.

(3) When any item of *works* is expected to be ready for inspection and/or survey check, the *Contractor* shall submit a copy of the completed RISC form with clear description of the *works* to be inspected and/or surveyed accompanied by the relevant drawings to the *Supervisor* in advance by the deadline specified below:

	Time for inspection and/or survey check		
	Monday to Friday (except General Holiday)	Saturday or Sunday	General Holiday (except Sunday)
Deadline for submission	24 hours before inspection and/or survey check	noon of the preceding Thursday	48 hours before inspection and/or survey check

Note: If the deadline for submission falls on Saturday or a General Holiday, the deadline shall automatically be advanced by 24 hours.

If any item of *works* is not completed by the time for inspection and/or survey check shown on the RISC form, then at the discretion of the *Supervisor* either a revised time for inspection and/or survey check shall be agreed or a new form giving a revised time for inspection and/or survey check shall be submitted by the *Contractor*, of which the deadline for submission shall nevertheless comply with this sub-clause. The *Supervisor* shall at his discretion refuse to inspect and/or survey any item of *works* without proper and timely submission of the RISC form in advance as required and any extra time or delay so caused shall not constitute any ground for compensation event.

(4) When the RISC forms could not be submitted prior to the time for inspection and/or survey check due to exceptional circumstances with prior agreement by the Supervisor, the Contractor shall submit the RISC forms no later than 3 calendar days following the inspection and/or survey check.

(5) For any item of *works* that requires inspections and/or survey checks of various construction activities, stages, elements and particular aspects or details, a separate RISC form shall be submitted for each of the inspections and/or survey checks of the relevant item. The various construction activities, stages, elements and particular aspects or details appropriate to each item of *works* shall be defined by the *Contractor* and agreed with the *Supervisor*. Unless with the agreement of the *Supervisor*, the *Contractor* shall only proceed to the next construction activity, stage, element and particular aspect or detail of any item of *works* after inspection and/or survey check and acceptance of same by the *Supervisor*.

(6) In addition to the above, the Contractor shall submit to the Supervisor before noon of each day (other than Saturday or General Holiday), a detailed list of the works to be carried out on the next working day. This shall include the exact location and the nature of each construction activity, stage, element and particular aspect or detail, and the time when the works are to be carried out to enable the Supervisor or his staff to attend for the purpose of inspecting and/or surveying such works.

(7) Under no circumstances should any item of works be covered up or put out of view unless with the agreement of the Supervisor. Should any item of works not be inspected and/or surveyed as a result of the Contractor's failure to take action in accordance with sub-clauses (1) to (6) of this clause, then such item of works will be treated as having been covered up without the agreement of Supervisor. Such item of works may be classified as a Defect.

(8) To enable the Supervisor to plan his inspections and/or survey checks on a weekly basis, the Contractor shall submit to the Supervisor before noon of each Friday, a short programme showing the works that he proposes to carry out in the following week. If that Friday is a General Holiday, the Contractor shall submit the short programme before noon of the preceding date which is not a General Holiday.

(9) To enhance the monitoring of the completeness and timeliness of inspections and/or survey checks, the Contractor shall maintain a register summary to keep track of the status of the RISC forms and submit the register summary to the Supervisor for record before noon of each Friday. If that Friday is a General Holiday, the Contractor shall submit the register summary before noon of the preceding date which is not a General Holiday.

(10) The Supervisor may notify the Contractor from time to time the names of his staff who may receive the RISC forms or are assigned to carry out inspection and/or survey check.

(11) The submission of the RISC form shall not relieve the Contractor of any of the duties, responsibilities, obligations or liabilities imposed upon him by any of the provisions of this contract nor restrict the Supervisor or his staff from carrying out any other inspection and/or survey check. The Contractor shall make due allowance in his programme of the works for the use of this RISC system. Notwithstanding other provisions of this contract, compliance with

the requirements of this clause shall not constitute any ground for compensation event.

1.58 Not Used.

Working hours

1.59 (1) Subject to the *Supervisor*'s written approval and agreement from relevant government departments, the *Contractor* may extend the working hours to avoid traffic congestion, etc. on proposed works on a high-speed road, the *Contractor*s should make reference to the Guidance Notes on Lane Working Time Schedule for High Speed Road Maintenance Contractor under the cover of HyD's letter ref. HNT609/HSR.G53 dated 13 August 1999.

(2) Working at night will be required for some of the *works*. The *Contractor* shall be responsible for obtaining the approval of the appropriate authorities as to the time when the work outside the normal working hours would be carried out and details of any necessary arrangements.

(3) The necessity for or request to work in extended hours or at night shall not constitute grounds for a compensation event by the *Contractor* for additional prices.

(4) The *Contractor*'s attention is drawn to the requirements of the Noise Control Ordinance regarding noise mitigation.

Agreement of site records

1.60 (1) The *Supervisor* shall record daily in the site diary information with regard to labour, plant, materials, works carried out, utilities and instructions issued to the contractor and any other facts that may affect the programme or quality of the *works*.

(2) The *Supervisor* shall, as soon as is reasonable after the information required by sub-clause (1) of this Clause has been recorded in the site diary, submit the site diary to the authorised agent or the *Contractor* for his agreement. Within 48 hours of receipt of the site diary, the authorised agent or the *Contractor* shall sign the site diary indicating his agreement to the information recorded or if he does not agree with any of the items recorded therein, shall draw reference to the points of disagreement in writing to the *Supervisor* where upon the *Supervisor* and the authorised agent or the *Contractor* shall ascertain jointly the facts and amend the site diary as necessary.

(3) If the authorised agent or the *Contractor* fails to comply with the requirements of this Clause, then the contents of the site diary as recorded by the *Supervisor* shall be taken to be correct.

Labour return

1.61 The *Contractor* shall enter daily, on a form as shown in PS Appendix B7 provided by the *Project Manager*, information in respect of labour employed on the Site for the *works*. Three copies of the form shall be prepared and two completed forms shall be submitted to the *Project Manager*. The remaining copy shall be retained by the *Contractor*.

Monthly return on construction site employment

1.62 GF 527 (as attached in ETWB Technical Circular (Works) No. 3/2003) and GF 527A (as attached in ETWB Technical Circular (Works) No. 4/2005) shall be completed by the *Contractor* and submitted to the *Project Manager* within the first four working days of the month following the reporting

statistics

month. The *Contractor* may be requested to substantiate the data provided in the forms, and correct for any discrepancy found. The guidelines for submission and completion shall refer to the relevant technical circulars. A copy of the form and the guidelines to complete the form are enclosed in PS Appendix B8.

Contractor to supply equipment

1.63 (1) The *Contractor* shall supply all hand and mechanical Equipment necessary for the construction, completion and maintenance of the *works*.

(2) All Equipment shall be maintained in good working order. The *Contractor* shall be required to remove and replace any item of equipment considered by the *Supervisor* to be unsatisfactory or unsuitable for use on the *works*.

(3) Any change in the method of performing the *works* as a consequence of such removal and replacement shall be at the cost of the *Contractor*, who shall have no compensation event against the *Client* on account of having to carry out the *works* by another method, for the cost of idleness or for bringing to the Site or removal of any Equipment.

Facilities to other contractors

1.64 The *Contractor* shall afford all reasonable facilities to and co-operate with relevant government departments and their contractors, and contractors of utility undertakings etc working on the Site. The *Contractor* shall programme and carry out the *works* in phases as may be necessary to cater for works by other contractors.

Reduce use of timber

1.65 Hardwood shall not be used for falsework, formwork, site hoardings including framing and panelling, and the shoring of trenches and pits. In addition, all components of site hoardings and sign boards shall be metal, and whenever feasible, with bolts-and-nuts jointing method.

Trade or proprietary names

1.66 (1) Where trade names or proprietary products are shown in the Contract documents, these are shown only as a means of describing the requirements of the article, equipment or fitting. Products having equivalent functions or performance may be submitted for the approval of the *Project Manager*.

(2) If the *Contractor* intends to use the intellectual property rights of another party in performing his obligations under the contract, appropriate licences should be obtained from the relevant owners.

(3) Before placing any order for material, manufactured articles or machinery, the *Contractor* shall submit for the approval of *Project Manager*, a list showing such items, and the names of the manufacturers and the manufacturers' publications giving details of the specifications and recommended methods of use etc.

Precautions during severe weather conditions

1.67 The *Contractor* shall monitor weather forecasts from the Hong Kong Observatory and shall ensure that he is aware of rainstorm warnings, flood warnings, landslip warnings, and the hoisting of tropical cyclone warning signals. Whenever any such warnings are announced, he shall maintain 24-hours appropriate precautions and all necessary arrangements for immediate action if required. In particular, whenever a tropical cyclone warning signal No. 8 or above, or a Rainstorm Yellow Warning or a Rainstorm Red

Warning or Rainstorm Black Warning, or a Landslip Warning comes into effect, apart from the requirement that the Site Agent could be readily contacted, the *Contractor* shall have a responsible representative ready at all times to receive instructions from the *Project Manager* or the *Supervisor*. The *Contractor* shall be able to mobilise adequate workmen and plan to carry out repair or remedial works immediately when such situation arises.

Site security 1.68 (1) Not used.

(2) The *Contractor* shall at the early stage of the Contract, co-ordinate with and seek advice from the Police and implement suitable and adequate security measures on Site to avoid construction site theft. The *Contractor* may contact the Regional Crime Prevention Office which are particular relevant to this Contract.

(3) Without limitation to the generality of sub-clause (2) above the *Contractor* shall at the early stage of the Contract formulate a security plan for approval by the *Supervisor*, after consulting the relevant Police crime prevention office.

Publicity 1.69 The *Contractor* shall refer all requests from the press or from other media for information concerning the *works* or the progress thereof directly to the *Project Manager*.

1.70 Not Used

Precautions against flooding and landslide during the wet season 1.71 (1) The *Contractor* shall prepare and submit to the *Supervisor* for approval a contingency plan to deal with flooding and landslide within and immediately outside the site during the wet season. The *Contractor* shall exercise the approved contingency plan and shall carry out drills on the implementation of the contingency plan as and when requested by the *Supervisor*.

(2) The *Contractor* should ensure that all temporary works are designed and maintained to minimise the risk of flooding in neighbouring areas. In circumstances where works interfere with an existing stormwater drainage system adequate flow diversions or pumping arrangements, at least equal to the capacity of the existing system, must be put in place. The *Contractor* should be particularly vigilant during times of heavy rain that arrangements are working satisfactorily.

(3) The *Contractor* should ensure that surface water is discharged through illicit connections into foul sewers and contribute to the exceptionally high flood flows and deposition of grit silt at sewerage treatment works which would cause treatment and pollution problems. The surface water must be directed to natural drainage channels or to stormwater drains.

Independent checking of the design, erection, use and removal of Temporary Works 1.72 (1) Pursuant to Clause F7 of the *additional conditions of contract*, the *Contractor* shall take full responsibility for the design of the Temporary Works, and the method statement for erection, use, and removal of all Temporary Works. The meaning of Temporary Works in this Clause, Clause F7 of *additional conditions of contract* shall also be extended to cover the construction methodology of the *works*.

(2) The design of the permanent works or any existing structures as shown on the Drawings and Specification do not include any design for the Temporary Works necessary for construction. The *Contractor* shall be responsible for verification of the permanent works and the existing structures to ensure that they are acceptable for the Temporary Works designed by the *Contractor*.

(3) Unless otherwise permitted by the *Project Manager*, the Temporary Works shall be designed and constructed to same standards as the Permanent Works. The *Contractor* shall make reference to the Structures Design Manual for Highways and Railways published by Highways Department and this Specification for the design and construction of the Temporary Works.

(4) Minimum requirement of the independent checking engineer is specified in Clause F7 of *additional conditions of contract*. In addition, the independent checking engineer for any Temporary Works related to strengthening works for existing retaining wall and highways structures should process at least 10 years of bridgeworks experience. The independent checking engineer for all other Temporary Works shall possess at least 10 years relevant experience appropriate to the Temporary Works to be certified.

(5) The design of Temporary Works and the method statements for erection, use, and removal of Temporary Works to be certified by the independent checking engineer includes but not limited to the following:-

(a) Temporary Works related to construction of the noise barriers, such as the noise barriers superstructure;

(b) Temporary Works related to any works on the existing structures;

(c) Temporary Works related to the removal of the existing sign gantry and construction of the sign gantry;

(d) Temporary Works for stockpiling and storage;

(e) The Site accommodation for the *Project Manager* and the *Supervisor* and the *Contractor*'s Accommodation;

(f) Temporary Works for all works over or adjacent to roads opened to traffic, footpaths and cycle tracks opened to public; and

(g) Temporary Works for all works which may affect existing utilities or structures.

(6) The construction methodology of the *works* to be certified by the independent checking engineer includes but not limited to the following:-

(a) Construction methodology related to the strengthening works for existing structures;

(b) Construction methodology related to construction of the noise barriers, such as the noise barriers superstructure;

(c) Construction methodology related to any works on the existing structures;

(d) Construction methodology related to the removal of the existing sign gantry and construction of the sign gantry; and

(e) Construction methodology for all works over or adjacent to roads opened to traffic, footpaths and cycle tracks opened to public;

(f) Construction methodology for all works which may affect existing utilities or structures.

(7) Any Temporary Works erected in close proximity to traffic shall be protected against impact from vehicles by suitably designed protective measures. The *Contractor* shall design such protective measures with regard to the conditions prevailing on the Site and the effect of any such impact. Such protective measures are to be designed as part of the Temporary Works subject to checking by an independent checking engineer in accordance with Clause F7 of *additional conditions of contract*.

(8) The form of the certificate required by Clause F7 of *additional conditions of contract* shall follow precisely the wording of the specimens in PS Appendix B9.

Design responsibility 1.73 (1) The *Contractor* shall be fully responsible for the design of the Temporary Works in accordance with Clause F7 of *additional conditions of contract*. The *Contractor* shall also be fully responsible for the design of permanent works in accordance with F2 of *additional conditions of contract* if they are specified in this contract to be designed by the *Contractor* or proposed by the *Contractor* as alternative design. The *Contractor* shall be responsible for obtaining all necessary approvals from relevant authorities for the *Contractor*'s Design in accordance with F2 of *additional conditions of contract*.

(2) Pursuant to Clause F2 of *additional conditions of contract*, the *Contractor* shall be responsible for the design of the permanent works as stated in the Specification or elsewhere in this contract, including the following:

(a) Noise Barriers Works in accordance with Section 27 of the Specification or elsewhere in this contract.

(3) The *Contractor* shall submit the following to the *Project Manager* for approval at least 90 days before commencement of the *Contractor*'s Design works, unless stipulated otherwise in the Contract:

(a) details of all components;

(b) three copies of fully detailed design drawings and calculations with ICE check certificate demonstrating the adequacy of all

components; and

(c) method statement of fabrication and installation.

(4) Submission by the *Contractor* of the information in sub-clause (3) of this Clause shall constitute an acknowledgement that all materials, site measurements and installation criteria have been determined and verified as appropriate and that the information contained in the submissions has been co-ordinated with the overall requirements of the *works*.

(5) The *Contractor* may adopt any information from the Drawings in his design for the parts of the *works* stated in the Contract as being required to be designed by him. Any such information adopted by the *Contractor* shall become the *Contractor*'s Design for which neither the *Client* nor the *Project Manager* shall be responsible.

Design requirements and procedure for Contractor's Design

1.74 (1) All *Contractor*'s drawings shall be prepared using SI units. The conversion of metric to imperial units and vice versa shall be carried out in accordance with the Hong Kong Government Metric Reference Guidebook – International System of Units.

(2) Each design package shall comprise *Contractor*'s drawings incorporating sufficient information to enable the *Project Manager* to gain a full appreciation of the *Contractor*'s design and proposed methods of construction and to assess that the *Contractor*'s Design satisfies the requirements of the Contract.

(3) The *Contractor* shall propose an Independent Checking Engineer (ICE) pursuant to Clause F2 of *additional conditions of contract* to check that the *Contractor*'s Design is in accordance with the Contract.

(4) The design review procedure shall be in accordance with Clause F2 and F7 of *additional conditions of contract*. The *Contractor* may submit uncertified copies of his *Contractor*'s drawings to the *Project Manager* concurrent with submission to the ICE. However, any such submission shall be for information only and without obligation upon the *Project Manager* to commence his review.

Working drawings for Contractor's Design

1.75 (1) Within 7 days of issuing the *Project Manager*'s approval, the *Project Manager* shall endorse the *Contractor*'s drawings for which the approval has been given as "working drawings". The *Contractor* may commence construction upon receipt of the approval and the endorsed working drawings.

(2) If, after receiving approval of the *Project Manager*, the *Contractor* subsequently amends any *Contractor*'s drawing, he shall again obtain the *Project Manager*'s approval for the amendments in accordance with the process specified above.

As-constructed drawings and design calculations for works designed

1.76 (1) The *Contractor* shall submit to the *Project Manager* as-constructed drawings and design calculations for any works designed by the *Contractor* in accordance with the following. The as-constructed drawings and the design calculations shall be submitted within four

by the Contractor

weeks after the *works* in concern have been completed.

(a) 2 sets of as-constructed drawings in both hard copy and soft copy. Soft copy shall be prepared in Microstation (.dgn) and Acrobat (.pdf) formats converted from the Microstation; requirements shall comply with HyD Technical Circular No. 2/2003.

(b) 2 sets of design calculations in both hard copy and soft copy. The design calculation shall include the Independent Checking Engineer Certificate; requirements shall comply with HyD Technical Circular No. 2/2003.

(c) 1 set of 35mm microfilm of the as-constructed drawings; requirements shall comply with HyD Technical Circular No. 2/2003.

(2) In addition the contractor shall consult CHE/N (the maintenance office for highway structures) on the as-built records that should be submitted upon completion of the *works*.

Restriction on use of public road

1.77 Unless otherwise agreed by the *Project Manager* in writing, the use of all public surfaced roads will be limited to pneumatic tyre and fully licensed vehicles only. Tracked vehicles will not be allowed to run on the roads. The *Contractor* shall be held responsible for any damage to roads caused by illegal movement of his vehicles and shall be required to make good immediately any such damage at his own expense.

1.78 Not Used

1.79 Not Used

Independent Checking Engineer (Clause F2 of additional conditions of contract)

1.80 Pursuant to Clause F2 of *additional conditions of contract*, the *Contractor* shall employ Independent Checking Engineer(s) (ICE) to issue the Check Certificate for the *Contractor*'s Design. The Check Certificate shall use the standard form appended in the latest edition of Structures Design Manual for Highways and Railways or as per other form as agreed with the Chief Highway Engineer/Bridges and Structures. Minimum requirement of the independent checking engineer is a professionally qualified engineer and a member of the Hong Kong Institution of Engineers or equivalent. In addition, the ICE for the *Contractor*'s Design of the Noise Barrier Works should possess at least 10 years bridgeworks experience. The ICE for all other *Contractor*'s Design shall possess at least 10 years relevant experience appropriate to the *Contractor*'s Design to be certified.

Method statement

1.81 (1) Unless otherwise specified, at least 6 weeks prior to the planned commencement date of each construction task detailed on the *Contractor*'s programmes in various parts of the *works* including Temporary Works, the *Contractor* shall submit a detailed written method statement to the *Supervisor* for approval which shall include but not limited to:

(a) Description of main construction task and sub-tasks;

(b) Sequence of sub-tasks;

(c) Quantities of the work and production rates to be achieved;

(d) Resources to be employed;

(e) Quality checks to be carried out and supervision to be exercised; and

(f) Safety precautions to be employed.

(2) Inspection and Testing Plans shall be submitted, together with the method statement, to the *Supervisor* for approval at least 6 weeks prior to commencement of the related works. The *Contractor* shall maintain a list of all proposed Inspection and Testing Plans for every activity requiring inspection. Inspection and Testing Plans shall include and identify at least the following:

(a) the sequence of inspection/testing activities;

(b) the inspection/testing requirements of either activities or materials;

(c) the acceptance criteria or relevant specification;

(d) the level of inspection required including the provision for witnessing by the *Project Manager* or his representative;

(e) any certification requirements or records to be kept; and

(f) records of any non-conformance identified during inspection or testing.

Insurance of works 1.82 (1) The *Contractor* shall provide insurance for the *works* in accordance with the insurance specimen in Annex 2 to Works Information.

(2) The *Contractor* shall extend the insurance coverage for all the proposed works outside the Site as specified but not limited to the implementation of the temporary traffic arrangement and control as described in Section 1 of the Specification.

Control of dogs on construction site 1.83 (1) No dog shall be kept by the *Contractor* or his employees, his agents or subcontractors or their employees, on the Site unless the dog is acceptable for licensing by the Agriculture, Fisheries and Conservation Department (AFCD), and is licensed under the Rabies Ordinance (Cap 421), implanted with a microchip and vaccinated against rabies. In addition, the keeper of the dog under the licences shall either be:

(a) an employee of the *Contractor* who shall be of a rank not lower than deputy site agent or equivalent as agreed by the *Project Manager*; or

(b) a security firm in its own name or an employee of the firm who shall not be of a rank lower than assistant manager level, whether the security firm is engaged by the *Contractor* solely

for the purpose of Site security.

(2) All licensed dogs kept on the Site must be neutered. The *Contractor* shall keep or cause the aforesaid security firm to keep on the Site a copy of the licence, together with a copy of the certificate issued by a registered veterinary surgeon confirming that the dogs kept on the Site have been neutered, for inspection by the *Supervisor* upon request. All licensed dogs on the Site shall be identified by suitable markings on their collars as agreed by the *Supervisor*, and shall be removed by the *Contractor* from the Site upon completion of the *works* under the Contract.

(3) The *Contractor* shall alert the AFCD and facilitate access, where appropriate, to the Site for removal of any unlicensed dogs from the Site.

(4) The *Contractor* shall observe and undertake, or cause his employees, his agents or subcontractors or their employees to observe and undertake, the licensing and control measures as set out in the current edition of the Code of Practice for the Keeping of dogs on Construction Sites in Hong Kong issued by the AFCD for and dogs kept on the Site. The *Project Manager* or his Representative shall have the power to order the removal of any person who fails to comply with the requirements from the Site.

Enhanced Requirements on Checking Qualifications of Contractor's Staff

1.84 (1) For the Contractor's management and supervisory staff (referred to as the "Contractor's staff" hereafter) where requirements on academic and professional qualifications are specified under this contract, the Project Manager of the Contractor shall check and certify that the copies of the qualification certificates provided for appointment or substitution of the proposed staff are true copies of their original qualification certificates. In a similar manner, the copies of the qualification certificates provided for appointment or substitution of the proposed Project Manager shall be checked and certified as true copies by a senior management staff in the Contractor's head office (e.g. a director of the Contractor's organisation). The above certified copies of the qualification certificates shall be included in the particulars to be submitted by the Contractor when the Contractor seeks the Project Manager's acceptance on the appointment or substitution of the Contractor's staff in concern.

(2) When requested by the Project Manager, the Contractor shall make arrangement to allow the Project Manager to inspect and check the original qualification certificates of the Contractor's proposed staff during the appointment or substitution process of the Contractor's staff.

1.85 Not Used

1.86 Not Used

1.87 Not Used

1.88 Not Used

1.89 Not Used

1.90 Not Used

MANDATORY PROVIDENT FUND SCHEME

Mandatory provident fund scheme requirements

1.91 The *Contractor* shall complete and return the Form "Notification of Commencement of Contract" as attached in PS Appendix B10 to Mandatory Provident Fund Scheme Authority at the early stage of Contract Date.

PAYMENT OF WAGES OF THE SITE PERSONNEL

General

1.92 (1) The *Contractor* shall ensure prompt payment of full wages to all the Site Workers employed by him and shall use all reasonable endeavours to have his subcontractors of all tiers to effect prompt payment of wages to all the Site Workers employed by them for the execution of the *works* in accordance with the Employment Ordinance.

(2) The *Contractor* shall include provisions in the contracts with his subcontractors requiring the implementation of the system of payment of wages to the Site Workers under the provisions of Clauses 1.92 to 1.103.

(3) For the purposes of the provisions of Clauses 1.92 to 1.103 of this Particular Specification, "bank" shall be an authorized institution within the meaning of Section 2 of the Banking Ordinance (Cap 155).

EMPLOYMENT OF THE SITE PERSONNEL

Written employment agreements

1.93 (1) Unless written employment contracts are already in place, the *Contractor* and his subcontractors of all tiers shall [have] enter into written employment contracts with all the Site Workers under their respective employments. The terms of the employment contract shall not be less favourable to the terms provided in the specimen employment contract (which may as necessary be revised by the *Project Manager* from time to time) given in Appendix I to the *additional conditions of contract*. The *Contractor* shall not allow any Site Workers without a properly executed written employment contract to work on the Site.

(2) A certified true copy of the written employment contract of each Site Workers shall be made available to the *Project Manager* or the *Project Manager*'s Representative for inspection if so requested by the *Project Manager* or the *Project Manager*'s Representative.

(3) Any Site Workers without a proper written employment contract with either the *Contractor* or his subcontractor shall not be permitted or caused to be permitted to enter to or remain in the Site.

(4) The *Contractor* shall advise and ensure his sub-contractors of all tiers to advise all the Site Workers under their respective employments that a copy of their respective written employment

contracts would be made available to the *Project Manager* for inspection if so requested by the *Project Manager* and that the personal data provided by means of the respective employment contracts will be used for the purposes and may be transferred to the transferees as listed out in the "Notes about Personal Data" to the specimen employment contract given in Appendix I to the *additional conditions of contract.*

Personal accident insurance of self-employed workers

1.94 (1) In accordance with Clause E6 of *additional conditions of contract*, the *Contractor* shall ensure that self-employed workers working on the Site shall each be covered by a personal accident insurance at a minimum coverage of HK$1,000,000. The *Contractor* shall not allow any self-employed worker without a valid personal accident insurance to work on the Site.

(2) The *Contractor* shall keep a certified true copy of the personal accident insurance of each self-employed worker on Site and shall be made available to the *Project Manager* or the *Project Manager*'s Representative for inspection if so requested by the *Project Manager* or the *Project Manager*'s Representative.

(3) Any self-employed workers without a proper personal accident insurance shall not be permitted or caused to be permitted to enter to or remain in the Site.

Lorry drivers

1.95 (1) Lorry drivers shall be either a Site Workers of the *Contractor* or his subcontractors, or a self-employed person with a personal accident insurance cover of not less than HK$1,000,000 for the execution of the *works*.

Payment of wages by designated bank accounts or personal cheques

1.96 (1) Subject to sub-clause (3) of this Clause, the *Contractor* shall within 14 days from the Contract Date submit to the *Project Manager* the name of the designated bank and all related arrangement details for payment of wages to all the Site Workers.

(2) The *Contractor* shall require all the Site Workers to open a wage payment account in the designated bank or in any other bank of their choice.

(3) Subject to evidence produced to the satisfaction of the *Project Manager*, Site Workers not able to open a personal bank account with any bank in Hong Kong shall have their wages paid by personal cash cheques with the subject Site Workers specified as payee and the cheques shall be immediately due on presentation. Payment of mandatory provident fund ("MPF") contributions for the subject Site Workers shall be made by bank transfers in accordance with Clauses 1.98 and 1.99. Copies of the following documents shall be submitted to the *Project Manager* for records:

(a) signed acknowledgement receipt of wage payment from the subject Site Workers;

(b) the cheque; and

(c) the certified true copy of the bank statement showing the cash withdrawal.

The above copies of the documents shall be submitted as soon as they are available and in any event items (i) and (ii) shall be submitted no later than 7 days from the date when payment is due for the subject Site Workers.

ATTENDANCE RECORDS

Attendance records

1.97 (1) Construction Industry Council ("CIC") has developed a computerized system (known as Construction Workers Registration System ("CWRS")) to facilitate contractors to establish and maintain daily site attendance records of registered construction workers by retrieving the data stored in workers' registration cards issued by the Registrar of Construction Workers ("Registrar") appointed under Construction Workers Registration Ordinance ("CWRO") (Cap. 583) and to furnish the records to the Registrar in accordance with section 58(7) of CWRO. The daily site attendance records of the Site Workers in the CWRS are to be adopted for the purposes of wage payment monitoring for his Site Workers. Prior to the operation or in case of breakdown of the attendance record system with the CWRS or subject to the approval of the *Project Manager*/*Supervisor*, the *Contractor* shall use log books to record the time of entry and departure of all Site Workers and self-employed workers in and out of the Site.

(2) The *Contractor* shall ensure that all Site Workers use the registration cards issued by the Registrar for entering and leaving the Site.

(3) The *Contractor* shall obtain both raw and verified attendance records from the CIC server and submit the raw and verified attendance records to the *Project Manager*/*Supervisor* for the purposes of wage payment monitoring.

(4) Any attendance record entry which is incomplete shall be clearly annotated with reasons as verified record to facilitate the easy identification by the *Project Manager*/*Supervisor* of any such incomplete record.

(5) The *Contractor* shall install the card reading devices for use with the CWRS at convenient locations with shelters within the Site to facilitate day-to-day recording. The exact locations shall be submitted for agreement by the *Project Manager*/*Supervisor*. The *Contractor* shall provide 2 number of portable card reading devices compatible with the registration card issued to registered construction workers by the Registrar for the use of the *Project Manager*/*Supervisor*, the *Supervisor* and his staff. The *Contractor*'s portable card reading devices shall be approved by the *Project Manager*/*Supervisor* and the equipment shall be securely kept by a designated staff. The *Contractor* shall provide a full time staff to man the operation of the card reading devices at each of the locations.

PAYMENT OF WAGES

Schedule of wages 1.98 (1) The *Contractor* shall compile daily records of the Site Workers working on the Site based on the data obtained from the attendance record system of the CWRS or log books kept by the *Contractor*. The use of log books shall only be used in case of breakdown of the attendance record system of the CWRS or subject to the approval of the *Project Manager*/*Supervisor*. The *Contractor* shall verify the data of all the Site Workers employed or engaged by him. The *Contractor* shall prepare separate daily records for each sub-contractor and distribute to these subcontractors the records of the Site Workers employed or engaged by them for verification. A copy of all the verified data and daily records together with the raw attendance record obtained from the CWRS shall be submitted to the *Project Manager*/*Supervisor*. A copy of the verified monthly attendance record obtained from the CWRS shall be submitted to the *Project Manager*/*Supervisor* within 7 days after each payment cycle.

(2) The *Contractor* shall prepare a schedule of wages of all the Site Workers employed or engaged by him based on the verified data from the attendance record system of the CWRS or log books kept by the *Contractor*. A copy of such schedule shall be submitted to the *Project Manager* for each payment cycle.

(3) The *Contractor* shall require his subcontractors, of all tiers, to prepare the respective schedules of wages of the Site Workers employed or engaged by them and the corresponding MPF contributions based on the verified data from the attendance record system of the CWRS or log books kept by the *Contractor*. A copy of these schedules shall be made available to the *Project Manager* for each payment cycle.

Payment of wages 1.99 (1) In accordance with the scheduled payment date stated in the employment contract, the *Contractor* shall ensure that sufficient funds have been reserved in the designated bank account for the payment of all wages as given on the verified schedules for the subject payment cycle and shall promptly instruct the designated bank to effect the payment to the respective wage payment accounts of all his Site Workers. The *Contractor* shall forward a certified true copy of the instruction records to the *Project Manager*.

(2) The *Contractor* shall ensure that his sub-contractors, of all tiers, have sufficient funds available in the respective designated bank accounts for the payment of the wages to their respective Site Workers as given on the verified schedule on the scheduled payment date stated in the employment contracts for the subject payment cycle and to ensure that his sub-contractors shall promptly instruct the designated bank to effect the payment to the wage payment accounts of all their Site Workers. The *Contractor* shall ensure that certified true copies of the instruction records are made available to the *Supervisor*.

(3) A flow chart illustrating the payment arrangements is in PS Appendix K.

MONITORING OF PAYMENT OF WAGES

Contractor's 1.100 (1) Within 7 days of the Contract Date, the *Contractor* shall nominate a

Labour Officer

staff responsible full time on-site for monitoring the payment of wages to all the Site Workers for the approval of the *Project Manager*. The nominated staff when approved shall provide all necessary assistance to the Assistant Clerical Officer (Labour Relations) in the monitoring process and in the handling of complaints on arrears of wages raised by any Site Workers. The staff nominated by the *Contractor* shall be referred to as the *Contractor*'s Labour Officer.

(2) The *Contractor*'s Labour Officer shall be a suitably qualified person with the following minimum qualifications: -

(a) attaining five subjects at grade 'E' or above in the Hong Kong Certificate of Education Examination (HKCEE) (including Chinese, English (Syllabus B) and Mathematics) or equivalent;

(b) full-time working experience on personnel management or human resources related duties and good job records;

(c) good command of both oral and written English and Chinese;

(d) good computer knowledge preferably with past experience on the operation of a computerized smart-card system; and

(e) preferably have completed a post secondary certificate programme on human resources or personnel management.

(3) The *Project Manager* shall have the power to withdraw his approval of the *Contractor*'s Labour Officer at any time. If such approval shall be withdrawn the *Contractor* shall, after receiving notice in writing of such withdrawal, remove the *Contractor*'s Labour Officer from the Site forthwith and shall replace him by another *Contractor*'s Labour Officer approved by the *Project Manager*.

Assistant Clerical Officer (Labour Relations)

1.101 (1) The *Contractor* shall afford all necessary assistance to the Assistant Clerical Officer (Labour Relations) ("ACO(LR)") in connection with the discharge of his duties which shall include but not necessarily limited to the following:

(a) to act as a one-point contact for the Site Workers on Site on any enquiries in relation to employment matters;

(b) to conduct briefing sessions to inform and to educate the Site Workers the benefits of the wage payment control measures being implemented under the Contract, in particular, the workers' obligations to report wage arrears;

(c) to conduct regular site visits to promote the monitoring system and to establish contacts with Site Workers to obtain feedback;

(d) to monitor payment of wages to assess whether they are made timely by the *Contractor* and all subcontractors;

(e) to oversee the setting up and maintenance of a record system on employment contracts, workers attendance, re-deployment, and wage payments;

(f) to undertake regular physical checks to verify the accuracy and reliability of the records and to identify irregularities, if any, for early intervention;

(g) to establish a simple and user-friendly complaint system, including the operation of a telephone hotline to receive enquiries from Site Workers on employment matters and to receive reports on wage defaults;

(h) to alert the *Project Manager*/*Supervisor* of anomalies and to refer the same to the *Contractor* for investigation and appropriate follow-up actions; and

(i) to report to the *Project Manager*/*Supervisor* and the Labour Department complaints on wage arrears as soon as they are received and to provide necessary assistance to Labour Department to facilitate investigation and/or dispute resolution where appropriate.

(j) to carry out random attendance checks, and to record and report the findings to the *Project Manager*/*Supervisor*.

(k) To attend interviews with the local labour unions and contractors associations, who visit the Site from time to time to promote the wage payment and monitoring process, to ascertain whether there are any difficulties with the monitoring process and offer assistance where necessary.

(2) The ACO(LR) shall have the authority to check the proper operation of the card reading devices of the CWRS, the records of the system and the records of instructions of payment of the wages. The *Contractor* shall provide copies of daily records and instruction records to and assist the ACO(LR) in the preparation of the monthly report on payment of the wages and the attendance records to the *Project Manager*/*Supervisor* based on the daily records and instruction records submitted by the Contractor.

(3) The *Contractor* shall display necessary information on the contact details of the ACO(LR) by erecting notices on the Site at prominent locations.

(4) The Contractor shall establish and maintain a separate telephone line in the Site accommodation for the *Project Manager*/*Supervisor* and the Supervisor to be used as a hotline for Site Workers to report matters of wage disputes to the ACO(LR).

Providing access and assistance to visitors

1.102 (1) Notwithstanding any other provisions under the Contract, from time to time representatives from established local labour unions and contractors associations may visit the Site to make propaganda for the wage payment arrangement and monitoring process for wage payment under this Contract to the Site Workers. These representatives may interview the *Contractor*'s Labour Officer and the ACO(LR) to ascertain whether there are any difficulties with the monitoring process and offer their assistance where necessary.

(2) Upon notification from the *Project Manager*, the *Contractor* shall provide access and all necessary assistance to these representatives visiting the Site for the said purposes.

(3) The *Contractor* shall ensure that his *Contractor*'s all risk and third party liability insurance policies are extended to cover these visitors covering the conditions shown the insurance specimen in Annex 2 to Works Information.

CASUAL WORKERS

Casual workers 1.103

(1) "Casual Workers" are those Site Workers who are expected to work on Site no more than an aggregated total of 7 working days throughout the duration of the Contract period. The provisions under Clauses 1.93, 1.96, 1.98 and 1.99 shall not apply to Casual Workers. Casual Workers are required to be issued a temporary smart-card and follow the attendance recording system in accordance with Clause 1.97. Casual Workers will be subject to the full provisions of Clauses 1.92 to 1.99 if they are required to work on site in excess of an aggregated total of 7 working days throughout the duration of the Contract period ("7-day restriction").

(2) Wages for the Casual Workers are to be paid daily by personal cash cheques with the subject Casual Worker specified as payee and the cheques shall be immediately due on presentation. The subject Casual Worker shall be required to sign a receipt of acknowledgement of each payment.

(3) Copies of the following documents shall be submitted to the *Project Manager* for records:

(a) signed acknowledgement receipt of wage payment from the subject Casual Worker;

(b) the cheque; and

(c) the certified true copy of the bank statement showing the cash withdrawal.

Copies of (a) and (b) above shall be submitted the day immediately following the subject payday referred to in sub-clause (2) of this Clause. Copies of (c) above shall be submitted as soon as they are available.

(4) The 7-day restriction under sub-clause (1) of this Clause may be relaxed subject to prior approval in writing from the *Project Manager* if the *Project Manager* is satisfied that there are reasonable grounds for not complying with the full requirements under Clauses 1.92 to 1.99 for the subject Casual Worker.

ENVIRONMENTAL MANAGEMENT

General 1.104

(1) The *Contractor* shall prepare an Environmental Management Plan (EMP) in accordance with this PS for implementation on the Site to reduce environmental nuisances and construction and demolition

(C&D) materials arising from Works, throughout the construction period. Irrespective of the measures proposed by the *Contractor* in the EMP, the *Contractor* shall not be absolved from his liability to satisfy in full all statutory requirements and requirements in this PS on Environmental Management.

(2) Organisation

(a) The *Contractor* shall assign a person as the Environmental Officer for overseeing all the environmental matters of the *works*. The minimum qualification of this person shall be -

(i) a corporate member of the Hong Kong Institution of Engineers (HKIE) in Environmental Discipline or equivalent; or

(ii) a corporate member of a professional institution (e.g. HKIE other than in the Environmental Discipline, or the Chartered Institution of Water and Environmental Management, or the Hong Kong Institute of Architects; or the Hong Kong Institute of Surveyors etc.) EITHER having at least 3 years of experience in environmental management, monitoring and auditing of construction activities, OR in possession of a diploma or above in environmental studies or environmental engineering awarded by a local or overseas tertiary educational institution; or

(iii) a senior staff of the *Contractor* having at least 5 years of construction site experience and completed the Environmental Officer Course run by the Construction Industry Council (CICTA), or an equivalent certificate course on environmental studies or environmental engineering run by a tertiary educational institution as agreed by the *Project Manager*. For the purpose of this Contract, a person who is attending or has applied to attend the Environmental Officer Course mentioned above shall fulfil the requirement of this sub-clause; or

(iv) a senior staff of the *Contractor*,

i. either possessing an engineering degree and having at least 10 years of construction site experience, or possessing a higher certificate in a construction related subject and having at least 15 years; and

ii. who has completed the "Environmental Management Course for Construction Managers" or equivalent organized by CICTA or similar training institutions as agreed by the *Project Manager*.

(v) The Environmental Officer shall be present full time on the Site but not necessarily be working solely on environmental duties. Such person can be a Safety Officer if the Contract has employed more than one Safety Officer. Details and curriculum

vitae of the person to be nominated as the Environmental Officer for the Contract shall be submitted to the *Project Manager* for approval.

(b) The duties of the Environmental Officer shall, for the purpose of environmental management under the Contract, include but without limitation to the followings:

(i) Prepare, implement and update the Environmental Management Plan;

(ii) Advise on measures to be taken in the interest of environmental protection, and implement such measures;

(iii) Liaise on all matters relating to environmental monitoring and auditing;

(iv) Carry out inspections of the Site for identifying potential hazards to the environment, and to report findings with recommendations for corrective actions;

(v) Participate in the weekly environmental walks (whether this is combined with the weekly safety walk or otherwise) with the nominated site staff of the *Supervisor*, and to supervise and monitor the environmental performance on the Site;

(vi) Check and ensure that any polluting or potentially polluting situation is promptly rectified;

(vii) Attend Site Safety and Environmental Management Committee (SSEMC) meetings and Site Safety and Environmental Committee (SSEC) meetings;

(viii) Compile the monthly environmental report for submission to the *Supervisor* at least five working days before the SSEMC meetings;

(ix) Arrange and provide the environmental training including the site specific induction training and toolbox talks for the staff and workers on the Site, and to organize environmental promotional activities; and

(x) Advise the *Contractor* on the implementation of an environmental management system.

(3) Performance Monitoring

(a) The *Contractor* shall arrange weekly environmental walk to be attended by the Environmental Officer, the *Contractor*'s Agent, and the *Supervisor* or his delegate to inspect the Site, checking that the environmental performance of the Site is satisfactory and in compliance with the requirements under the Contract and EMP. The places to be inspected in the weekly environmental walk shall be determined by the *Supervisor* or his delegate.

(b) The *Contractor* may arrange the weekly environmental walk to be carried out along with the weekly safety walk or other site inspections subject to the agreement of the *Supervisor*. The weekly environmental walks conducted under this Clause are entirely without prejudice to and do not relieve any of the *Contractor*'s responsibility to carry out regular inspections to upkeep the environmental performance of the Site as required by the statute or other Clauses under this Contract.

(c) The *Contractor* shall prepare and agree with the *Supervisor* a comprehensive checklist for use in weekly environmental walk. The checklist will form the basis for assessing the environmental performance of the *Contractor* on the Site. Any defects or deficiencies identified in the weekly environmental walk shall be duly recorded in a summary table, a proforma of which is attached at PS Appendix L1. More than one table shall be used for recording the defects or deficiencies if the weekly environmental walk for the week is carried out by more than one inspection team.

(d) Immediately after the weekly environmental walk, the summary table shall be agreed and signed by the *Contractor*'s representative and the *Supervisor* or his delegate attending the weekly environmental walk, and a copy should be kept by the *Supervisor*. The *Contractor* shall take prompt action to rectify the deficiencies identified and shall report the status of rectification actions in the forthcoming weekly environmental walk or the Site Safety and Environmental Management Committee meeting whichever comes first.

(e) The following items should be included in the agenda for discussion at every SSEMC meeting and SSEC meeting, or other established channels for performance monitoring as agreed by the *Supervisor*:

(i) Review the sufficiency of the measures in the EMP and proposals for improvement;

(ii) Monitor the *Contractor*'s environmental performance and achievement with reference to EMP;

(iii) Assess the effectiveness of EMP taking into account the *Contractor*'s environmental performance and achievement; and

(iv) Monitor the follow-up action by the *Contractor* on the defects and deficiencies identified in weekly inspections.

(4) Environmental Training

(a) The *Contractor* shall ensure that all site management staff in his employment on the *works* shall have attended and completed the "Environmental Management Course for Construction Managers" run by CICTA or similar training institutions as agreed by the *Supervisor*. For the purpose of this

sub-clause, site management staff include the *Contractor*'s agent, project managers etc. but excluding the Environmental Officer. If anyone of the site management has not attended the course, the *Contractor* shall arrange such staff to attend the required environmental course within 14 days from the date of employment of such staff on the Site, and to complete the training within three months from the said date.

(b) Not used.

(c) Pursuant to the relevant PS Clause for Site Safety, the *Contractor* shall ensure that the site specific induction training cover environmental management in addition to safety for all staff and workers employed for the *works* or in connection with the Contract, whether in the employment of the *Contractor* or his subcontractors. The environmental part of the induction training shall be delivered by the Environmental Officer or Environmental Supervisor or the Assigned Person, as applicable. The training content should cover subjects such as organization structure, duties and responsibilities, measures, targets, in-house rules and regulations etc. The duration of the site specific induction training under the relevant PS Clause for Site Safety shall be extended by at least 15 minutes to cover the necessary subjects on environmental management.

(d) Pursuant to the relevant PS Clause for Site Safety, the *Contractor* shall provide toolbox talks for workers on environmental nuisance abatement and waste management in addition to safety and health. The frequency of training and the contents of the tool box training shall be subject to the approval of the *Supervisor* as in the relevant PS Clause for Site Safety.

(5) Reporting

The *Contractor* shall submit a monthly report on environmental management for discussion in the SSEMC meeting. The monthly report shall be prepared by the Environmental Officer as mentioned in clause 1.104(2)(a) and duly endorsed by the *Contractor*'s agent containing the following information:

(a) A summary of pollution incidents and remedies, comprising:

(i) Complaints

(ii) abatement notices issued by EPD

(iii) offences spotted by EPD during inspections

(iv) summonses of environmental offences

(b) A list of major forthcoming activities in the next two months which will likely have environmental impacts or cause nuisances to the surroundings, together with the proposed control or mitigation measures;

(c) The training programme for the next month and records of

training arranged/conducted in the previous month pursuant to Clause 1.104(4) above;

(d) The updated organization chart on environmental management;

(e) A summary of defects and deficiencies identified during inspections and weekly environmental walks, together with the follow-up actions and remedies taken to prevent similar recurrences.

Air pollution abatement

1.105 (1) The *Contractor* shall ensure that dusty materials, including building debris and construction materials which are dusty by their nature, are properly covered by tarpaulin or other approved means. When dusty construction activities such as demolition work, drilling work or artificial hard materials etc. are carried out in close proximity to the public, the *Contractor* shall provide dust abatement measures to the satisfaction of the *Supervisor*. Such measures shall include where appropriate screens or enclosures, water spraying system or the fitting of vacuum cleaning devices to pneumatic or power driven drilling, cutting and polishing machines, etc.

(2) Where the public will be affected by exhaust fumes or smoke emission from any Constructional Plant or construction activities (e.g. welding) in the Site, such Constructional Plant or construction activities shall be shielded by a screen. Such screen shall be at least 1.8m in height, incombustible and shall be approved by the *Supervisor*.

(3) Not used

(4) All dump trucks entering or leaving the Site shall be provided with mechanical covers in good service condition in accordance with the specification given in Part B of PS Appendix L2.

(5) For Constructional Plant driven by internal combustion engines, the *Contractor* shall ensure that the smoke emission from the plant shall not exceed Shade 1 on the Ringelmann Chart continuously for 30 seconds at any time.

(6) Ultra-low-sulphur diesel (ULSD) (defined as diesel fuel containing not more than 0.005% by weight of sulphur) shall be used in all diesel-operated Constructional Plant on the Site pursuant to Clause D19 of *additional conditions of contract*. The *Contractor* shall demonstrate his compliance by maintaining a summary record of ULSD pursuant to Clause D19 of *additional conditions of contract*. A proforma of the summary record is attached at PS Appendix L3.

(7) Where there is practical difficulty in implementing any air pollution abatement measures specified above, the *Contractor* shall submit alternative proposals for the approval of the *Supervisor* before the work commences. The *Project Manager* or the *Supervisor* shall have the power to order the removal of any person who, or Constructional Plant or equipment which, fails to comply with the requirements under this PS Clause off Site.

Noise pollution abatement 1.106

(1) The *Contractor* shall adopt the following noise abatement practices:

(a) Use non-percussive equipment such as hydraulic crusher, sawing, coring machines etc. for demolition and concrete breaking work;

(b) Close all hoods, cover panels and inspection hatches of powered mechanical plant such as generators, air compressors etc. during operation; and

(c) Fit mufflers or silencers, and dampening layer with steel collars to hand held pneumatic breakers.

(2) Where percussive breakers are used, the *Contractor* shall enclose/wrap the breaker tip with sound insulating material to reduce the noise. This requirement is not applicable to works under emergency, or with prior agreement from the *Supervisor* that the provision of such is not necessary or not practicable under a given site condition.

(3) Where the noise level measured at the noise sensitive receivers (NSR) exceeds 75 dB(A) (L_{eq} 30 min) for domestic premises, and 70 dB(A) (L_{eq} 30 min) for all educational institutes or 65 dB(A) during the school examinations (including public examination) periods, the *Contractor* shall provide an acoustic screen or enclosure to shield the public or NSR from the noisy activity at source or adopt quiet process/plant including the use of 'quality powered mechanical equipment' pursuant to sub-clause (4) below, except for works under emergency or with the prior agreement of the *Supervisor* that the provision of such is not necessary. The acoustic screen or enclosure shall be made of incombustible, sound insulating material with performance such that the noise level measured at the NSR do not exceed the limits as mentioned in this sub-clause. The detailed information including the size and form of the screen or enclosure shall be proposed by the *Contractor* and submitted to the *Supervisor* for approval before the work commences. The acoustic screen or enclosure shall be securely fixed at the base to avoid overturning. Notwithstanding any approval given, the *Contractor* shall be fully liable for his design in all respects.

(4) The Environmental Protection Department (EPD) has published a category of Quality Powered Mechanical Equipment (QPME) in the website: *http://www.epd.gov.hk/cgi-bin/npg/qpme/list.pl?lang=eng*.

(5) The categories of QPME include, but are not limited to:

(a) Crane, mobile

(b) Excavator, wheel/tracked

(c) Generator

A list of plant models under the QPME is also given by the EPD at the website: *http://www.epd.gov.hk/cgi-bin/npg/qpme/search_gen.pl?lang=eng&st=sim&smtype=1*.

(6) Where a QPME is used, the plant should be registered with EPD, and the label issued by EPD from such registration shall be affixed on the plant at all times and kept legible. The *Contractor* shall also establish a register to record all QPME used on the Site.

(7) The *Project Manager* or the *Supervisor* shall have the power to inspect the QPME if he has doubt on its compliance with the QPME requirements. The Constructional Plant shall be non-QPME for the purpose of this sub-clause if it does not have the registration label issued by EPD so affixed.

Wastewater pollution abatement

1.107 (1) The *Contractor* shall minimize the generation of wastewater from the Site through the following means:

(a) Prevent surface run-off from washing across the Site and spilling over to areas outside of the Site;

(b) Minimize water consumption;

Waste management

1.108 (1) All C&D materials arising from or in conjunction with the *works* shall be sorted on the Site to recover reusable and/or recyclable materials. For the avoidance of doubt, C&D materials mean both inert and non-inert materials generated from construction and demolition activities. The inert portion of the C&D materials includes soil, building debris, broken rock, concrete, etc., and the non-inert portion comprises timber, paper, plastics, general refuse, etc.

(2) Unless otherwise stated, all surplus C&D materials arising from or in connection with the *works* shall become the property of the *Contractor* when it is removed from the Site. The *Contractor* shall promptly remove all sorted and processed surplus materials arising from or in connection with the *works* from the Site to minimise temporary stockpiling on the Site

(3) On-site Sorting of C&D Materials

(a) The *Contractor* shall devise a system for on-site sorting of C&D materials. The system shall include the identification of the source of generation, estimated quantity, arrangement for on-site sorting and/or collection, temporary storage areas, frequency of collection by recycling contractors or frequency of removal off the Site, etc.

(b) The *Contractor* shall sort the materials at source into:

(i) metals;

(ii) paper and plastics;

(iii) chemical waste; and

(iv) materials suitable for disposal at public fill reception facilities, sorting facilities and landfills/outlying islands transfer facilities. Disposal at the sorting facilities should

first be approved by the *Supervisor*.

(c) Not Used.

(d) Equipment and material packaging (i.e. paper and cardboard) shall be recovered, properly stockpiled in dry and covered condition to prevent cross contamination by other C&D materials. The *Contractor* shall pay particular attention to avoid cross contamination in the course of collecting paper for recycling.

(e) The *Contractor* shall ensure the materials disposed of at public fill reception facilities, sorting facilities, and landfills/outlying islands transfer facilities, comply with their respective requirements under Schedule 6 of the Waste Disposal (Charges for Disposal of Construction Waste) Regulation (Cap. 354) and be fully liable for all non-compliance.

(f) The *Contractor* shall identify and provide sufficient space for temporary storage of C&D materials to facilitate collection and/or sorting on the Site. The space provided shall commensurate with the estimated quantity for each type of C&D materials generated on the Site.

(g) Except for those inert C&D materials to be reused on the Site, the *Contractor* shall remove all other C&D materials off the Site as soon as practicable in order to optimize the use of the on-site storage space.

(h) The *Contractor* shall make arrangements with potential recycling contractors to facilitate that recyclable materials sorted from the Site are collected with reasonable care.

(i) The *Contractor* shall establish a system for proper handling and storage of chemical waste generated from the Site, and arrange collection and disposal of such chemical waste by specialist contractors.

(j) The *Contractor* shall carry out thorough sorting of C&D materials generated from demolition works for recovering of broken concrete, reinforcement bars, mechanical and electrical fittings, hardware as well as other fittings/materials that have established recycling outlets.

(4) Waste Flow Table (WFT)

(a) The *Contractor* shall establish a mechanism to record the quantities of C&D materials generated each month, using the monthly summary “Waste Flow Table” (WFT) as given at PS Appendix L4 attached. The *Contractor* shall complete the monthly summary WFT, and submit it to the *Supervisor* together with the updated sections of EMP (if any) by not later than the 15th day of each month following the month reported on, or if it is a General Holiday, the day following the General Holiday.

(5) Not Used

(6) Recording

(a) The *Contractor* shall record the quantities of all the recyclable materials before removal off the Site by the recycling contractors, and include the details in the WFT for submission to the *Supervisor* pursuant to Clause 1.108(4)(a).

(b) The *Contractor* shall establish a comprehensive register of the Disposal Delivery Form pursuant to Clause 25.25 for recording the disposal of C&D materials. The register shall also cover the recyclable materials removed by the recycling contractors off the Site.

1.109 Not Used

1.110 Not Used

Environmental Management Plan

1.111 The *Contractor* shall submit the EMP for the Contract to the *Supervisor* in accordance with the Clause D20 of additional conditions of contract. The EMP shall be signed before submission by both the *Contractor*'s agent and the Environmental Officer. The EMP shall include the following details categorised into four parts namely, general, nuisance abatement, waste management and records:

PART A - GENERAL

(1) Environmental Management Policy: A policy statement setting out the management's approach, commitments, targets and measures for environmental and waste management of the Contract; it shall –

(a) be signed by the managing director of the *Contractor*, or where the *Contractor* is a joint venture (where incorporated or unincorporated) the managing director of each participant of the joint venture;

(b) state the *Contractor*'s policy and strategy in promoting and implementing the *Contractor*'s environmental management including waste management on the Site;

(c) commit to provide sufficient resources and facilities for the implementation of environmental nuisance abatement and waste management;

(d) commit to obtain and renew the necessary environmental licences, registrations and permits, and comply with the relevant statutory requirements and licensing standards.

(e) commit to prevent pollution;

(f) state the targets (if any) to be achieved in reducing or minimizing environmental nuisances and generation of C&D materials;

(g) be communicated to all levels of persons involved in the *works*;

and

(h) be dated and reviewed periodically for continuous improvement.

(2) Organizational Structure for Environmental Management: The organizational structure shall be illustrative graphically, showing the manpower resources provided to implement the EMP; it shall include:

(a) The Environmental Officer appointed pursuant to Clause 1.104(2)(a) above;

(b) Not Used

(c) The line of communication and authorities on environmental management matters;

(d) The name of the person (if different from the Environmental Officer in sub-clause (a) above) to be responsible for updating the EMP, the monthly summary WFT under Clause 1.108(4); and

(e) The names and telephone numbers of persons (including subcontractors and workers' representatives if appropriate) together with their trade for each specific area of the *works* which they are responsible for the supervision in implementation of the EMP on the Site.

(3) Duties and Responsibilities: The duties and responsibilities of individuals and/or different levels of staff (including subcontractors) involved in the implementation of the EMP; it shall -

(a) define clearly who shall be responsible for the provision of resources and facilities for the implementation of the EMP;

(b) define clearly the responsibilities among individuals and different levels of staff on environmental management; and

(c) state the powers vested in the Environmental Officer to enable him to discharge his/her duties in the proper implementation of the EMP on the Site.

(4) Environmental Training

(a) The requirements and arrangements for training on environmental and waste management for all staff covering:

(i) site managerial staff;

(ii) site supervisory staff; and

(iii) workers.

(b) The training materials on environmental and waste management to be included in the site-specific induction

training pursuant to Clause 1.104(4)(c).

(c) The topics of toolbox training to be provided to workers on environmental nuisance abatement and waste management.

(5) In-house Rules and Regulations

(a) The in-house rules and regulations on environmental nuisance abatement and waste management including those specific rules and regulations laid down by the existing occupiers of the premises (in case of alteration works) for the *works* to be carried out in areas that are occupied, partially occupied and/or controlled by the occupiers.

(b) The arrangements made by the *Contractor* to ensure compliance with the in-house rules and regulations on environmental nuisance abatement and waste management are to be documented, reviewed, amended and communicated to all levels of staff working on the Site.

(c) The means and disciplinary action to ensure implementation and enforcement of the in-house rules and regulations on environmental nuisance abatement and waste management.

(6) Committees: The arrangement for follow-up actions on environmental issues in the SSEMC meetings and SSEC meetings.

(7) Performance Monitoring: The arrangement to establish procedures for monitoring the environmental performance on the Site including identification, recording and reporting of non-compliance with the EMP and their rectification; they shall include:

(a) Planning and review of the frequency, coverage and extent of inspections conducted by different levels of site management and supervisory staff, and senior management from the headquarters;

(b) Development of comprehensive checklists for use in the inspections, and a system for reporting of non-compliance identified and monitoring of the corrective actions taken by the appropriate staff; and

(c) Compliance with all requirements in the Contract including licenses or permits.

(8) Promotion: The methods of promoting and maintaining the awareness on environmental nuisance abatement and waste management amongst all persons on the Site; they shall include:

(a) Display of the company's environmental policy, non-compliance statistics, posters and signs at prominent locations;

(b) Talks and campaigns, and distribution of bulletins or newsletters drawing attention to particular environmental issues; and

(c) Procedure for recognition and commending those site personnel, teams or subcontractors with good performance on environmental nuisance abatement and waste management.

(9) Review of Requirements

(a) Arrangement for identification of potential environmental nuisance and review of the effectiveness of environmental nuisance abatement and waste management measures;

(b) Location plan and inventory of nearby sensitive receivers or features (including residents, watercourse and natural environment) likely to be affected by the construction activities;

(c) Location plan and inventory of nearby major environmental protection fixtures/pollutant discharge points and environmental monitoring stations; and

(d) Emergency response plan of environmental incidents.

PART B – ENVIRONMENTAL NUISANCE ABATEMENT

(10) Air pollution abatement: Arrangement and details of the air pollution abatement measures; it shall include:

(a) List of activities on the Site that will cause air pollution;

(b) Specification and technical details of air pollution abatement measures pursuant to sub-clause (a) above for covering the activities as specified in Clause 1.105 where appropriate;

(c) Construction details of the dust/smoke screen to be provided for the *works* where the public will be affected, pursuant to Clause 1.105;

(d) System for monitoring the effectiveness and efficiency of the air pollution abatement measures implemented on the Site; and

(e) System for recording and controlling the use of ULSD for all diesel-operated Constructional Plant on the Site.

(11) Noise pollution abatement: Arrangement and details of the noise pollution abatement measures; it shall include:

(a) List of activities on the Site that will cause noise pollution;

(b) Specification and technical details of the noise pollution abatement measures for each activity pursuant to sub-clause (a) above;

(c) Construction details of the acoustic screens or enclosures to be provided for the *works* where the public will be affected, pursuant to Clause 1.106 where appropriate;

(d) System for monitoring the effectiveness and efficiency of the

noise pollution abatement measures implemented on the Site; and

(e) Inventory of noise emission labels for QPME (if any) used on the Site.

(12) Wastewater pollution abatement: Arrangements and details of the wastewater pollution abatement measures; they shall include:

(a) List of activities on the Site that will cause wastewater pollution;

(b) Means to prevent surface run-off from washing across the Site and egress of surface run-off from the Site where applicable; and

(c) System for monitoring the effectiveness and efficiency of the wastewater pollution abatement measures implemented on the Site.

PART C – WASTE MANAGEMENT

(13) Waste Reduction Measures: Arrangement and details of the waste reduction measures; it shall include:

(a) List of work processes or activities that will generate C&D materials during the *works*;

(b) Measures to reduce/minimize the generation of C&D materials for the Contract through proper planning of works and good site management such as minimizing over-ordering, avoiding cross contamination to reusable and/or recyclable materials collected, and forward planning for removal, storage and collection of disposal materials;

(c) Not Used

(d) Pre-identification of reusable/recyclable materials, such as hard rock, broken concrete, metallic waste, timber, paper/cardboard packaging etc., to facilitate on-site sorting of the materials.

(14) Waste Targets

(a) The targets set for the Contract, which should cover the followings:

(i) All metallic waste to be recovered for collection by recycling contractors;

(ii) All cardboard and paper packaging (for plant, equipment and materials) to be recovered, properly stockpiled in dry and covered condition to prevent cross contamination;

(iii) All chemical waste to be collected and properly disposed of by specialist contractors; and

(iv) All demolition debris to be sorted to recover broken concrete, reinforcement bars, mechanical and electrical fittings, hardware as well as other fittings/materials that have established recycling outlets.

(15) On-site sorting of C&D Materials

(a) The system devised for on-site sorting for each type of C&D materials arising from or in connection with the *works*, and the resources and facilities to be provided for effective implementation of the system on the Site pursuant to Clause 1.108(3);

(b) The layout plan showing the temporary storage of C&D materials on the Site; and

(c) The arrangement for collection of recyclable materials by recycling contractors where appropriate.

(16) Waste Flow

(a) The arrangement for verifying the quantities of C&D materials reused, recycled and/or removed from the Site for the preparation of the monthly summary WFT at PS Appendix L4, pursuant to Clause 1.108(4).

PART D – RECORDS

(17) Sample Proforma for Recording: Sample proforma shall be provided for the following:

(a) Inventory of licences, registration particulars and permits;

(b) Inspection reports for weekly environmental walks;

(c) Inventory of complaints, inspections by enforcement authorities, abatement notices, environmental offences and convictions;

(d) Summary record of non-compliance of air, noise and wastewater pollution incidents;

(e) Summary record of delivery notes for ordering of ULSD from oil companies and the replenishment of such fuel to individual diesel-operated construction plant and equipment on the Site;

(f) Monthly Summary of Waste Flow Table;

(g) Summary record of chemical waste disposal;

(h) Summary record of trip ticket system;

(i) Records of environmental training;

(j) Not Used.

(k) Summary record of quality powered mechanical equipment (QPME) used on the Site including their servicing period;

1.112 Not Used

CONSULTATION

Consultation

1.113 The *Contractor* shall assist the *Project Manager* handling all consultation with public. This includes but not limited to attending all consultation meetings, such as District Offices and District Councils; and meetings with individual District Councillors, representatives from the relevant Estate Owners' Committees, Area Committees, local organisations, or alike as requested by the *Project Manager*, presenting the *works* and the *Contractor*'s proposals, and handling all comments, complaints, and enquiries. The *Contractor* shall be responsible for the presentation materials, which shall be submitted for agreement with the *Project Manager* at least 2 weeks before such meetings. The *Contractor* is not entitled to extension of time or reimbursement in cost due to their involvement in such meetings.

1.114 Not Used

PUBLIC COMPLAINTS AND ENQUIRIES CUM INFORMATION CENTRE

Public Complaints and Enquiries Cum Information Centre

1.115 (1) The *Contractor* shall set up and maintain at the Principal Office for the *Project Manager* and the *Supervisor* or Secondary Office erected under this Contract, as directed by the *Project Manager*, one 24-hour Public Complaints and Enquiries Cum Information Centre (the Centre) to attend to calls, enquiries and complaints received and request for information relating to the *works*, including routine maintenance works. The telephone number shall be prominently displayed on Site or on the project signboards.

(2) The Centre shall be manned by at least one operative and shall be equipped with

(a) telecommunication facilities with call waiting and call recording function;

(b) a dedicated fax cum scanner machine with its own telecommunication line; and

(c) a computer notebook.

(3) The Centre is set up to handle incoming calls/faxes/email or other communications from the Integrated Call Centre (ICC) operated by the Efficiency Unit of the Government of Hong Kong SAR or the public and also request for information and assistance from the *Project Manager* and the *Supervisor* and other Government Departments such as Traffic Police and Transport Department.

(4) The Centre is set up to handle incoming calls/faxes/email or other communications from the Integrated Call Centre (ICC) operated by the Efficiency Unit of the Government of Hong Kong SAR or the public and also request for information and assistance from the

Project Manager and the *Supervisor* and other Government Departments such as Traffic Police and Transport Department.

(5) Attendance to communications from ICC/public

When a communication is received from the ICC/public, the Centre shall contact the re1evant *Contractor*'s staff as soon as practicable who shall carry out the following :-

(a) if the incident is obviously outside the ambit of the *works*, the *Contractor*'s staff shall report back to the ICC/public, the *Project Manager* and the *Supervisor* as soon as practicable and not later than 30 minutes after the referral from ICC/public.

(b) if it is not obviously outside the ambit of the *works*, the *Contractor*'s staff shall conduct immediate investigation on site to look into the subject matter of the complaint and report back to the *Project Manager* and the *Supervisor* on his findings within 2 hours of the referral from the ICC/public. This communication with the *Project Manager* and the *Supervisor* shall also be reported to the Centre for record.

(c) if the subject matter is safety related, the *Contractor* shall make all necessary arrangement to ensure safety of the public and the road users and liaise with the relevant authorities for lane closure if necessary.

(d) if the subject matter falls within the responsibility of the *Contractor*, he shall arrange to repair defects referred to by the ICC/public and report completion of works as provided for in the relevant PS Clauses. Such relevant replies to the ICC/public shall be vetted and agreed by the *Project Manager* before sending out to any other parties. In addition, he shall submit a reply form to the *Project Manager* detailing the actions taken or will be taken in the format specified by the *Project Manager* within 5 days of receiving the communication from ICC/public. If the reply is interim only, he shall submit another rep1y in the format specified by the *Project Manager* within 2 days after completion of the work.

(e) if the subject matter falls outside the responsibility of the *Contractor*, he shall seek instruction from the *Project Manager* on the necessary actions to be taken and shall submit a reply form to the *Project Manager* detailing the actions taken or will be taken in the format specified by the *Project Manager* within 5 days of receiving the communication from ICC/public if the reply is interim only, he shall submit another reply in the format specified by the *Project Manager* within 2 days after completion of the *works*. All such relevant replies shall be vetted and agreed by the *Project Manager* before sending out to any other parties.

(f) where applicable, the *Contractor* shall provide within 5 days of request by the *Project Manager* draft responses and briefing material to correspondence received from the general public which has been received by the Centre, the *Project Manager* or

the ICC in respect of any matter referring to the *works*.

(g) The *Contractor* shall notify the *Project Manager* and the *Supervisor* within 1 hour of each communication from ICC/public received during normal working hours, and within 1 hour of the beginning of the next working day, for communication received outside normal working hours unless otherwise specified. The *Project Manager* and the *Supervisor* shal1be kept fully informed of action being taken to address the communication from ICC/public.

(6) Request for assistance and information

The Centre shall respond to request for assistance and information from the *Project Manager* and the *Supervisor* and other Government departments satisfactorily by calling back/email/fax within the time limit as stipulated below :-

Type of Assistance/ Information Requested	Time Limit for Response
Contact a *Contractor*'s staff	Call back originator within 15 minutes of call relayed to the designated staff
Emergency contact telephone nos. of various organizations	Immediate as far as practicable and in any event within 15 minutes of request
Programme for the day's work/ any other days' work within 3 months	Call back or fax or email to originator as per originator's request within 15 minutes
Progress for any day's work and from commencement of Contract Period	Call back or fax or email to originator as per originator's request within 15 minutes
Maintenance history of a road feature/location within specified period	Call back or fax or email to originator as per originator's request within 4 hours
Details of Noise Control Permit and Excavation Permits held in relation to specific works	Call back or fax or email to originator as per originator's request within 4 hours
Location of any item of minimum construction plant	Call back or fax or email to originator as per originator's request within 15 minutes
Inventory records of the road network	Call back or fax or email to originator as per originator's request within 15 minutes
Handling/ assistance in	As soon as practicable

emergency situation	
Other request for information/ assistance relating to the *works*	Respond within the time reasonably requested by the originator

(7) For the avoidance of doubt, the information described above includes information on all the *works*, including routine maintenance works, and the Centre shall be able to respond to requests for such information within the same time limit. The *Contractor* shall provide a liaison officer to make arrangements for *Contractor* to provide assistance in emergency situations and shall have authority to undertake such arrangements.

(8) All communications received by the *Contractor* relating to the *works* whether written or verbal shall be logged in a register by the *Contractor* recording inter alia the following information in respect of each communication received from the *Project Manager* or any other source :-

(i) Date of receipt of communication where written

(ii) Date of receipt of communication where verbal

(iii) Details of communication

(iv) Note on whether communication requires a response

(v) Date(s) of issue of written information to the *Project Manager* with reference number and transmittal method(s)

(vi) Any follow up actions that shall be required by either the *Contractor* or the *Project Manager*

(9) The register specified in the preceding paragraph shall also include the following information summarised for each calendar month and submitted to the *Project Manager* for checking and agreement by the 7th of each calendar month or any date as agreed by the *Project Manager* :-

(i) Number of communications received from the ICC/public in the month

(ii) Number of cases not yet closed, which is carried over from previous months

(iii) Number of requests for assistance/information

(iv) Number of communications with the *Project Manager* or the *Supervisor* relating to ICC/public cases failing to meet the target time for reply or interim reply (one no. counted for each failure)

(v) Number of responses to requests for assistance/information failing to meet the target time for response

(10) The *Project Manager* and the *Supervisor* shall have access at any time to verify and inspect all registers.

1.116 Not Used

1.117 Not Used

UNIFORM

Site Uniform 1.118 The Contractor shall ensure that all workers involved in site works, except workers who are engaged for the execution of the Works on the Site for an aggregated period of 7 days or shorter, wear a set of site uniform when they are on the Site. The site uniform, in the form of polo shirt (short-sleeve or long-sleeve to suit the weather conditions) and trousers, shall adopt the design of the anti-heat stress uniform of the Hong Kong Polytechnic University or a similar design submitted to and not disapproved by the Supervisor within 5 working days after the date of delivery of the design to the Supervisor. The Contractor may show the Contractor's logo or a sub-contractor's logo on the site uniform.

Maintenance 1.119 (1) If the vest for the Casual Workers is to be shared among different Casual Workers, the *Contractor* shall wash the vest and ensure that it is in clean and good conditions each time before it is provided to another Casual Worker.

(2) The *Contractor* shall replace as soon as practicable uniform items provided to Site Personnel and self-employed workers that are damaged, torn, worn, expired or lost.

Design 1.120 (1) The *Contractor* shall within 7 days from the *starting date* submit the design of the uniform items for Site Personnel and self-employed workers to the *Project Manager*. In submitting the design to the *Project Manager* the *Contractor* may propose a standard uniform design developed and adopted by the construction industry that can be or has been used in other contracts provided the design complies with the requirements of the Contract.

(2) The *Project Manager* shall within 2 weeks after receiving the design submitted by the *Contractor* in accordance with sub-clause (1) of this Clause, either approve the design or provide his comments on the design and specify a time within which the *Contractor* shall re-submit his design. The *Project Manager* shall within 1 week after receiving any design re-submitted by the *Contractor* in accordance with this sub-clause, either approve the design or provide his further comments on the design and specify a time within which the *Contractor* shall re-submit his design. The *Contractor* shall submit or re-submit, as appropriate, his design within the time specified by the *Project Manager*.

(3) The design of the uniforms for Site Personnel and self-employed workers shall comply with the following requirements-

General

(i) All items shall have sufficient choices of sizes to reasonably fit

all personnel.

(ii) Clothing shall have no loose or dangling parts which can be easily entangled by machineries.

Clothing

(iii) The fabric for clothing shall be suitable for the designed weather condition.

(iv) The fabric for polo shirt for workers shall be CVC fabric (65% cotton and 35% polyester) with minimum weight of 200 grams per square metre (gsm) or of such fabric as approved by the *Project Manager* and shall comply with the requirements in PS Appendix W1.

(v) Washable retroreflective strips or bands shall be incorporated onto the polo shirts and winter jackets for workers. The layout of the retroreflective strips or bands shall comply with BS EN 471.

(vi) The fabric for summer clothing should preferably be of light color and be thin and absorbent to minimise heat absorption and enhance heat and perspiration dissipation. Subject to sub-clause (3)(ii) of this Clause, summer clothing shall be loose-fitting to enhance perspiration dissipation.

(vii) Trousers for workers shall be 100% cotton or of such fabric as approved by the *Project Manager* and shall comply with the requirements in PS Appendix W2. Trousers shall have semi-elastic adjustable waist size with belt loops and at least two front pockets and two back pockets. The horizontal dimension of belt loops for trousers shall be at least 20mm wide.

(viii) A logo specified by the *Client* or the *Project Manager* shall be integrated into the design and be shown in reasonable size on the top right front side of the uniform top. The Intellectual Property Rights of the specified logo are owned by the *Client* and unless otherwise permitted by the *Client* in writing, the specified logo shall not be reproduced or used in any way other than that stipulated in the Contract. For the avoidance of doubt, the *Client*'s departmental logo or the logo of the *Project Manager*/ *Supervisor* organisation shall not be shown on uniform items for Site Personnel and self-employed workers but the *Contractor* may show the *Contractor*'s logo on the uniform items for Site Personnel and self-employed workers.

(ix) For the uniform for workers, the *Contractor* shall provide independent test certificates for the fabric used for the uniform and for the sample garments to demonstrate to the satisfaction of the *Project Manager* their compliance with the requirements in PS Appendix W1 and PS Appendix W2 as appropriate. The *Contractor* shall submit new independent test certificates whenever there is a change in fabric or uniform supplier. In any case, the *Project Manager* may order tests to be conducted in accordance with the requirements in PS Appendix W1 and

Ps Appendix W2 for the fabric or garments. The expense and costs of such tests shall be borne by the *Contractor*.

Safety helmet

(x) Safety helmets shall comply with the "Guidance Notes on the Selection, Use and Maintenance of Safety Helmets" published by the Labour Department. Safety helmets shall have detachable chin straps and nape strap adjustable by adjustment wheel dial or ratchet knob system.

(xi) The *Contractor* shall propose the colour code of helmets for the approval of the *Project Manager*. The safety helmets of the following three categories of Site Personnel shall be of different colours –

- workers
- *Contractor*'s supervisory, technical and managerial staff
- *Project Manager*'s and *Supervisor*'s staff

(4) The *Contractor* shall, if instructed by the *Project Manager*, submit sample of the uniform items to the *Project Manager*.

1.121 Not Used

TREE RISK ASSESSMENT AND MITIGATION MEASURES

Scope of Work 1.122 Arboricultural services to be carried out by the *Contractor* under this Contract shall cover the following scope of work –

(1) To carry out tree risk assessment for the Trees within the Site through on-the-ground /and aerial inspections

(2) To recommend mitigation measures for risk abatement;

(3) To record inspections and recommendations of each Tree by completing a comprehensive inspection report with details approved by the *Project Manager* in consultation with Highways Department/ Landscape Unit and Leisure and Cultural Services Department;

(4) To carry out and supervise arboricultural works approved by the *Project Manager* in consultation with Highways Department/ Landscape Unit and Leisure and Cultural Services Department, and record detail of works being done;

(5) To provide Qualified Arborists ("QA") to carry out tree assessment and supervise all arboricultural works to the Trees within the Site ; and

(6) To deploy QA to complete all inspection reports and work records.

(7) To deploy QA to complete at least one assessment and provide tree assessment report of Form II for each tree for every 3 months in view of possible damage to the tree during construction period. After

typhoon or rain storm black warning, the QA should conduct inspection to all trees within three days and carry out remedial action such as tree felling or pruning immediately within one week, or an earlier time as requested by HyD and/or LCSD.

(8) To deploy QA to monitor the work on site during any tree remedial work and provide the report with photos before and after the work.

Detailed Service Requirements 1.123 Arboricultural services to be carried out by the *Contractor*, unless otherwise specified, under this Contract shall cover the following requirements –

(1) Service Period

The *Contractor* shall carry out and complete the tasks to the satisfaction of *Project Manager* during the Service Period which is the period from *starting date* to *defect date*.

(2) Trees in the Contract

The *Contractor* shall carry out assessment and arboricultural works for each Tree within the Site.

(3) Provision of Qualified Arborists (QA)

a. The *Contractor* shall deploy QA to carry out tree risk assessment and supervise works concerning tree surgery, rigging, climbing, cabling and pests and disease treatments. Each QA shall possess the following qualifications in either Degree Entry or Non-Degree Entry respectively –

	Degree Entry	Non-Degree Entry
Academic Training	• Recognized Bachelor Degree or above in Arboriculture, Horticulture or Biological Science or equivalent to the standards of Level 5 or above in Hong Kong Qualifications Framework (HKQF);	• Recognized diploma in Arboriculture, Horticulture or Biological Science or equivalent to the standards of Level 3 in Hong Kong Qualifications Framework (HKQF);

Professional Qualifications	• Certified Arborist qualification of International Society of Arboriculture; or • Technician Member, Professional Member or Fellow of the Arboricultural Association of the UK; or • European Tree Technician of the European Arboricultural Council; or • General Member of the National Arborists Association of Australia or equivalent	• Certified Arborist qualification of International Society of Arboriculture; or • Technician Member, Professional Member or Fellow of the Arboricultural Association of the UK; or • European Tree Technician of the European Arboricultural Council; or • General Member of the National Arborists Association of Australia or equivalent
Working experience	• Have at least three (3) years of working experience in arboriculture prior to the Tender Closing Date	• Have at least four (4) years of working experience in arboriculture prior to the Tender Closing Date

b. In case the QA terminates his/her employment with the *Contractor* or the *Project Manager* requests the *Contractor* to replace the QA in view of his/her poor performance and/or for whatever reasons, the *Contractor* shall find another QA with the above qualifications for replacement within fourteen (14) days.

(4) Tree Risk Assessment and Inspection

a. Assessment Cycle

i. QA shall complete at least one assessment for each Tree for twelve (12) months (one assessment cycle) or other period less than 12 months as specified by *Project Manager*.

ii. The *Contractor* shall arrange QA to conduct extra tree health and hazard assessment for relevant Trees under the emergency situations including but not limited to failure of major branches or trunk, blooming of fungal fruiting bodies, accident causing tree injury, acute pest problem and other situation causing imminent danger.

iii. As regards emergency inspection after inclement weather, the *Contractor* shall arrange QA to conduct and complete inspections within one week to all Trees as

listed after cancellation of Typhoon No. 8 or above and Black Rainstorm Warning. The inspections shall focus on the damages, the stability and potential hazard. The QA shall also propose appropriate remedial work and recommendation, if any, in connection with the inspections.

b. Assessment and Inspection Work

i. QA shall conduct visual assessment of each Tree on-the-ground and, if necessary, aerial inspection of each Tree through secured tree climbing or other methods in compliance with working safety at height. The QA shall also assess internal decay of the main trunk and other locations being considered with structural defects.

ii. The QA shall use proper equipment such as wood decay detection equipment and drilling resistance measuring equipment, for checking the condition of Tree's internal decay and take photos for major defects or irregularities spotted. The *Contractor* shall provide all tools and equipment as approved by the *Project Manager*, Highways Department/ Landscape Unit and Leisure and Cultural Services Department for its QA in carrying out the work.

iii. The QA shall not cause any unnecessary damages to any tree parts or excessive soil compaction during inspection.

iv. The *Contractor* shall arrange its own transport and provide all necessary equipment and machineries for its QA to carry out assessments.

v. The *Contractor* shall cordon-off the site during the aerial inspection and make all arrangements for site cordon-off or road closure, if necessary.

vi. The *Contractor* shall ensure that all their staff shall wear proper safety clothing and use all equipment correctly.

vii. The *Contractor* shall also take every safety precaution to eliminate danger to its staff, the general public and property of others during tree inspection.

c. Assessment and Inspection Report

i. As regards routine assessment and emergency inspection for situation other than caused by inclement weather, QA shall complete and sign a full comprehensive assessment report for each Tree. The *Contractor* shall vet and endorse the comprehensive assessment report before submission to the *Project Manager*, Highways Department/ Landscape Unit and Leisure and Cultural Services Department. The comprehensive assessment report shall include but not limited to the following

information –

- Tree general information
  - Register number and location;
  - Date of inspection;
  - Tree descriptions: species, diameter at breast height, height, crown spread, etc.
- Observations
  - Tree form descriptions;
  - Trunk defects, such as co-dominant stem, included bark, crack, bark peeling off, cavity, presence of fungal bodies, sap exudation, signs of borers or other pest infestation, etc.;
  - Leaf abnormality, such as abnormal size, colour, and defoliation, signs of pest and disease infection, etc.;
  - Branch defects, such as die-back, broken, crack, presence of fungal bodies, signs of borers or other pest infestation, etc.;
  - Signs of root defects, such as soil crack, root plate movement, presence of fungal bodies on root plate, etc.;
  - Site conditions; and,
  - Other observations
- Views on risk caused by defects with justifications;
- Recommended measures for risk abatement; and
- Attachments for witnessing the content of the report (photographs, tomograph, resistograph, digital microprobe drill chart, etc.)

ii. During tree assessment or inspection, the QA shall also check the condition of installations, if any, such as cable, bracing rod and propping structure so as to identify any visual damage and other irregularities.

d. Recommendations on Mitigating Measures

i. Based on the result of each assessment report, the QA shall submit to the *Project Manager*, Highways Department/ Landscape Unit and Leisure and Cultural Services Department, a proposal for mitigating measures to each Tree in the assessment or the inspection report.

The proposals shall be realistic, practicable and workable and without causing permanent blockage of vehicular or pedestrian traffic.

ii. In addition to word description, the QA may use annotated plans, photos, sketches, or drawings for clear indication of the following pruning works or installations –

Pruning/Installation	**Required details**
Crown Reduction	• Overall crown reduction: recommended percentage; or • Selected branch reduction: location of major limbs (direct connection to the trunk), position of the cut and percentage of reduction.
Crown Thinning	• Location of major limbs (if any) to be removed and position of the cut
Crown Lifting	• Location of major limbs (if any) to be removed and position of the cut
Cabling	• Exact location of each limb involved • Position of each cable and its connection points on approximate position of related limbs • Suggested specification of cable and end-connection between the limb and the cable • Exact location and the limb position of another tree if involved in cabling
Bracing	• Exact location of each limb involved • Position of each bracing rod, its passing through angle and the approximate installation position of related limbs • Suggested specification of bracing rod

Guying	• Exact location of each limb involved • Position of each cable and its connection point on approximate position of related limb and that of the ground/other surface • Suggested specification of cable and end-connection between the limb and the cable • Proposed specification of the ground support
Propping	• Exact location of each limb involved • Position of each prop and its connection point on approximate position of related limb and that of the ground/other surface • Suggested detailed design of the propping structure, specification of the prop and its foundation • The connection details between the prop and the main trunk or the limb, and between the prop and the ground/other surface

iii. As regards recommended chemical treatment for nutrition problem or pests and diseases, the QA shall specify chemicals to be used with potential chemical suppliers, application method and schedule.

iv. After submission, the QA shall present the reports and the proposals as and when called upon to do so by the *Project Manager*, Highways Department/ Landscape Unit and Leisure and Cultural Services Department. . The QA shall, if necessary, revise the recommendations by incorporating further detailed elaboration as required by and up to the satisfaction of the *Project Manager*, Highways Department/ Landscape Unit and Leisure and Cultural Services Department. The amended proposals shall be re-submitted to the *Project Manager*, Highways Department/ Landscape Unit and Leisure and Cultural Services Department by one week.

(5) Arboricultural Works

a. Schedule for Work Completion

The *Contractor* shall complete all arboricultural works within 60 days after the satisfaction of the *Project Manager*, Highways Department/ Landscape Unit and Leisure and Cultural Services Department on the Recommendations on Mitigating Measures. Except for emergent arboricultural works which should be carried out immediately to mitigate the

hazard.

b. Emergency Works

i. The *Contractor* shall be responsible for providing QA, other workers and equipment for carrying out remedial works under emergency situations.

ii. The *Contractor* shall provide the QA, other workers and equipment to finish all remedial operations within two working days from the day of the sustained damage. If the *Contractor* shall be unable to finish the remedial works within the two-day period, the *Contractor* shall submit a schedule of remedial works with a proposed date for completion for *Project Manager* in consultation with Highways Department/ Landscape Unit and Leisure and Cultural Services Department's approval.

c. Pruning

i. The QA shall be responsible for carrying out and supervising pruning works, including crown cleaning, crown lifting, crown thinning and crown reduction.

ii. In carrying out pruning, the *Contractor* shall ensure its QA to comply with either of the following standards (the latest version) –

- American National Standard for Tree Care Operations (ANSI A300(Part 1)) – Tree, Shrub, and Other Woody Plant Maintenance – Standard Practices (Pruning); or

- British Standard for Tree Work (BS 3998:1989); or;

- Other international standards approved by the *Project Manager*, Highways Department/ Landscape Unit and Leisure and Cultural Services Department.

iii. Except for removal of defective major limbs of which approval from the *Project Manager*, Highways Department/ Landscape Unit and Leisure and Cultural Services Department shall be sought, the QA shall carry out crown cleaning to remove dead, diseased, or broken branches and twigs during tree assessment or emergency inspection.

iv. The QA shall carry out and supervise crown lifting, crown thinning and crown reduction in accordance with the recommendations approved by the *Project Manager*, Highways Department/ Landscape Unit and Leisure and Cultural Services Department.

v. As regards emergency pruning against imminent danger, the QA shall immediately seek verbal consent from the

Project Manager, Highways Department/ Landscape Unit and Leisure and Cultural Services Department prior to the pruning work.

vi. The *Contractor* shall provide chainsaws, rigging ropes, lifting platform and all necessary equipment for its QA and other workers to carry out pruning works.

vii. The QA shall arrange secured lowering of cut limbs that likely cause damage or injury when free falling onto the ground.

viii. The pruning operation shall not cause damage to the nearby planting or structure. All cut branches carrying diseases or insects have to be bagged in a recyclable polythene bag and properly disposed to avoid contamination.

d. Work Record

i. The *Contractor* shall provide a record of work completion for each Tree by using the form approved by the *Project Manager*, Highways Department/ Landscape Unit and Leisure and Cultural Services Department. All such records shall be signed by QA.

e. Safety Requirements

i. The *Contractor* shall provide proper climbing equipment and protective clothing for its QA and at all times ensure that all workers operating the chainsaws and other machineries shall have been trained properly and wear safety boots, goggles, protective clothing and reflective safety vests of a conspicuous colour.

ii. The *Contractor* shall ensure that all personnel working on highway structures (including footbridges, non-expressway bridges and bridge columns) shall be made known that they have a statutory duty to take reasonable care of their own health and safety as well as that of others their actions or omissions at work may affect. The hazards associated with personnel working on highway structures include working at height, falling objects and traffic accidents. All personnel working on highway structures shall wear reflective belts, safety belts or safety harnesses, lifelines which are connected to stable anchorage points.

iii. The *Contractor* shall be responsible to take every safety precaution to eliminate danger to its workmen, the general public and property of others. The *Contractor* shall follow the safety requirements as mentioned in one of the following standards (the latest version) –

- American National Standard for Tree Care Operations (ANSI A300(Part 1)) – Tree, Shrub, and

Other Woody Plant Maintenance – Standard Practices (Pruning); or

- British Standard for Tree Work (BS 3998:1989); or;
- Other international standards approved by the *Project Manager*, Highways Department/ Landscape Unit and Leisure and Cultural Services Department.

iv. Before commencing tree pruning or climbing works, the *Contractor* shall ensure that there is no overhead high voltage cable near the working area and that pruning operation will not cause damage to the nearby planting or structure. Safety measures must be taken to ensure the safety of the public. The *Contractor* shall ensure that neither unrelated person nor animal stays nearby during operation and coordinate with relevant authorities regarding any special pruning equipment and/or lighting requirements which may interrupt traffic. Warning signs shall be put up to keep public away from the pruning operation.

v. The *Contractor* shall ensure that all chainsaws or other machineries used shall meet safety requirements in Hong Kong.

vi. The *Project Manager*, Highways Department/ Landscape Unit and Leisure and Cultural Services Department reserves the right to direct the immediate removal of any unsafe machines from work sites and any subsequent monetary claim to the *Contractor* concerned in case of its delay in completion of Services due to consequence of such removal of machines shall be to the *Contractor*'s disadvantage and liability.

vii. The *Contractor* shall ensure safety and health of its employees and comply with the Occupational Safety and Health Ordinance, Cap. 509 and its sublegislations.

f. Collection and Disposal of Refuse

i. The *Contractor* shall be responsible for the picking up and removal of tree debris, cut pieces of wood, refuse generated from tree assessment/inspection and arboricultural works. The *Contractor* shall also clear away all debris or refuse, such as joss sticks, incensed idols, construction debris, etc. within the dripline.

ii. The *Contractor* shall provide recyclable polythene bags and transport the refuse collected to the controlled tips at its own cost using wheeled handcarts or any other appropriate equipment in properly covered containers.

iii. No damage should be caused to the planting in the working sites when carrying out cleansing duties by the

Contractor.

iv. The *Contractor* shall strictly prohibit its employees or agents from sweeping or dumping the refuses onto the adjoining pavements, surface channels or gullies, or other area or sites at which refuse dumping is not allowed.

g. Work Arrangement along Carriageways

i. The *Contractor* shall at its own expense make all arrangements for road closure so as to carry out tree assessment/inspection and arboricultural works.

ii. The *Contractor* shall arrange sub-contractor for lane closure from approved list of the Highways Department and obtain the necessary approvals from Commissioner for Transport, the Commissioner of Police, the Director of Highways and any other relevant authority for temporary traffic arrangements. The *Contractor* shall ensure that its workers shall strictly follow the requirements for temporary traffic arrangements as stipulated in the latest version of Code of Practice for the Lighting, Signing and Guarding of Road Works issued by the Highways Department.

iii. The *Contractor* shall also comply with any other conditions and restrictions imposed by the Commissioner for Transport, the Commissioner of Police and the Director of Highways.

Reference

1.124 (1) In addition to the above, the *Contractor* can make reference to the "Guidelines for Tree Risk Management and Assessment Arrangement on an Area Basis and on a Tree Basis" included in PS Appendix X.

1.125 Not Used

1.126 Not Used

1.127 Not Used

1.128 Not Used

1.129 Not Used

1.130 Not Used

FABRICATION WORKS

Fabrication Works

1.131 (1) The *Contractor* shall submit to the *Project Manager* for acceptance a plan detailing the location and facilities of the proposed fabrication yard, the area for storage of the fabricated elements prior to transportation to the Site, the means of the transportation of the fabricated works from the proposed fabrication yard to the Site, the arrangement for the *Project Manager* or *Supervisor* or *Client* to supervise or inspect the works in the proposed fabrication yard, the

method statement, the quality assurance system and any further submissions and requirements as may be imposed by the *Supervisor*. The *Contractor* shall be responsible for arranging, coordinating, and agreeing with the relevant authorities or owners of the proposed fabrication yard. The *Contractor* shall allow all these necessary costs in his Prices incurred by the transportation of the fabricated works and in relation to or arising from the formation, use, maintenance and subsequent clearance and evacuation of the proposed fabrication yard.

(2) When the proposed fabrication yard are proposed to be in Mainland China or other place outside the Hong Kong Special Administrative Region, the *Contractor* shall make all necessary arrangements to facilitate the *Project Manager* or *Supervisor* to supervise the works in the proposed fabrication yard on a full-time supervision basis. Furthermore, the *Contractor* shall allow all these necessary costs in his Prices incurred by the *Project Manager* or *Supervisor* or *Client* in relation to and arising from the supervision or inspection including, but not limited to, travel arrangement, site accommodation, overnight accommodation, subsistence allowance, payment of China taxes on corporate income and out of pocket expenses for the procurement of a tax agent, etc..

(3) The *Contractor* shall not commence any works on the proposed fabrication yard in Mainland China or other place outside the Hong Kong Special Administrative Region without the prior acceptance of the *Supervisor*.

B5 DIESEL

B5 Diesel 1.132 (1) B5 diesel shall be used as fuel for all non-road based construction machinery on the Site. The non-road based construction machinery includes all types of machines and plants which are powered by internal combustion engines, such as air compressors, mobile generators, excavators, crawler cranes, bulldozers, rollers, etc.

(2) The B5 diesel shall be Euro V diesel containing a fatty acid methyl ester content of 5%, as determined in accordance with the document published by the European Committee for Standardization commonly known as EN 14078:2003. The Euro V diesel shall be light diesel oil complying with the specifications set out in Chapter 311L Schedule 1 — Air Pollution Control (Motor Vehicle Fuel) Regulation of the Laws of Hong Kong, all applicable legislation in Hong Kong and the requirements of all relevant regulatory authorities in Hong Kong which are in effect on the date that the B5 diesel is supplied.

(3) The Contractor shall maintain a proper record of the use of B5 diesel on Site for inspection by the Project Manager whenever necessary.

(4) Notwithstanding Clause (1) above, the Contractor may propose for the Supervisor's acceptance for the use of Euro V diesel instead of B5 diesel should there be a supply shortage or other difficulties in delivering of B5 diesel to the Site.

MANAGEMENT OF SUB-CONTRACTORS

General 1.133

(1) The Contractor shall ensure the submission and quarterly updating of the Sub-contractor Management Plan (SMP) in the form and contents as prescribed in the *contract*.

(2) The *Contractor* shall monitor and ensure the implementation of and the compliance with the SMP.

SMP 1.134

The *Contractor* shall, in accordance with *additional conditions of contract* Clause C5, prepare and submit to the *Supervisor* 2 copies of the Sub-contractor Management Plan signed by the Site Agent. The SMP shall contain detailed information as required by the Guideline on Scope and Contents of the Sub-contractor Management Plan at Appendix B to the *additional conditions of contract*.

Quarterly updated SMP 1.135

(1) The *Contractor* shall, in accordance with *additional conditions of contract* Clause C5, submit to the *Supervisor* 2 copies of the quarterly updated Subcontractor Management Plan signed by the Site Agent. The quarterly updated SMP shall contain updated detailed information as required by the Guideline on Scope and Contents of the Sub-contractor Management Plan at Appendix B to the *additional conditions of contract*.

(2) If there is no change to the previous SMP, the *Contractor* shall, in accordance with the *additional conditions of contract* Clause C5, declare such status in writing instead of submitting the same SMP again. The declaration shall be signed by the Site Agent and for the purpose of this Particular Specification, the declaration shall be considered as a quarterly updating of the SMP.

MANAGEMENT OF SUB-CONTRACTORS

Systematic Risk Management 1.136

During the course of the *contract*, the *Contractor* shall, when required by the *Supervisor*, participate in the Systematic Risk Management activities and prepare risk reports in accordance with ETWB TCW No. 6/2005 and ETWB Risk Management User Manual. These activities may include attending risk management workshops, meeting with relevant stakeholders, preparing the *Contractor*'s own risk treatment plans, risk management plans and risk registers, etc. Unless otherwise specifically provided for in the contract, the *Contractor* shall not be entitled to any additional payment for performing such management activities. Nevertheless, for the avoidance of doubt, it is clarified that any instructions, orders, directions or decisions of the *Supervisor* given or made in accordance with the *contract* or any variation to the *works* ordered in accordance with the *contract* due to a particular identified risk treatment activity shall be dealt with under the relevant provisions in the *contract*.

1.137 Not Used

1.138 Not Used

1.139 Not Used

1.140 Not Used

1.141 Not Used

1.142 Not Used

1.143 Not Used

MODULAR INTEGRATED CONSTRUCTION (MIC) AND DESIGN FOR MANUFACTURE AND ASSEMBLY (DFMA)

Modular Integrated Construction (MiC) and Design for Manufacture and Assembly (DfMA)

1.144 The *Contractor* are encouraged to adopt innovation in their design and/or method statement proposals by application of new approaches and tools that support the construction of the project to enhance time efficiency, quality and sustainability, improve safety and project functionality. Off-site construction using modular integrated construction (MiC) and Design for Manufacture and Assembly (DfMA) for both temporary and permanent works, are encouraged to be adopted to improve the effectiveness of project management, enhance cost-effectiveness and productivity as well as safeguard occupational health and safety. Detailed proposal providing manufacture locations, MiC / DfMA components, method of installation, the associated temporary works, supervision and resource plan, quality assurance plan, etc. shall be submitted for *Project Manager*'s acceptance. The Project Manager's or the Supervisor's acceptance of the Contractor's design or of his work does not change the Contractor's responsibility to Provide the Works or his liability for his design. For off-site construction, the *Contractor* shall refer to the details stated in the Publication of Construction 2.0 – Time to change, which was promulgated in September 2018 by Development Bureau (DevB) and is available to download at: http://www.hkc2.hk/en/c20.html.

For those parts of *Contractor's* design which are feasible for adoption of MiC / DfMA technique, the *Contractor* shall submit the design in accordance with relevant contract specification. DfMA technology for construction of Noise Barrier Works shall refer PS 27.03.

The Project Manager's acceptance / non-acceptance of the Contractor's design and request for revision to the Contractor's design would not constitute a compensation event and would not incur any change to the Prices and /or Completion Date.

PUBLIC RELATIONS (PR) WORKS

Public Relations (PR) Works

1.145 (1) The *Contractor* shall assist the *Project Manager* and the *Supervisor* in handling the PR issues including, but not limited to, the following:

(i) Inform and explain to the public the objectives, implementation, schedule and impacts of the project;

(ii) Promote the project to the public and the parties concerned with a positive message;

(iii) Gain support and minimize objections from the community and the parties concerned;

(iv) Prepare and supply all necessary drawings, photomontages,

documents, consultation papers, presentations, display materials for public consultations; and

(v) Provide assistance and information to facilitate all PR activities as instructed by the Project Manager and the Supervisor and/or his PR Consultants.

(2) The *Contractor*'s attention is drawn to the fact that the PR Works as mentioned in Clause 1.145 (1) above, may be executed outside the Works Limit of the sites and within the territory of HKSAR.

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